

DUBAI RAPID LINK CONSORTIUM

DUBAI METRO PROJECT OFFICE

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		DM001-E-MAC-CDP-DR-DCC-374901	A1

**Dubai Metro, UAE
Rashidiya Main Depot**

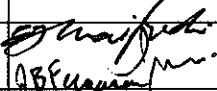
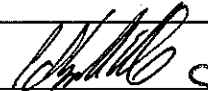
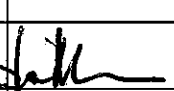
**Light Repair Workshop
(Building 2)**

DCP3 Submission

01 July 2006

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Note: App. and Auth. mean "Approved" and "Authorized" respectively.

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Rev A1	Page 2 of 13

Table	
1	Introduction.....4
2	Drawings.....4
Appendices10	
Appendix A1– Architectural Specification for Building 2.....11	
Appendix B1– Design Calculation for Building 2.....12	
Appendix B2– Design Report for Building 2.....13	

1 Introduction

The Dubai Metro Project is under construction by the Dubai Rapid Link Consortium (DURL). The Civil Works are contracted to the Japan Turkey Metro Joint Venture (JTMJV), and the Rail Systems Works are contracted to Mitsubishi Heavy Industries (MHI). The rolling stock will be provided by Kinki Sharyo.

This DCP3 submission for Light Repair Workshop (Building) 2 in Rashidiya Depot is made by the Designers for the JTMJV, and as such concentrates on the scope of the Civil Works Contract, which includes the Architecture, structure and M&E Design.

2 Drawings

The following drawings are submitted along with this report.

	<u>Drawing Title</u>	<u>Drawing No.</u>	<u>Rev.</u>
Structural Drawings			
1	Main Depot Rashidiya – Building 2 Light Repair Workshop: Drawing List	DM001-E-MAC-DEG-DD-DCC- 370600	A1
2	Main Depot Rashidiya – General Notes and Typical Details Sheet 1	DM001-E-MAC-DEG-DD-DCC- 370401	A1
3	Main Depot Rashidiya – General Notes and Typical Details Sheet 2 of 2	DM001-E-MAC-DEG-DD-DCC- 370402	A1
4	Main Depot Rashidiya – Building 2 Light Repair Workshop: Foundation Layout Plan	DM001-E-MAC-DEG-DD-DCC- 370601	A2
5	Main Depot Rashidiya – Building 2 Light Repair Workshop: Ground Slab Layout Plan	DM001-E-MAC-DEG-DD-DCC- 370605	A2
6	Main Depot Rashidiya – Building 2 Light Repair Workshop: Stringers Rail Layout Plan	DM001-E-MAC-DEG-DD-DCC- 370611	A2
7	Main Depot Rashidiya – Building 2 Light Repair Workshop: Roof Layout Plan	DM001-E-MAC-DEG-DD-DCC- 370615	A2
8	Main Depot Rashidiya – Building 2 Light Repair Workshop: Section A-A	DM001-E-MAC-DEG-DD-DCC- 370621	A2
9	Main Depot Rashidiya – Building 2 Light Repair Workshop: Section B-B	DM001-E-MAC-DEG-DD-DCC- 370622	A1
10	Main Depot Rashidiya – Building 2 Light Repair Workshop: Section C-C	DM001-E-MAC-DEG-DD-DCC- 370623	A1
11	Main Depot Rashidiya – Building 2 Light Repair Workshop: Section D-D	DM001-E-MAC-DEG-DD-DCC- 370624	A1

	<u>Drawing Title</u>	<u>Drawing No.</u>	<u>Rev.</u>
12	Main Depot Rashidiya – Building 2 Light Repair Workshop: Footing Details	DM001-E-MAC-DEG-DD-DCC- 370631	A1
13	Main Depot Rashidiya – Building 2 Light Repair Workshop: Typical Slab Reinforcement – Zone 1 & 2	DM001-E-MAC-DEG-DD-DCC- 370635	A1
14	Main Depot Rashidiya – Building 2 Light Repair Workshop: Slab Reinforcement at GL 2-1 to 2-2, 2-13 to 2-14, & 2-25 to 2-26	DM001-E-MAC-DEG-DD-DCC- 370636	A1
15	Main Depot Rashidiya – Building 2 Light Repair Workshop: Beam Reinforcement	DM001-E-MAC-DEG-DD-DCC- 370641	A1
16	Main Depot Rashidiya – Building 2 Light Repair Workshop: Staircase Plan & Sections – Sheet 1	DM001-E-MAC-DEG-DD-DCC- 370651	A1
17	Main Depot Rashidiya – Building 2 Light Repair Workshop: Staircase Plan & Sections – Sheet 2	DM001-E-MAC-DEG-DD-DCC- 370652	A1
18	Main Depot Rashidiya – Building 2 Light Repair Workshop: Bracing Connection Details	DM001-E-MAC-DEG-DD-DCC- 370655	A1
Architectural Drawings			
19	Main Depot Rashidiya – Building 2 Light Repair Workshop: Drawing List	DM001-E-MAC-DC3-DD-DCC- 370060	A1
20	Main Depot Rashidiya – Building 2 Light Repair Workshop: G.A. Plan Layout Level 1	DM001-E-MAC-DC3-DD-DCC- 370061	A1
21	Main Depot Rashidiya – Building 2 Light Repair Workshop: Setting Out Plan Level 1 – 1/3	DM001-E-MAC-DC3-DD-DCC- 370062	A1
22	Main Depot Rashidiya – Building 2 Light Repair Workshop: Setting Out Plan Level 1 – 2/3	DM001-E-MAC-DC3-DD-DCC- 370063	A1
23	Main Depot Rashidiya – Building 2 Light Repair Workshop: Setting Out Plan Level 1 – 3/3	DM001-E-MAC-DC3-DD-DCC- 370064	A1
24	Main Depot Rashidiya – Building 2 Light Repair Workshop: Setting Out Plan Level 2 – 1/3	DM001-E-MAC-DC3-DD-DCC- 370065	A1
25	Main Depot Rashidiya – Building 2 Light Repair Workshop: Setting Out Plan Level 2 – 2/3	DM001-E-MAC-DC3-DD-DCC- 370066	A1
26	Main Depot Rashidiya – Building 2 Light Repair Workshop: Setting Out Plan Level 2 – 3/3	DM001-E-MAC-DC3-DD-DCC- 370067	A1

	<u>Drawing Title</u>	<u>Drawing No.</u>	<u>Rev.</u>
27	Main Depot Rashidiya – Building 2 Light Repair Workshop: Roof Plan	DM001-E-MAC-DC3-DD-DCC- 370068	A1
28	Main Depot Rashidiya – Building 2 Light Repair Workshop: G.A. Elevations	DM001-E-MAC-DC3-DD-DCC- 370069	A1
29	Main Depot Rashidiya – Building 2 Light Repair Workshop: G.A. Elevations	DM001-E-MAC-DC3-DD-DCC- 370070	A1
30	Main Depot Rashidiya – Building 2 Light Repair Workshop: G.A. Section (1/2)	DM001-E-MAC-DC3-DD-DCC- 370071	A1
31	Main Depot Rashidiya – Building 2 Light Repair Workshop: G. A. Sections (2/2)	DM001-E-MAC-DC3-DD-DCC- 370072	A1
32	Main Depot Rashidiya – Building 2 Light Repair Workshop: Detailed Sections	DM001-E-MAC-DC3-DD-DCC- 370081	A1
33	Main Depot Rashidiya – Building 2 Light Repair Workshop: Detailed Sections	DM001-E-MAC-DC3-DD-DCC- 370082	A1
34	Main Depot Rashidiya – Building 2 Light Repair Workshop: Detailed Sections	DM001-E-MAC-DC3-DD-DCC- 370083	A1
35	Main Depot Rashidiya – Building 2 Light Repair Workshop: Typical Details Roof Cladding Sheet 1 of 2	DM001-E-MAC-DC3-DD-DCC- 370341	A1
36	Main Depot Rashidiya – Building 2 Light Repair Workshop: Typical Details Roof Cladding Sheet 1 of 2	DM001-E-MAC-DC3-DD-DCC- 370342	A1
37	Main Depot Rashidiya – Building 2 Light Repair Workshop: Typical Details Roof Cladding Sheet 2 of 2	DM001-E-MAC-DC3-DD-DCC- 370346	A1
38	Main Depot Rashidiya – Building 2 Light Repair Workshop: Typical Details External Wall Cladding 1/4	DM001-E-MAC-DC3-DD-DCC- 370348	A1
39	Main Depot Rashidiya – Building 2 Light Repair Workshop: Typical Details Internal Partition 1/3	DM001-E-MAC-DC3-DD-DCC- 370355	A1
40	Main Depot Rashidiya – Building 2 Light Repair Workshop: Typical Details Internal Partition 2/3	DM001-E-MAC-DC3-DD-DCC- 370356	A1
41	Main Depot Rashidiya – Building 2 Light Repair Workshop: Typical Details Internal Partition 3/3	DM001-E-MAC-DC3-DD-DCC- 370357	A1
42	Main Depot Rashidiya – Building 2 Light Repair Workshop: Typical Details Roller Shutter Details	DM001-E-MAC-DC3-DD-DCC- 370370	A1

	<u>Drawing Title</u>	<u>Drawing No.</u>	<u>Rev.</u>
M & E Drawings			
43	Main Depot Rashidiya – Building 2 Light Repair Workshop: List of Drawings for Electrical System	DM001-E-MAC-CEM-DD-DCC- 362060	A1
44	Main Depot Rashidiya – Building 2 Light Repair Workshop: Electrical System Symbols 1/2	DM001-E-MAC-CEM-DD-DCC- 362061	A1
45	Main Depot Rashidiya – Building 2 Light Repair Workshop: Electrical System Symbols 2/2	DM001-E-MAC-CEM-DD-DCC- 362062	A1
46	Main Depot Rashidiya – Building 2 Light Repair Workshop: LV Electrical Schematic	DM001-E-MAC-CEM-DD-DCC- 362070	A1
47	Main Depot Rashidiya – Building 2 Light Repair Workshop: Lighting Control System Diagram	DM001-E-MAC-CEM-DD-DCC- 362071	A1
48	Main Depot Rashidiya – Building 2 Light Repair Workshop: Building Management System Diagram	DM001-E-MAC-CEM-DD-DCC- 362072	A1
49	Main Depot Rashidiya – Building 2 Light Repair Workshop: Small Power & Ancillary Services for Level 1 – 1/4	DM001-E-MAC-CEM-DD-DCC- 362080	A1
50	Main Depot Rashidiya – Building 2 Light Repair Workshop: Small Power & Ancillary Services for Level 1 – 2/4	DM001-E-MAC-CEM-DD-DCC- 362081	A1
51	Main Depot Rashidiya – Building 2 Light Repair Workshop: Small Power & Ancillary Services for Level 1 – 3/4	DM001-E-MAC-CEM-DD-DCC- 362082	A1
52	Main Depot Rashidiya – Building 2 Light Repair Workshop: Small Power & Ancillary Services for Level 1 – 4/4	DM001-E-MAC-CEM-DD-DCC- 362083	A1
53	Main Depot Rashidiya – Building 2 Light Repair Workshop: Lighting/Emergency Lighting for Level 1 – 1/4	DM001-E-MAC-CEM-DD-DCC- 362100	A1
54	Main Depot Rashidiya – Building 2 Light Repair Workshop: Lighting/Emergency Lighting for Level 1 – 2/4	DM001-E-MAC-CEM-DD-DCC- 362101	A1
55	Main Depot Rashidiya – Building 2 Light Repair Workshop: Lighting/Emergency Lighting for Level 1 – 3/4	DM001-E-MAC-CEM-DD-DCC- 362102	A1
56	Main Depot Rashidiya – Building 2 Light Repair Workshop: Lighting/Emergency Lighting for Level 1 – 4/4	DM001-E-MAC-CEM-DD-DCC- 362103	A1

	<u>Drawing Title</u>	<u>Drawing No.</u>	<u>Rev.</u>
57	Main Depot Rashidiya – Building 2 Light Repair Workshop: Lightning Protection/Earthing for Level 1 – 1/4	DM001-E-MAC-CEM-DD-DCC- 362120	A1
58	Main Depot Rashidiya – Building 2 Light Repair Workshop: Lightning Protection/Earthing for Level 1 – 2/4	DM001-E-MAC-CEM-DD-DCC- 362121	A1
59	Main Depot Rashidiya – Building 2 Light Repair Workshop: Lightning Protection/Earthing for Level 1 – 3/4	DM001-E-MAC-CEM-DD-DCC- 362122	A1
60	Main Depot Rashidiya – Building 2 Light Repair Workshop: Lightning Protection/Earthing for Level 1 – 4/4	DM001-E-MAC-CEM-DD-DCC- 362123	A1
61	Main Depot Rashidiya – Building 2 Light Repair Workshop: Building Management System for Level 1	DM001-E-MAC-CEM-DD-DCC- 362140	A1
62	Main Depot Rashidiya – Building 2 Light Repair Workshop: List of Drawings for Fire System	DM001-E-MAC-CEM-DD-DCC- 362170	A1
63	Main Depot Rashidiya – Building 2 Light Repair Workshop: Fire Alarm System Symbols	DM001-E-MAC-CEM-DD-DCC- 362171	A1
64	Main Depot Rashidiya – Building 2 Light Repair Workshop: Fire Alarm Schematic	DM001-E-MAC-CEM-DD-DCC- 362180	A1
65	Main Depot Rashidiya – Building 2 Light Repair Workshop: Fire Alarm System for Level 1 – 1/4	DM001-E-MAC-CEM-DD-DCC- 362190	A1
66	Main Depot Rashidiya – Building 2 Light Repair Workshop: Fire Alarm System for Level 1 – 2/4	DM001-E-MAC-CEM-DD-DCC- 362191	A1
67	Main Depot Rashidiya – Building 2 Light Repair Workshop: Fire Alarm System for Level 1 – 3/4	DM001-E-MAC-CEM-DD-DCC- 362192	A1
68	Main Depot Rashidiya – Building 2 Light Repair Workshop: Fire Alarm System for Level 1 – 4/4	DM001-E-MAC-CEM-DD-DCC- 362193	A1
69	Main Depot Rashidiya – Building 2 Light Repair Workshop: List of Drawings for Sanitary and Drainage System	DM001-E-MAC-CEM-DD-DCC- 362220	A1
70	Main Depot Rashidiya – Building 2 Light Repair Workshop: Sanitary and Drainage System Symbols	DM001-E-MAC-CEM-DD-DCC- 362221	A1
71	Main Depot Rashidiya – Building 2 Light Repair Workshop: Schematic Diagram Sanitary and Drainage System	DM001-E-MAC-CEM-DD-DCC- 362230	A1
72	Main Depot Rashidiya – Building 2 Light Repair Workshop: System	DM001-E-MAC-CEM-DD-DCC- 362231	A1

	<u>Drawing Title</u>	<u>Drawing No.</u>	<u>Rev.</u>
73	Main Depot Rashidiya – Building 2 Light Repair Workshop: Detail of Fire Water Lifting Sump Pit and Oil Interceptor	DM001-E-MAC-CEM-DD-DCC- 362232	A1
74	Main Depot Rashidiya – Building 2 Light Repair Workshop: Drainage Piping Elevation & Detail of Backflow Preventer	DM001-E-MAC-CEM-DD-DCC- 362233	A1
75	Main Depot Rashidiya – Building 2 Light Repair Workshop: Water Supply System for Level 1 – 1/4	DM001-E-MAC-CEM-DD-DCC- 362240	A1
76	Main Depot Rashidiya – Building 2 Light Repair Workshop: Water Supply System for Level 1 – 2/4	DM001-E-MAC-CEM-DD-DCC- 362241	A1
77	Main Depot Rashidiya – Building 2 Light Repair Workshop: Water Supply System for Level 1 – 3/4	DM001-E-MAC-CEM-DD-DCC- 362242	A1
78	Main Depot Rashidiya – Building 2 Light Repair Workshop: Water Supply System for Level 1 – 4/4	DM001-E-MAC-CEM-DD-DCC- 362243	A1
79	Main Depot Rashidiya – Building 2 Light Repair Workshop: Sanitary and Drainage System for Level 1 – 1/4	DM001-E-MAC-CEM-DD-DCC- 362250	A1
80	Main Depot Rashidiya – Building 2 Light Repair Workshop: Sanitary and Drainage System for Level 1 – 2/4	DM001-E-MAC-CEM-DD-DCC- 362251	A1
81	Main Depot Rashidiya – Building 2 Light Repair Workshop: Sanitary and Drainage System for Level 1 – 3/4	DM001-E-MAC-CEM-DD-DCC- 362252	A1
82	Main Depot Rashidiya – Building 2 Light Repair Workshop: Sanitary and Drainage System for Level 1 – 4/4	DM001-E-MAC-CEM-DD-DCC- 362253	A1
83	Main Depot Rashidiya – Building 2 Light Repair Workshop: List of Drawings for Air-Conditioning System	DM001-E-MAC-CEM-DD-DCC- 362290	A1
84	Main Depot Rashidiya – Building 2 Light Repair Workshop: Air- Conditioning System Symbols 1/2	DM001-E-MAC-CEM-DD-DCC- 362291	A1
85	Main Depot Rashidiya – Building 2 Light Repair Workshop: Air- Conditioning System Symbols 2/2	DM001-E-MAC-CEM-DD-DCC- 362292	A1
86	Main Depot Rashidiya – Building 2 Light Repair Workshop: Schematic Piping Diagram for Air-Conditioning System	DM001-E-MAC-CEM-DD-DCC- 362300	A1
87	Main Depot Rashidiya – Building 2 Light Repair Workshop: LV Electrical Schematic	DM001-E-MAC-CEM-DD-DCC- 362301	A1
88	Main Depot Rashidiya – Building 2 Light Repair Workshop: Air- Conditioning System for Level 1– 1/4	DM001-E-MAC-CEM-DD-DCC- 362310	A1

	<u>Drawing Title</u>	<u>Drawing No.</u>	<u>Rev.</u>
89	Main Depot Rashidiya – Building 2 Light Repair Workshop: Air- Conditioning System for Level 1– 2/4	DM001-E-MAC-CEM-DD-DCC- 362311	A1
90	Main Depot Rashidiya – Building 2 Light Repair Workshop: Air- Conditioning System for Level 1– 3/4	DM001-E-MAC-CEM-DD-DCC- 362312	A1
91	Main Depot Rashidiya – Building 2 Light Repair Workshop: Air- Conditioning System for Level 1– 4/4	DM001-E-MAC-CEM-DD-DCC- 362313	A1
92	Main Depot Rashidiya – Building 2 Light Repair Workshop: Air- Conditioning System for Roof Plan	DM001-E-MAC-CEM-DD-DCC- 362320	A1
93	Main Depot Rashidiya – Building 2 Light Repair Workshop: List of Drawings for Fire Protection System	DM001-E-MAC-CEM-DD-DCC- 362340	A1
94	Main Depot Rashidiya – Building 2 Light Repair Workshop: Fire Protection System Symbols	DM001-E-MAC-CEM-DD-DCC- 362341	A1
95	Main Depot Rashidiya – Building 2 Light Repair Workshop: Diagram of Fire Protection System	DM001-E-MAC-CEM-DD-DCC- 362350	A1
96	Main Depot Rashidiya – Building 2 Light Repair Workshop: Fire Protection System for Level 1 – 1/4	DM001-E-MAC-CEM-DD-DCC- 362360	A1
97	Main Depot Rashidiya – Building 2 Light Repair Workshop: Fire Protection System for Level 1 – 2/4	DM001-E-MAC-CEM-DD-DCC- 362361	A1
98	Main Depot Rashidiya – Building 2 Light Repair Workshop: Fire Protection System for Level 1 – 3/4	DM001-E-MAC-CEM-DD-DCC- 362362	A1
99	Main Depot Rashidiya – Building 2 Light Repair Workshop: Fire Protection System for Level 1 – 4/4	DM001-E-MAC-CEM-DD-DCC- 362363	A1

Appendices

Appendix A - Architecture

Appendix A1 – Architectural Specification for Building 2, Rashidiya Depot

Appendix B - Structural Engineering

Appendix B1 – Design Calculation for Building 2, Light Repair Workshop in Rashidiya Depot (Document No. MIPL/CAL/0008)

Appendix B2 – Design Report for Building 2, Light Repair Workshop in Rashidiya Depot (Document No. MIPL/RPT/0012)

Appendix A1– Architectural Specification for Building 2

Rashidiya Depot

Appendix B1– Design Calculation for Building 2
Light Repair Workshop in Rashidiya Depot
(Document No. MIPL/CAL/0008)

Appendix B2– Design Report for Building 2
Light Repair Workshop in Rashidiya Depot
(Document No. MIPL/RPT/0012)

CLIENT Government of Dubai Roads & Transport Authority	CONSORTIUM Dubai Rapid Link Consortium  MITSUBISHI HEAVY INDUSTRIES LTD.  Mitsubishi Corporation  KAJIMA CORPORATION  yapi merkezi
ENGINEER  	DESIGNER  ATKINS WS Atkins & Partners Overseas Supported by AEDAS

**OBYASHI-KAJIMA-YAPI MERKEZI
JT METRO JV**

**DUBAI METRO PROJECT
ARCHITECTURAL SPECIFICATION
RASHIDIYA DEPOT
BUILDING 2
DCP 3 SUBMISSION**

Prepared by Aedas Architects Ltd.



Date: 28 Jun 2006

Table of Content

<u>Subject</u>	<u>Page No.</u>
H31 Metal profiled/ flat sheet cladding/ covering	3
J30 Liquid applied tanking/damp proofing	9
J40 Flexible sheet tanking/ damp proofing	12
K10 Plasterboard dry linings/ partitions/ ceilings	15
K40 Demountable suspended ceilings	22
L10 Windows/ rooflights/ screens/ louvres	28
L20 Doors/ shutters/ hatches	31
L35 Fixed utilitarian access systems.....	35
M12 Resin flooring.....	39
M40 Stone/ concrete/ quarry/ ceramic tiling/ mosaic	42
M50 Rubber/ plastics/ cork/ lino/ carpet tiling/ sheeting.....	46
M60 Painting/clear finishing	51
M61 Intumescent coatings for fire protection of steelwork.....	55
N10 General fixtures/ furnishings/ equipment	58
N13 Sanitary appliances and fittings.....	62
N25 Permanent access and safety equipment.....	67
P10 Sundry insulation/ proofing work.....	71
P12 Fire stopping systems.....	72
P20 Unframed isolated trims/ skirtings/ sundry items.....	77
P21 Door/ window ironmongery	78
R10 Rainwater drainage systems	86
Z10 Purpose made joinery	92
Z11 Purpose made metalwork	94
Z12 Preservative/ fire retardant treatment.....	98
Z20 Fixings/ adhesives	100
Z22 Sealants	102
Z31 Powder coatings	103

Remark: This is a live document and will constantly be updated by the time of each design control point submission. Current blanks regarding building product specification are to be filled in by DCP 4 submission unless otherwise agreed.

H31 METAL PROFILED/ FLAT SHEET CLADDING/ COVERING

TYPES OF CLADDING/ COVERING SYSTEM

120A METAL CLADDING - EXTERNAL

- Support structure: Cold rolled steel Zed purlins. Refer to structural drawings for centres.
 - Bearing width (minimum): 58mm.
 - Pitch: 3°.
- External sheets: Kalzip Aluminium Standing Seam.
 - Manufacturer: Corus Building Systems, Haydock Lane, Haydock, St Helens, Merseyside WA11 9TY. Tel: 01942 295500. Fax: 01942 272136. Email: enquiries@Corus.co.uk. Web address: <http://www.Corus.co.uk>. Product reference: Kalzip 65/400.
 - Material: Aluminium alloy BS EN AW 3004 (AlMn1Mg1) clad with aluminium alloy BS EN AW 7072 (AlZn1) to minimum 4% thickness both sides.
0.2% proof stress (Rp0.2) = 185 N/mm² minimum.
Ultimate tensile strength (Rm) = 220 N/mm² minimum .
 - Thickness (nominal): 0.9 mm (nominal). Tolerance on thickness to be +0.05 mm and – 0.025 mm. This is a Class 2 thickness tolerance as per BS EN 508-2:2000 .
 - Finish side 1 (outer): Natural stucco embossed . Colour: TBC .
 - Finish side 2 (inner): _____ . Colour: _____ .
 - Additional requirements: Kalzip sheets to be supplied in single length sheets. Purlin centres and/or gauge of Kalzip sheets to be determined from Kalzip load/span tables.
- Accessories: As per Corus Building Systems – Kalzip standard details and recommendations.
- Primary cladding/ covering sheet fasteners: As per Corus Building Systems – Kalzip standard details and recommendations .
 - Fastener profile location: _____ .
 - Number of fasteners per sheet width:
Eaves and end laps: _____ .
Intermediate supports: _____ .
- End laps size (minimum): N/A .
- Sealing laps:
 - End laps: No end laps – single length sheets .
 - Side laps: Kalzip sheets are mechanically seamed over head of support clips with a Kalzip zipping machine .
- Stitching laps:
 - End laps: No end laps – single length sheets .
 - Side laps: Kalzip sheets are mechanically seamed over head of support clips with a Kalzip zipping machine .
- Spacers: Support clips, steel reinforced polyamide Kalzip E- Clips type E.180 .
 - Fasteners: 6.0 mm diameter stainless steel torque controlled fasteners, reference SDK3-S-377-6.0x30. Two fasteners per E.Clip positioned diagonally opposite in the holes of the base of the clip. The use of long length Kalzip sheets (e.g. over 40 m) may require a greater number or different type of fastener .
- Thermal insulation: As clause 271 .
- Vapour control layer: As clause 261 .

- Acoustic insulation: None .
- Lining sheets: As clause 241A .
- Additional requirements: Guided fall arrest system as clause N25/210 .

GENERAL REQUIREMENTS

170 DESIGN

- Cladding/ covering system: Complete detailed design and submit before commencement of fabrication.
 - Standard: To BS 5427-1.
- Related works: Coordinate in detailed design.

172 THERMAL PERFORMANCE/ BRIDGING

- Requirement: Complete the thermal design of the cladding/ covering system to avoid excessive thermal bridging.
 - Standard: MCRMA Technical Paper 14.

175 PRODUCT SAMPLES

- General: Before commencing detailed design, submit labelled samples of the following: Metal cladding.

176 FASTENER SAMPLES

- General: During detailed design, submit labelled samples of each type of fastener.

DESIGN/ PERFORMANCE REQUIREMENTS

185 PERFORMANCE COMPLIANCE

- Verification: Before commencing fabrication, submit evidence based on laboratory testing or computer modelling.
 - Verifying authority: Cladding manufacturer.

187 DEFLECTION OF METAL CLADDING/ COVERING

- Roof covering: Maximum permitted deflection under distributed loads as a multiple of span and due to:
 - Dead load: $L/500$.
 - Dead and imposed loads: $L/200$.
 - Dead and wind loads: $L/90$.
- Wall cladding: Maximum permitted deflection under distributed loads as a multiple of span and due to:
 - Dead and wind loads: $L/120$.

192 SOUND TRANSMITTANCE OF CLADDING/ COVERING SYSTEM

- Minimum weighted sound reduction index (Rw) within 100 to 3150 Hz frequency range to BS 5821-3: _____.
 - Location: Through External wall.

194 INTERNAL SOUND ABSORPTION OF CLADDING/ COVERING SYSTEM

- Minimum sound absorption coefficients (alpha S) to BS EN ISO 354:
One third octave band centre frequency (Hz) alpha S
100-500 _____

196 INTEGRITY OF CLADDING/ COVERING

- Requirement: Determine profiles, sizes and thicknesses of sheets, the sizes, number and spacing of fixings, configuration and location of spacer systems and incorporation of other accessories and fittings to ensure cladding/ covering system will resist factored dead, imposed and design live loads, and accommodate deflections and thermal movements without damage, in accordance with BS 5427 -1.
- Primary fasteners: Not to be subjected to any bending moment.
- Wind loads: Calculate to BS 6399-2, Standard Method and BS 5427-1 appropriate to location, exposure, height, building shape and size, taking account of existing and known future adjacent structures.
- Imposed roof load (no access): As determined from BS 6399 -3 and BS 5427 -1.
- Permanent imposed roof loads: Walkways, ventilator fans, smoke release vents etc. as shown on drawings.
- Temporary imposed roof loads: Maintenance access equipment and personnel.
- Impact loads: To BS 8200:
 - Location and category: Cladding generally, category C.

198 WATER PENETRATION

- Requirement: Under site exposure conditions, moisture must not penetrate onto internal surfaces, or into cavities not designed to be wetted.

200 AVOIDANCE OF INTERSTITIAL CONDENSATION

- Requirement: Determine interstitial condensation risk of cladding/ covering system using the method described in BS 5250 Appendix D. If necessary, provide a vapour control layer to ensure that damage and nuisance from interstitial condensation does not occur.
- Outdoor psychrometric conditions (notional): To BS 6229, table A.1 as follows: Winter Summer
 - Temperature -5°C 18°C
 - Relative humidity 90% 65%
 - Vapour pressure 0.36 kPa 1.34 kPa
- Indoor psychrometric conditions (notional): As follows:
 - Temperature: 20°C for offices, 15° for sheds .
 - Relative humidity: 40% for offices, 35% for sheds .
 - Vapour pressure: 0.93 kPa for offices, 0.60 kPa for sheds .
- Calculated amount of winter interstitial condensate (maximum): 0.35 kg/m² .
- Winter interstitial condensate:
 - Calculated amount (maximum): 0.35 kg/m².
 - Calculated annual net retention: Nil.

202 AVOIDANCE OF SURFACE CONDENSATION

- Requirement: Determine surface condensation risk of cladding/ covering system using the method described in BS EN ISO 13788. If necessary, revise thermal insulation to provide satisfactory temperature factor (f_{min}). ensure that damage and nuisance from surface condensation does not occur.
-

FIXING CLADDING/ COVERING

215 PAINTING STRUCTURE

- Sequence: Paint outer surface of supporting structure before fixing cladding/ covering.

219 FASTENERS

- Unspecified fasteners: Recommended for the purpose by the cladding/ covering manufacturer.

221 FITTINGS AND ACCESSORIES

- Unspecified fittings and accessories: Recommended for the purpose by the cladding/ covering manufacturer.

223 PREVENTION OF ELECTROLYTIC ACTION

- Isolating tape: Type recommended by cladding/ covering manufacturer.
 - Location: To contact surfaces of supports and sheets of dissimilar metals.

241A ALUMINIUM LINING

- Manufacturer: Corus Building Systems, Haydock Lane, Haydock, St Helens, Merseyside WA11 9TY. Tel: 01942 295500. Fax: 01942 272136. Email: enquiries@Corus.co.uk. Web address: <http://www.Corus.co.uk>.
 - Product reference: Kalzip Liner TR35/200S.
- Finish/ Colour: White lining enamel (for internal use)..
- Primary sheet fasteners: 5.5 mm diameter carbon steel self drilling fasteners, reference SD-3-T15-5.5x25. One fastener to be installed in every other valley where Kalzip E.Clips are not to be positioned
- End laps size (minimum): 100 mm to coincide with purlin position.
- End and side lap sealing: Not required.

261 VAPOUR CONTROL MEMBRANE

- Material: Kalzip VCL Clear.
 - Vapour resistance (minimum): 530 MNs/g.
- Continuity: No breaks and with the minimum of joints.
 - Penetrations and abutments: Seal to vapour control membrane with tape. Achieve full bond.
 - Laps: Not less than 150 mm, seal with tape. Achieve full bond.
- Tape: Double sided sealant with vapour resistivity not less than the vapour control membrane.
 - Size (width and thickness): 15 mm wide x 2 mm thick.
- Repairs and punctures: Seal with lapped patch of vapour control membrane and continuous band of sealant tape along edges

271 MINERAL WOOL THERMAL INSULATION

- Standard: To BS EN 13162.
- Manufacturer: Corus Building Systems, Haydock Lane, Haydock, St Helens, Merseyside WA11 9TY. Tel: 01942 295500. Fax: 01942 272136. Email: enquiries@Corus.co.uk Web address: <http://www.Corus.co.uk>.
 - Product reference: Kalzip Insulation Plus – 37.
- Thickness (minimum): 180 mm laid in two layers of 90 mm lightly compressed to overall thickness of 165 mm to achieve maximum U-value of 0.25 W/m2K.
- Installation: Continuous and not compressed between outer and lining sheets. Secure to prevent future movement or dislodgement.

300 PROFILE FILLERS GENERALLY

- Material: _____.
- Manufacturer: Submit proposals.

- Product references: _____.
- Colour: _____.
- Thickness: _____.
- Fixing method: _____.
 - Requirement: To close cavities/ regulate air paths within the external envelope. Tight fit with no unintended gaps.

305 FIRE RESISTING PROFILE FILLERS

- Types: To accurately match sheet profile.
- Fixing method: Adhesive recommended by profile filler manufacturer.

310 PURPOSE MADE COLD FORMED METAL ACCESSORIES

- Material: Aluminium alloy BS EN AW 3004 or Galvanised steel to BS EN 10147.
 - Thickness/ gauge: 0.9 mm or 0.7mm.
 - Finish/ Colour: TBC.
 - Fasteners:
 - Type: As per Corus Building Systems – Kalzip standard details and recommendations.
 - Location: Refer to drawings.
 - Fixing centres: _____.

410 FIXING SHEETS GENERALLY

- Cut edges: Clean true lines.
- Penetrations: Openings to minimum size necessary.
 - Edge reinforcement: _____.
- Sheet orientation: Exposed joints of side laps away from prevailing wind unless shown otherwise on drawings.
- Sheet ends, laps and raking cut edges: Fully supported and with fixings at top of lap.
- Fasteners: Drill holes. Position at regular intervals in straight lines, centred on support bearings.
 - Position of fasteners in oversized drilled holes: Central.
 - Fasteners torque: Sufficient to correctly compress washers.
- Debris: Remove dust and other foreign matter before finally fixing sheets.
- Completion: Check fixings to ensure watertightness and that sheets are secure.
- Cut edges: Paint to match face finish.

460 ACCOMMODATION OF THERMAL MOVEMENT

- Sheet type/ location: _____.
- Method: As per Corus Building Systems – Kalzip standard details and recommendations.

470 STRUCTURAL MOVEMENT JOINTS

- Type: Cover flashing fixed on one side over gap between sheets.
- Location: Coincident with structural movement joint.
- Width of gap: To match structural movement joint requirements.
- Requirement: Weathertight.

480 FLASHINGS/ TRIMS GENERALLY

- Lap joint treatment:

- Vertical and sloping flashings/ trims: End laps to be same as for adjacent sheeting.
- Horizontal flashings/ trims: End laps to be 150 mm, sealed and where possible arranged with laps away from prevailing wind.
- Method of fixing: To structure in conjunction with adjacent sheeting. Otherwise to sheeting.
 - Fasteners: As per Corus Building Systems – Kalzip standard details and recommendations.

540 ABUTMENTS

- Junctions with flashings: Weathertight and neatly dressed down.



550 SEALING LAPS ON EXTERNAL SHEETS

- Sealant tape: Types recommended by sheet manufacturer.
 - Position: Below fixing positions in straight unbroken lines, parallel to and slightly back from edge of sheet.
- Seal quality: Effective, continuous and not overcompressed.
- End laps: Sealant tape positions:
 - Single line tape: Immediately below line of fasteners.
 - Second line tape (where specified): Slightly set back from edge of external sheet.
- Side laps: Sealant tape positions:
 - Single line tape: Outside line of fasteners.
 - Second line tape (where specified): On other side of fasteners.

554 WATER VAPOUR SEALING AT LAPS AND PENETRATIONS IN METAL LININGS - SEALANT TAPE

- Sealant tape: As per Corus Building Systems – Kalzip standard details and recommendations.
 - Position : Below fixing positions in straight unbroken lines, parallel to and slightly back from edge of sheet.
- Seal quality: Effective, continuous and not overcompressed.

555 WATER VAPOUR SEALING AT LAPS AND PENETRATIONS IN METAL LININGS - ALUMINIUM TAPE

- Aluminium foil tape: As per Corus Building Systems – Kalzip standard details and recommendations.
 - Position: Centrally and parallel to edge of oversheet in straight unbroken lines.
 - Joints in tape: Minimum overlap of 50 mm.
- Seal and adhesion quality: Effective and continuous.

560 SAFETY SIGNS

- Fixing locations of signs: As indicated on drawings.
- Manufacturer: Submit proposals.
 - Product reference: _____.
- Material: Aluminium with polyester powder coated base, screen printed graphics.
- Signs description: Hazard sign as BS 5499-5, code 8.C.0072 with supplementary text sign lettering 'DANGER fragile roof'. General sign with mandatory instruction to BS 5499-5 code 44.A.0103 with supplementary text sign, lettering 'Use crawling boards'.

J30 LIQUID APPLIED TANKING/DAMP PROOFING

LIQUID APPLIED TANKING/ DAMP PROOFING

To be read with Preliminaries/General conditions

TYPES OF TANKING OF DAMP PROOFING

110 COLD APPLIED TANKING

- Substrate: As drawings .
- Primer: As recommended by coating manufacturer .
- Coating: Bituminous
- Manufacturer: Submit proposals .
Product reference: _____ .
 - Number of coats: As recommended by coating manufacturer .
 - Coverage per coat (minimum): Consult with coating manufacturer .
- Reinforcement: As recommended by coating manufacturer .
- Blinding: As recommended by coating manufacturer .

130 COLD APPLIED DAMP PROOFING

- Substrate: As drawings .
- Primer: As recommended by coating manufacturer .
- Coating: Bituminous
 - Manufacturer: Submit proposals .
Product reference: _____ .
 - Number of coats: As recommended by coating manufacturer .
 - Coverage per coat (minimum): Consult with coating manufacturer .
- Reinforcement: As recommended by coating manufacturer .
- Blinding: As recommended by coating manufacturer .

EXECUTION

205 SUITABILITY OF SUBSTRATE

- Substrates generally:
 - Smooth, even textured, clean, dry and frost free.
 - Within tolerances for level and surface regularity
 - Vertical and horizontal surfaces: Correctly prepared and free from irregularities.
- Moisture content and stability of substrate: Must not impair integrity of finished tanking/damp proofing.
- Preliminary work: Complete including:
 - Chases.
 - External angles.
 - Formation of upstands and kerbs.
 - Movement joints.
 - Penetrations/ Outlets.

207 PRIMERS

- Application: Uniform, continuous coverage.

COATINGS

210 COATING APPLICATION

- . Adjacent surfaces exposed to view in finished work: Protect.
- Coatings:
 - Apply in dry atmospheric conditions when primer is tacky / dry - if used onto bituminous surfaces .
 - Uniform, continuous coverage. Do not allow to pool in hollows.
 - Firmly adhered to substrate and free from imperfections.
 - Prevent damage to finished coatings.
- Penetrations: Impervious.
- Final covering: Apply as soon as possible after coating has hardened.

220 COLD APPLIED COATINGS

- Thinning: Not permitted unless recommended by manufacturer.
- Successive coats:
 - Allow to dry before applying next.
 - Apply at right angles to previous.

230 BITUMINOUS COATINGS

- Surfaces treatment: Lightly dampen
- Air and surface temperatures (minimum): 5°C.
- Weather conditions: Do not apply if there is a risk of rain during application and drying unless effective temporary cover is provided over working area.

240 REINFORCEMENT

- Laying: As recommended by coating manufacturer .
- Edge overlap (minimum): As recommended by coating manufacturer
- Finish: Smooth and free from imperfections.

260 JUNCTIONS WITH DPCS

- DPCs: Clean, all edges fully exposed.
- Fully coat dpc and overlap adjacent surfaces by minimum 75 mm .

270 BLINDING

- Coatings: As recommended by coating manufacturer .
- Surplus material: Remove when coatings are completely dry.

COMPLETION

280 PROTECTION OF COATINGS

- Surface: Clean and free from contaminants.
- Board manufacturer: Submit proposals .
 - Product reference: _____ .
 - Thickness: As recommended by board manufacturer .
- Placement: As recommended by board manufacturer .
 - Edge overlap (minimum): 75mm .
 - Perimeter treatment: As recommended by board manufacturer .
- Contact: Secure. Full and continuous with coating.

290 BACKFILLING

- Timing: Carry out as soon as possible after tanking and protection are complete.



J40 FLEXIBLE SHEET TANKING/ DAMP PROOFING

To be read with Preliminaries/ General conditions.

110 HARDCORE BEDS

- Preparation: Blind with soft sand or pulverized fuel ash.
 - Thickness (minimum): As drawings.
- Finish on completion: Smooth, consolidated bed free of sharp projections.

TYPES OF TANKING/ DAMP PROOFING

120 LOOSE LAID POLYETHYLENE DAMP PROOF MEMBRANE

- Substrate: As drawings.
- Membrane manufacturer: Contractor's choice to PIFA Standard 6/83A or Agrément certified.
 - Product reference: _____.
- Thickness/ Gauge: 250 micrometres (1000 gauge) if Agrément certified or to the PIFA Standard 6/83A.
- Joints: Lapped minimum 150 mm.
 - Surfaces to be joined: Clean and dry.
- Sealing: Continuous mastic strip between overlaps; edge of top sheet sealed with jointing tape.

130 LOOSE LAID POLYETHYLENE OVERSITE DAMP PROOF MEMBRANE

- Manufacturer: Submit proposals.
 - Product reference: _____.
- Thickness/ Gauge: 250 micrometres (1000 gauge) if Agrément certified or to the PIFA Standard 6/83A.
- Joints: Lapped minimum 300 mm.
- Covering: Concrete to BS8500-2 and BS EN 206-1, Designated mix GEN 1 or Standard mix ST2.
 - Thickness (minimum): 50 mm.
 - Timing: Immediately after placing membrane.

WORKMANSHIP

310 WORKMANSHIP GENERALLY

- Condition of substrate:
 - Clean and even textured, free from voids and sharp protrusions.
 - Moisture content: Compatible with dpm/ tanking.
- Air and surface temperature: Do not apply sheets if below minimum recommended by membrane manufacturer.
- Condition of membrane at completion:
 - Neat, smooth and fully supported, dressed well into abutments and around intrusions.
 - Completely impervious and continuous.
 - Undamaged. Prevent puncturing during following work.
- Permanent overlying construction: Cover membrane as soon as possible.

320 INSPECTION

- Give notice: Before covering any part of membrane with overlying construction.

330 PRIMERS

- Types and application: As recommended for the purpose by membrane manufacturer.
- Curing: Allow to dry thoroughly before covering.

340 HOT APPLIED BONDING COMPOUNDS

- Type: Hot applied oxidized bitumen to BS 3690-2.
 - Grade: As recommended by membrane manufacturer for the conditions and type of surface .
- . Application: Continuous even coating to provide full bonding over whole surface. Do not overheat.

345 COLD APPLIED BONDING COMPOUNDS

- Type and application: As recommended for the purpose by the membrane manufacturer.

350 ANGLES IN BONDED DAMP PROOF MEMBRANE/ TANKING

- Preformed rot proof fillet to internal angles:
 - Size (minimum): 50 x 50 mm.
 - Bedding: Bitumen mastic or bonding compound.
- Reinforcing strip to all angles:
 - Material: As damp proofing membrane.
 - Width (minimum): 300 mm.
 - Timing: Apply before main sheeting.
- . Dressing of main sheeting onto adjacent surfaces (minimum): 100 mm.

360 JUNCTIONS WITH PROJECTING DPCS/ CAVITY TRAYS

- Condition of adjoining surfaces: Clean and dry.
- Projecting dpcs/ cavity trays: Lap and fully bond/ seal with sheeting.
 - Laps (minimum): 150 mm.
 - Bonding/ Sealing: Mastic tape.

365 JUNCTIONS WITH FLUSH DPCS/ CAVITY TRAYS

- Preparation of adjacent dpcs/ cavity trays:
 - Expose edge where concealed.
 - Adjoining surfaces: Clean and dry.
 - Joints: Bond/ seal sheeting to wall.
 - Bonding/ Sealing: Mastic tape .

370 PIPES, DUCTS, CABLES, ETC

- Preformed collars: Recommended for the purpose by membrane manufacturer .
 - Sealing: Fully bond to penetrations and sheeting.
- Completed junctions: Impervious.

380 PROTECTION OF EXTERNAL TANKING

- Manufacturer: Submit proposals.
 - Product reference: _____.
 - Thickness: As recommended by the selected manufacturer.

- Surface: Clean and free from contaminants.
- Contact with membrane: Secure, full and continuous.



K10 PLASTERBOARD DRY LININGS/ PARTITIONS/ CEILINGS

To be read with Preliminaries/ General conditions.

TYPES OF DRY LINING

125 METAL STUD PARTITION SYSTEM TO 1HR FR COMPARTMENT WALL between Siding & Cleaning Shed and Office Accommodation

- Manufacturer: British Gypsum Ltd, East Leake, Loughborough, Leicestershire LE12 6JT
Tel: 0870 545 6123 Fax: 0870 545 6356 Email: BGTechnical.Enquiries@bpb.com Web: www.british-gypsum.com.
 - Product reference: British Gypsum GypWall™ CLASSIC, including Isowool APR 1200 insulation.
- Studs:
 - Centres: 600 mm.
- Head condition: Refer to British Gypsum.
 - Deflection allowance: Refer to drawings.
- Insulation: Isowool APR 1200 glass mineral w.
 - Thickness: 50 mm .
- Linings: 2 x 12.5 mm Gyproc SoundBloc to each side.
- Finishing: Jointing as clause 670.
 - Primer/ Sealer: 1 coat of Gyproc Drywall Primer.
 - Accessories: Rigid beads/ stops, as clause 692.
- Other requirements: Access panels, as clause 430.

126 METAL STUD PARTITION SYSTEM TO OFFICE ACCOMMODATION - inherently providing 1HR FR, 52dB - MAXIMUM HEIGHT 4900mm

- Manufacturer: British Gypsum Ltd, East Leake, Loughborough, Leicestershire LE12 6JT
Tel: 0870 545 6123 Fax: 0870 545 6356 Email: BGTechnical.Enquiries@bpb.com Web: www.british-gypsum.com.
 - Product reference: British Gypsum GypWall™ CLASSIC, including Isowool APR 1200 insulation.
- Studs:
 - . Type: 92 mm Gypframe 92 S 50 'C' Studs.
 - Centres: 600 mm.
- Head condition: Refer to British Gypsum.
 - Deflection allowance: Refer to drawings.
- Insulation: Isowool APR 1200 glass mineral wool.
 - Thickness: 25 mm .
- Linings: First layer 12.5mm Gyproc Wallboard square edged; face layer 12.5mm Gyproc Wallboard tapered edge.
- Finishing: Taped seamless finish, as clause 670.
 - Primer/ Sealer: 1 coat of Gyproc Drywall Primer.
 - Accessories: Rigid beads/ stops, as clause 692.
- Other requirements: Access panels, as clause 430.

127 METAL STUD PARTITION SYSTEM TO KITCHENETTES & CHANGING AREAS - inherently providing 1HR FR, 52dB - MAXIMUM HEIGHT 4900mm

- Manufacturer: British Gypsum Ltd, East Leake, Loughborough, Leicestershire LE12 6JT
Tel: 0870 545 6123 Fax: 0870 545 6356 Email: BGTechnical.Enquiries@bpb.com Web: www.british-gypsum.com.
 - Product reference: British Gypsum GypWall™ CLASSIC, including Isowool APR 1200 insulation.
- Studs:
 - Type: 92 mm Gypframe 92 S 50 'C' Studs.
 - Centres: 600 mm.
- Head condition: Refer to British Gypsum.
 - Deflection allowance: Refer to drawings.
- Insulation: Isowool APR 1200 glass mineral wool.
 - Thickness: 25 mm .
- Linings: First layer 12.5mm Gyproc Wallboard square edged; face layer 12.5mm Gyproc Moisture Resistant Board tapered edge.
- Finishing: Taped seamless finish, as clause 670.
 - Primer/ Sealer: 2 coats of Gyproc Drywall Sealer.
 - Accessories: Rigid beads/ stops, as clause 692.
- Other requirements: Access panels, as clause 430.

155 WALL LINING SYSTEM (METAL STUDS) TO OFFICE ACCOMMODATION

- Manufacturer: British Gypsum Ltd, East Leake, Loughborough, Leicestershire LE12 6JT
Tel: 0870 545 6123 Fax: 0870 545 6356 Email: BGTechnical.Enquiries@bpb.com Web: www.british-gypsum.com.
 - Product reference: British Gypsum GypLyner™ IWL System.
- Studs:
 - Type: 60 mm Gypframe 60 I 70 'I' Studs.
 - Centres: 600 mm.
- Cavity between wall and studs: As drawings.
- Unbraced height (maximum): 3000-3900 mm.
- Head condition: Refer to British Gypsum.
 - Deflection allowance: Refer to drawings.
- Insulation: Isowool APR 1200.
 - Thickness: 50 mm.
- Vapour control layer: Not required.
- Linings: 1 x 15 mm Gyproc DuraLine.
- Access units: Gyproc Profilex Access Panels: non fire-rated – as clause 430.
- Finishing: Taped seamless finish, as clause 670.
 - Primer/ Sealer: 1 coat of Gyproc Drywall Primer.
 - Accessories: Trims: Edge reveals; corner reveals and door reveals. Beads/ Stops: As clause 692.
- Other requirements: Fire stopping around services as section P31..

265 ENCASEMENT SYSTEM (FRAMELESS) TO STEEL COLUMNS & BEAMS IN B1 ACCOMMODATION OFFICE

- Manufacturer: Manufacturer: British Gypsum Ltd, East Leake, Loughborough, Leicestershire LE12 6JT Tel: 0870 545 6123 Fax: 0870 545 6356 Email: BGTechnical.Enquiries@bpb.com Web: www.british-gypsum.com.



- Product reference: British Gypsum FireCase.
- Structural members: As drawings.
 - Extent of protection: Columns: Four sides. Beams: Three sides.
- Fire performance:
 - Protection to structural steel: BS 476-21, 60 minutes.
- Support system: Sizes and spacings of intermediate backing strips/ noggings/ metal angles and fixings as recommended by board manufacturer.
- Linings: Glasroc Firecase. Thickness: 15mm.
- Finishing: Taped seamless finish, as clause 670.
 - Primer/ Sealer: 1 coat of Gyproc Drywall Primer.
 - Accessories: [Rigid beads/ stops, as clause 692.
- Other requirements: Fixing: Four-sided protection to steel columns: Cladding is commenced from the base of the column. Boards are positioned and staple-fixed board-to-board using a Gyproc Impulse Staple Gun (or similar), or alternatively, using Gyproc Firecase Screws of appropriate length, at 150 mm centres. Board joints on adjacent sides are staggered by a minimum 600 mm. For double layer linings, board joints are staggered between layers by a minimum 300 mm.
- Three-sided protection to steel columns incorporating steel angles: Gypframe GA1 Steel Angle is fixed to the wall both sides of the column, at maximum 600 mm centres. Additional steel angles are incorporated where the column flange is at right angles to the wall structure. Boards are positioned and screw-fixed to the steel angles at 150 mm centres. Board-to-board fixings are made by staples using a Gyproc Impulse Staple Gun (or similar), or alternatively, using Gyproc Firecase Screws of appropriate length, at 150 mm centres. For double layer linings, board joints are staggered between layers by a minimum 300 mm.
- Three-sided protection to steel beams incorporating steel angles: Gypframe GA1 Steel Angle is fixed to the wall both sides of the column, at maximum 600 mm centres. Additional steel angles are incorporated where the column flange is at right angles to the wall structure. Boards are positioned and screw-fixed to the steel angles at 150 mm centres. Board-to-board fixings are made by staples using a Gyproc Impulse Staple Gun (or similar), or alternatively, using Gyproc Firecase Screws of appropriate length, at 150 mm centres. For double layer linings, board joints are staggered between layers by a minimum 300 mm. For single layer encasements, fascia board joints are backed with Glasroc FireCase s strips, cut to a minimum 60 mm width and stapled or screw-fixed behind fascia board ends so as to half-lap the joints.
- Three-sided protection to steel columns and beams incorporating Glasroc FireCase s soldiers to support single layer linings: Glasroc FireCase s soldiers are pre-cut to fit neatly into the steel section. The soldiers are located into both sides of the section at 1200 mm maximum centres as boarding progresses. At fascia board joints, two soldiers are fitted side by side so that each one finishes flush with the board end. Cladding is fixed to each joint soldier and also any intermediate soldiers using three staples, fixed using a Gyproc Impulse Staple Gun (or similar), or alternatively, using three Gyproc Firecase Screws of appropriate length. Boarding is continued as previously, staggering board joints.

GENERAL/ PREPARATION

305 COMPLIANCE WITH PERFORMANCE REQUIREMENTS

- Testing/ Assessment: Submit UKAS accredited laboratory reports for the following: Fire resistance: Partitions (including deflection heads and doorsets) and suspended ceilings (including access units)..
- Materials, components and details: As used in testing/ assessment reports. If iscrepancies arise, give notice.

335 ADDITIONAL SUPPORTS

- Framing: Accurately position and securely fix to give full support to:
 - Partition heads running parallel with, but offset from main structural supports.
 - Fixtures, fittings and service outlets. Mark framing positions clearly and accurately on linings.
 - Board edges and lining perimeters, as recommended by board manufacturer to suit type and performance of lining.

375 NEW WET LAID BASES

- Dpcs: Install under full width of partitions/ freestanding wall linings.
 - Material: Bituminous sheet or plastics.

COMPONENTS

401 GYPSUM PLASTERBOARD

- Type: To BS 1230-1, type 1.
 - Core density (minimum): 650 kg/m³.
- Exposed surface and edge profiles: Suitable to receive specified finish.

403 GYPSUM PLASTERBOARD (MOISTURE RESISTANT)

- Type: To BS 1230-1, type 3 and 4.
 - Core density (minimum): 710 kg/m³.
 - Core: Moisture resistant.
 - Paper facings: Moisture resistant.
- Exposed surface and edge profiles: Suitable to receive specified finish.

404 GYPSUM PLASTERBOARD (IMPROVED FIRE PROTECTION)

- Type: To BS 1230-1, type 5.
 - Core density (minimum): 800 kg/m³.
 - Core: Including fibres for improved cohesion.
- Exposed surface and edge profiles: Suitable to receive specified finish.

408 GYPSUM PLASTERBOARD (IMPACT RESISTANT)

- Type: To BS 1230-1, type 5.
 - Core density (minimum): 900 kg/m³.
 - Paper facings: Heavy duty.
- Exposed surface and edge profiles: Suitable to receive specified finish.

409 GYPSUM PLASTERBOARD (IMPROVED SOUND INSULATION)

- Type: To BS 1230-1, type 1.
 - Core density (minimum): 820 kg/m³.
- Exposed surface and edge profiles: Suitable to receive specified finish.

430 ACCESS PANELS TO PLASTERBOARD WALL LININGS

- Type: Gyproc Proflex Standard Panel. Manufacturer: British Gypsum Ltd, East Leake, Loughborough, Leicestershire LE12 6JT Tel: 0870 545 6123 Fax: 0870 545 6356 Email: BGTechnical.Enquiries@bpb.com Web: www.british-gypsum.com .
 - Sizes: 300 x 300 mm, 450 x 450 mm or 550 x 550 mm .
- Frame: Picture frame. Zinc coated mild steel, with powder coated white finish .

- Panel: Zinc coated mild steel, with powder coated white finish .
- Lock: Budget lock .

INSTALLATION

435 DRY LININGS GENERALLY

- General: Use fixing, jointing, sealing and finishing materials, components and installation methods recommended by board manufacturer.
- Cutting plasterboards: Neatly and accurately without damaging core or tearing paper facing.
 - Cut edges: Minimize and position at internal angles wherever possible. Mask with bound edges of adjacent boards at external corners.
- Fixings boards: Securely and firmly to suitably prepared and accurately levelled backgrounds.
- Finishing: Neatly to give flush, smooth, flat surfaces free from bowing and abrupt changes of level.

455 METAL FRAMING FOR PARTITIONS/ WALLS LININGS

- Setting out: Accurately aligned and plumb.
 - Frame/ Stud positions: Equal centres to suit specified linings, maintaining sequence across openings.
 - Additional studs: To support vertical edges of boards.
- Fixing centres at perimeters (maximum): 600 mm.
- Openings: Form accurately.
 - Doorsets: Use sleeved or boxed metal studs and/ or suitable timber framing to achieve strength grade requirements for framing assembly and adequately support weight of door.
 - Services penetrations: Allow for associated fire stopping.

505 INSTALLING MINERAL WOOL INSULATION

- Fitting insulation: Closely butted joints and no gaps. Use fasteners to prevent slumping or displacement.
- Services:
 - Electrical cables overlaid by insulation: Sized accordingly.
 - Ceilings: Cut insulation around electrical fittings, etc.

510 SEALING GAPS AND AIR PATHS

- Location of sealant: To perimeter abutments and around openings.
 - Pressurized shafts and ducts: At board-to-board and board-to-metal frame junctions.
- Application: To clean, dry and dust free surfaces as a continuous bead with no gaps.
 - Gaps greater than 6 mm between floor and underside of plasterboard: After sealing, fill with jointing compound.

530 CAVITY FIRE BARRIERS WITHIN PARTITIONS/ WALL LININGS

- Metal framed systems:
 - Material: Wire reinforced mineral wool 50 mm (minimum) thick.
 - Installation: Form accurately and fix securely with no gaps to provide a complete barrier to smoke and flame.
- Adhesive fixed wall lining systems:

- Material: Adhesive compound.
- Installation: Form in a continuous line with no gaps to provide a complete barrier to smoke and flame.

555 FIRE STOPPING AT PERIMETERS OF DRY LINING SYSTEMS

- Material: Tightly packed mineral wool or intumescent mastic/ sealant.
- Application: To perimeter abutments to provide a complete barrier to smoke and flame.

560 JOINTS BETWEEN BOARDS

- Tapered edged plasterboards:
 - Bound edges: Lightly butted.
 - Cut/ unbound edges: 3 mm gap.
- Square edged plasterboards: 3 mm gap.
- Square edged fibre reinforced gypsum boards: 5 mm gap.

565 VERTICAL JOINTS

- Joints: Centre on studs.
 - Partitions: Stagger joints on opposite sides of studs.
 - Two layer boarding: Stagger joints between layers.

570 HORIZONTAL JOINTS

- Surfaces exposed to view: Horizontal joints not permitted. Seek instructions where height of partition/ lining exceeds maximum available length of board.
- Two layer boarding: Stagger joints between layers by at least 600 mm.
- Edges of boards: Support using additional framing.
 - Two layer boarding: Support edges of outer layer.

575 PLANK PLASTERBOARD

- First layer in two layer boarding: Square edged with long edges at right angles to studs.

590 FIXING PLASTERBOARD TO METAL FRAMING/ FURRINGS

- Partitions/ Wall linings: Fix securely and firmly at the following centres (maximum):
 - Single layer boarding: To all framing at 300 mm centres. Reduce to 200 mm centres at external angles.
 - Multi-layer boarding: Face layer at 300 mm centres, and previous layers around perimeters at 300 mm centres.
- Ceilings: 230 mm. Reduce to 150 mm at board ends and at lining perimeters.
- Position of screws from edges of boards (minimum): 10 mm.
 - Screw heads: Set in a depression. Do not break paper or gypsum core.

595 DEFLECTION HEADS

- Fixing boards: Do not fix to head channels.

FINISHING

650 LEVEL OF DRY LINING ACROSS JOINTS

- Sudden irregularities: Not permitted.
- Joint deviations: Measure from faces of adjacent boards using methods and straightedges (450 mm long with feet/ pads) to BS 8212, clause 3.3.5.

- Tapered edge joints: Permissible deviation (maximum) across joints when measured with feet resting on boards: 3 mm.
- External angles: Permissible deviation (maximum) for both faces: 4 mm.
- Internal angles: Permissible deviation (maximum) for both faces: 5 mm.

670 SEAMLESS JOINTING TO PLASTERBOARDS

- Cut edges of boards: Lightly sand to remove paper burrs.
- Filling and taping: Fill joints, gaps and internal angles with jointing compound and cover with continuous lengths of paper tape, fully bedded.
- Protection of edges/ corners: Reinforce external angles, stop ends, etc. with specified edge/ angle bead.
- Finishing: Apply jointing compound. Feather out each application beyond previous application to give a flush, smooth, seamless surface.
- Nail/ screw depressions: Fill with jointing compound to give a flush surface.
- Minor imperfections: Remove by light sanding.



692 RIGID BEADS/STOPS

- Type: Galvanized steel to BS 6452-1.

695 INSTALLING BEADS/ STOPS

- Cutting: Neatly using mitres at return angles.
- Fixing: Securely using longest possible lengths, plumb, square and true to line and level, ensuring full contact of wings with substrate.
- Finishing: After joint compounds/ plasters have been applied, remove surplus material while still wet from surfaces of beads exposed to view.

K40 DEMOUNTABLE SUSPENDED CEILINGS

To be read with Preliminaries/General conditions.

TYPES OF CEILING SYSTEM

115 UNIT SUSPENDED CEILING SYSTEM TO KITCHENTTES & CHANGING ROOMS

- Structural soffits: Refer to drawings .
- Ceiling system manufacturer: Armstrong World Industries Ltd, Building Products Division, Armstrong House, 38 Market Square, Uxbridge, UB8 1NG. Tel: Freephone 0800 371 849. Fax: 01895 274 287. Web: www.armstrong-ceilings.co.uk .
 - Product reference: Mineral Performance Ceramaguard .
- Ceiling type:
 - Grid form: Prelude TL/NC 24 Butt Cut .
 - Grid exposure: 24mm exposed grid system .
- Grid spacings: 600 x 600 mm .
- Access: Infill units fully demountable .
- Ceiling module: _____ .
- Ceiling soffit above finished floor level: Refer to drawings .
- Integrated services fittings: _____ .
- Other requirements: _____ .

116 UNIT SUSPENDED CEILING SYSTEM TO ACCOMMODATION OFFICES

- Structural soffits: Refer to drawings .
- Ceiling system manufacturer: Armstrong World Industries Ltd, Building Products Division, Armstrong House, 38 Market Square, Uxbridge, UB8 1NG. Tel: Freephone 0800 371 849. Fax: 01895 274 287. Web: www.armstrong-ceilings.co.uk .
 - Product reference: Prima Plain .
- Ceiling type:
 - Grid form: Prelude TL 24 Butt Cut / Prelude XL² 24 Joggled / with Peakform Main Runners / with Prelude Sixty main runners .
 - Grid exposure: 24mm exposed grid system .
- Grid spacings: 600 x 600 mm .
- Access: Infill units fully demountable .
- Ceiling module: _____ .
- Ceiling soffit above finished floor level: Refer to drawings .
- Integrated services fittings: _____ .
- Other requirements: _____ .

GENERAL/ PERFORMANCE

205 COMPLIANCE WITH PERFORMANCE REQUIREMENTS

- . Testing/ assessment: Submit UKAS accredited laboratory reports for the following: Electrical continuity and earth bonding .
- Materials, components and details: Use those used in the test/ assessment reports. If discrepancies arise, give notice.

210 ENVIRONMENT

- Environmental classification to BS 8290-1: Normal .

COMPONENTS

245 STANDARDS

- Components: To BS 8290-2.
 - Aluminium bars, tubes and sections: To relevant parts of BS EN 515, BS EN 573, BS EN 755 and BS EN 12020.

255 TIMBER EDGE BATTENS _____

- Type: Planed softwood to BS EN 942, class J10.
 - Moisture content at time of fixing: 15% ± 2%.
 - Finished size: 21 x 44 mm .
 - Finish (apply before ceiling grid is installed): Prime, undercoat and final coat of matt black to visible surfaces .

260 PERIMETER TRIMS _____

- Type: 32 x 19 mm steel angle trim colour and finish to match exposed grid .
 - Fixings to perimeters: BZP raised head screws .
 - Fixing centres (maximum): 450 mm .

280 ACCESS UNITS _____

- Manufacturer: British Gypsum Ltd, East Leake, Loughborough, Leicestershire LE12 6JT
Tel: 0870 545 6123 Fax: 0870 545 6356 Email: BGTechnical.Enquiries@bpb.com Web: www.british-gypsum.com .
 - Product reference: Gyproc Proflex Standard Panel .
- Size: 550 x 550 mm .
- Frame type: Picture .
- Fire performance: As suspended ceiling.
- Lock type: Budget lock .
- Operation: Concealed steel hinges, downward opening .
- Finish: Zinc coated mild steel, with powder coated finish .
- Colour: White .

INSTALLATION

305 SETTING OUT

- General: Accurate, continuous, even, and jointed at regular intervals.
 - Soffits: Free from undulations, lipping and distortions in grid members.
- Infill and access units, integrated services: Fitted correctly and aligned.
- Edge/ perimeter infill units size (minimum): Half standard width or length.
- Corner infill units size (minimum): Half standard width and length.
- Grid: Position to suit infill unit sizes
 - Permitted deviations from nominal sizes: Allow for.
- Infill joints and exposed suspension members: Straight, aligned and parallel to walls.
- Suitability of construction: Give notice where building elements and features to which the ceiling systems relate are not square, straight or level.

310 BRACING

- General: Secure, with additional bracing and stiffening to give a stable ceiling system resistant to design loads and pressures.

315 PROTECTION

- Loading: Do not apply loads for which the suspension system is not designed.
- Ceiling materials: Remove and replace correctly using special tools and clean gloves, etc. as appropriate.

320 TOP FIXING

- Building structure: Verify suitability.
 - Concrete: Drill and insert suitable expanding anchors.
 - Aerated concrete: Fix through from the top of the concrete unit and provide a system of primary channels.
 - Structural steel: Drill, or use suitable proprietary clips/ adaptors.
 - Metal roof decking: Fix to sides of corrugations.
 - Timber: Fix to side of joists at least 50 mm from bottom edge. If ceiling system is intended for fire protection, fix into top third of joists.
 - Hollow structural members: Submit fixing proposals.
- Cartridge or powder activated methods: Do not use.

325 INSTALLING HANGERS

- Wire hangers: Straighten and tension before use.
- Installation: Install vertical or near vertical without bends or kinks. Do not allow hangers to press against fittings, services, insulation covering ducts/ pipes.
- Obstructions: Where obstructions prevent vertical installation, either brace diagonal hangers against lateral movement, or hang ceiling system on an appropriate rigid sub-grid bridging across obstructions and supported to prevent lateral movement.
- Extra hangers: Provide as required to carry additional loads.
- Fixing:
 - Wire hangers: Tie securely at top with tight bends to loops to prevent vertical movement.
 - Angle/ strap hangers: Do not rivet for top fixing.
- Spacings: _____ .

330 INSTALLING TIMBER EDGE BATTENS

- Fixings:
 - Type: _____ .
 - Centres (maximum): 450 mm .

335 PERIMETER TRIMS

- Jointing: Neat and accurate, without lipping or twisting.
 - External and internal corners: Mitre joints.
 - Intermediate butt joints: Minimize. Use longest available lengths of trim. Align adjacent lengths.
- Fixing: Fix firmly to perimeter wall or other building structure.
 - Fixing centres: 450 mm .

340 EXPOSED GRIDS

- Grid fixings: _____ .
- Main runners: Install level. Adjust with supporting hangers taut. Do not adjust by kinking or bending hangers.
 - Spliced joints: Stagger.

- Wire hangers passing through main runners: Sharply bend and tightly wrap loops.
- Angle/ strap hangers: Do not rivet for bottom fixing.
- Angular displacement of long axis of one runner in relation to next runner in line with it: Not visually apparent.
- Cross members supported by main runners or other cross members: Install to form a right angle with intersecting runner.
- Cross tees: Flat and coplanar with flange of main runner after panel insertion.
 - Cross tees over 600 mm long, cut and resting on perimeter trim: Provide an additional hanger.
- Holding down clips: Locate to manufacturer's recommendations.
 - Fire protecting/ resisting ceiling systems: Use clip type featured in the fire test/ assessment.



385 UPSTANDS AND BULKHEADS

- Vertical ceiling systems: Support and brace to provide alignment and stability.
- High upstands: Provide support at base of upstand.

390 OPENINGS IN CEILING MATERIALS

- General: Neat and accurate. Suit sizes and edge details of fittings. Do not distort ceiling system.

395 INTEGRATED SERVICES

- General: Position services accurately, support adequately. Align and level in relation to the ceiling and suspension system. Do not diminish performance of ceiling system.
- Small fittings: Support with rigid backing boards or other suitable means. Do not damage or distort the ceiling.
 - Surface spread of flame rating of additional supporting material: Match ceiling material.
- Services outlets:
 - Supported by ceiling system: Provide additional hangers.
 - Independently supported: Provide flanges to support ceiling system.

401 CEILING MOUNTED LUMINAIRES

- Support: Independent .
 - Independently supported luminaires: Adjustable suspension to line and level of ceiling.
 - Ceiling supported luminaires: Modifications and/ or extra support required: _____ .
- Surface mounted luminaires: Do not inhibit designed grid expansion in fire.
- Modular fluorescent recessed luminaires: Compatible with ceiling module. Extension boxes must not foul ceiling system.
- Recessed rows of luminaires: Provide flanges for support of grid and infill units, unless mounted above grid flanges. Retain in place with lateral restraint.
- Fire protecting/ resisting ceiling systems: Luminaires must maintain protection integrity of ceiling system.
- Access: Provide access for maintenance of luminaires.

406 TRUNKING

- Recessed trunking: Provide flanges for support of grid and infill units, unless mounted above grid flanges. Retain in place with lateral restraint.

411 MECHANICAL SERVICES

- Fan coil units:
 - Inlet and outlet grilles: Trim ceiling grid and infill units to suit.
 - Space beneath: Sufficient for ceiling system components.
 - Suspension and connections: Permit accurate setting out and levelling of fan coil units.
- Air grilles and diffusers:
 - Setting out: Accurate and level.
 - Linear air diffusers: Provide flanges for support of grid and infill units. Provide for displacement of ceiling grid. Retain in place with lateral restraint.
 - Grille/ diffuser ceiling joints: Provide smudge rings and edge seals.
- Smoke detectors and PA speakers:
 - Infill units: Scribe to suit.
 - Independent suspension: Required .
 - Flexible connections: Required.
- Sprinkler heads: Carefully set out and level.

500 ELECTRICAL CONTINUITY AND EARTH BONDING

- Substantial conductive parts of the ceiling system: Electrically continuous and fully earth bonded to carry prospective earth fault currents.
 - Standard: To BS 7671.
- Sequence: Complete earth bonding as soon as possible after completion of each independent area of suspension system.
- Testing: After completion of the ceiling system, associated services and fittings, test conductive parts of suspension system required to carry earth fault current, or used as bonding connections. Give notice before testing.
 - Electrical continuity: Measure from various distant conductive points of ceiling system and to earth bar in distribution board serving the area.
 - Test current: Sufficient to indicate probable electrical performance under fault conditions.
 - Test instrument: Type providing a pulse of about 25 A at safe voltage for safe duration, and indicating resistance in ranges 0-2 ohms and 0-20 ohms.
 - Resistance of measuring conductors: Deduce from test instrument readings.
 - Test readings: Record and certify. Add results to resistance of other parts of the path forming the earth fault loop.

COMPLETION

505 TOOLS

- Access tools: At Practical Completion, supply one set of the following: Operating poles for access panels & suction pad lifters for ceiling tiles .

520 USER INSTRUCTIONS

- Contents: Include the following:
 - Correct methods for removing and replacing infill units and other components.
 - Cleaning methods and materials.
 - Recommendations for redecoration.
 - Ceiling systems intended for fire protection: Limitations placed on subsequent alterations and maintenance procedures, to ensure that their fire performance is not impaired.

- Maximum number, position and value of point loads that can be applied to ceiling system after installation.

530 SPARES

- General: At Practical Completion supply the following: 25 tiles for ceiling system K40/115.



L10 WINDOWS/ ROOFLIGHTS/ SCREENS/ LOUVRES

To be read with Preliminaries/ General conditions.

PRELIMINARY INFORMATION/ REQUIREMENTS

110 EVIDENCE OF PERFORMANCE

- Certification: Provide independently certified evidence that all incorporated components comply with specified performance requirements.

120 SITE DIMENSIONS

- Procedure: Before starting work on designated items take site dimensions, record on shop drawings and use to ensure accurate fabrication.
- Designated items: _____.

COMPONENTS

310 STEEL WINDOWS

- Manufacturer: Submit proposals.
 - Product reference: _____.
- Finish as delivered: Polyester powder coated.
- Glazing details: Insulating glass units incorporating low emissivity glass ($en = 0.05$), argon filled.
- Ironmongery/ Accessories: As schedule.
- Fixing: As clause 781.

330 ALUMINIUM WINDOWS

- Manufacturer: Submit proposals.
 - Product reference: _____.
- Finish as delivered: Polyester powder coating.
- Glazing details: Insulating glass units incorporating low emissivity glass ($en = 0.05$), argon filled.
- Ironmongery/ Accessories: As schedule.
- Fixing: As clause 782.

650 METAL LOUVRES

- Manufacturer: Submit proposals.
 - Product reference: _____.
- Material: Aluminium.
 - Finish as delivered: Powder coated.
- Fire resistance rating: Not applicable.
- Number of louvre banks: Minimum two.
- Louvre blade pitch and angle: Pitch: 75 mm.
- Blanking panels: As drawings.
- Accessories/ Other requirements: Removable section for plant access. Stainless steel insect screens.
- Fixing: Bolted to structural steelwork as clause 781.

INSTALLATION

710 PROTECTION OF COMPONENTS

- General: Do not deliver to site components that cannot be installed immediately or placed in clean, dry floored and covered storage.
- Stored components: Stack vertical or near vertical on level bearers, separated with spacers to prevent damage by and to projecting ironmongery, beads, etc.

740 CORROSION PROTECTION

- Surfaces to be protected: Aluminium alloy components in contact with iron and steel unless galvanized, concrete, mortar or plasters, especially when embedded.
- Protective coating: Two coats of bitumen solution to BS 6949 or an approved mastic impregnated tape.
 - Timing of application: Before fixing components.



750 BUILDING IN

- General: Not permitted unless indicated on drawings.
 - Brace and protect components to prevent distortion and damage during construction of adjacent structure.

765 WINDOW INSTALLATION GENERALLY

- Installation: Into prepared openings.
- Gap between frame edge and surrounding construction:
 - Minimum: 5mm.
 - Maximum: 10mm.
- Distortion: Install windows without twist or diagonal racking.

781 FIXING OF STEEL FRAMES

- Standard: As section Z20.
- Fasteners: Stainless steel wood screws.
 - Spacing: When not predrilled or specified otherwise, position fasteners not less than 50 mm and not more than 190 mm from ends of each jamb, adjacent to each hanging point of opening lights and at maximum 900 mm centres.

782 FIXING OF ALUMINIUM FRAMES

- Standard: As section Z20.
- Fasteners: Stainless steel wood screws.
 - Spacing: When not predrilled or specified otherwise, position fasteners not more than 250 mm from ends of each jamb, adjacent to each hanging point of opening lights, and at maximum 600 mm centres.

790 FIRE RESISTING FRAMES

- Gap between back of frame and reveal: Completely fill with intumescent mastic or tape.

800 BACKFILLING OF STEEL FRAME SECTIONS

- Windows fixed direct into openings: After fixing, fill back of steel frame with waterproof cement fillet.

810 SEALANT JOINTS

- Sealant:
 - Manufacturer: Submit proposals.
Product reference: _____.

- Colour: From selected manufacturers range.
- Application: As section Z22 to prepared joints. Finish triangular fillets to a flat or slightly convex profile.

820 IRONMONGERY

- Fixing: Assemble and fix carefully and accurately using fasteners with matching finish supplied by ironmongery manufacturer. Do not damage ironmongery and adjacent surfaces.
- Checking/ Adjusting/ Lubricating: Carry out at Completion and ensure correct functioning.



L20 DOORS/ SHUTTERS/ HATCHES

To be read with Preliminaries/ General conditions.

PRELIMINARY INFORMATION/ REQUIREMENTS

110 EVIDENCE OF PERFORMANCE

- Certification: Provide independently certified evidence that all incorporated components comply with specified performance requirements.

115 FIRE RESISTING DOORS/ DOORSETS/ ASSEMBLIES

- Evidence of fire performance: Provide certified evidence, in the form of a product conformity certificate, directly relevant fire test report or engineering assessment, that each door/ doorset/ assembly supplied will comply with the specified requirements for fire resistance if tested to BS 476-22, BS EN 1634-1 or BS EN 1634-3. Such certification must cover door and frame materials, glass and glazing materials and their installation, essential and ancillary ironmongery, hinges and seals.

150 SITE DIMENSIONS

- Procedure: Before starting work on designated items take site dimensions, record on shop drawings and use to ensure accurate fabrication.
- Designated items: _____ .

COMPONENTS

430 WOOD DOORSETS INTERNAL NON RESISTING

- Fire resistance rating: Not applicable..
- Sound insulation rating: Minimum 29dBA.
- Door leaf:
 - Core: Heavy duty.
 - Facings: MDF board.
 - Lippings: 3 mm hardwood lipping all round.
 - Finish as delivered: Prepared and primed, M60/130.
- Frame and architraves:
 - Wood species: European redwood.
 - Appearance class to BS EN 942: J40.
 - Finish as delivered: Prepared and primed, M60/130.
- Preservative treatment: Not applicable.
- Glazing details: Refer to door schedule.
- Moisture content on delivery: 9-13% for buildings with heating providing room temperatures in the range 12-21 °C.
- Ironmongery: As ironmongery schedule.
- Perimeter seals: Acoustic seals as door schedule.
- Fixing: Frames prepared for fixings at works.

431 WOOD DOORSETS INTERNAL FIRE RESISTING

- Fire resistance rating: To BS EN 1634-1, 30 minutes integrity.
- Sound insulation rating: Minimum 29dBA.
- Door leaf:
 - Core: Heavy duty.
 - Facings: MDF board.

- Lippings: 3 mm hardwood lipping all round.
 - Finish as delivered: Prepared and primed, M60/130.
- Frame and architraves:
 - Wood species: European redwood.
 - Appearance class to BS EN 942: J40.
 - Finish as delivered: Prepared and primed, M60/130.
- Preservative treatment: Not applicable.
- Glazing details: Refer to door schedule.
- Moisture content on delivery: 9-13% for buildings with heating providing room temperatures in the range 12-21 °C.
- Ironmongery: As ironmongery schedule.
- Perimeter seals: Fire & acoustic seals as door schedule.
- Fixing: Frames prepared for fixings at works.



480 DOORSETS EXTERNAL - STEEL

- Manufacturer: Submit proposals.
 - Product reference: _____.
- Door leaf: 50mm thick, 1.2mm thick zinc coated mild steel or stainless steel.
 - Finish as delivered: Polyester powdercoated.
 - Frame and architraves: _____.
 - Finish as delivered: Polyester powdercoated.
- Glazing details: Not applicable.
- Ironmongery: As ironmongery schedule.
- Perimeter seals: EPDM weatherseal.
- Other requirements: As ironmongery/door schedules.
- Fixing: Consult with selected manufacturer for recommendations.

481 DOORSETS INTERNAL - STEEL, FD 60S FIRE RESISTING AND SMOKE CONTROL

- Manufacturer: Submit proposals.
 - Product reference: _____.
- Door leaf: 50mm thick Lock Form construction from 1.2mm thick zinc coated mild steel with fire tested core.
 - Finish as delivered: Polyester powdercoated.
- Frame and architraves: _____.
 - Finish as delivered: Polyester powdercoated.
- Glazing details: Not applicable.
- Ironmongery: As ironmongery schedule.
- Perimeter seals: 2 No. Black polyurethane, self-adhesive door silencers, Silicone self-adhesive tear drop gasket.
- Other requirements: As ironmongery/door schedules.
- Fixing: Consult with selected manufacturer for recommendations.

610 ROLLER SHUTTERS _____

- Manufacturer: Stertil UK Ltd, Stertil House, Unit A, Brackmills Business Park, Caswell Road, Northampton NN4 7PW. Tel: 0870 770 0471, Fax: 01604 765 181, Email: info@stertiluk.com Web: www.stertiluk.com .
 - Product reference: Stokvis Rolashutter® .
- Performance: _____ .

- Arrangement: Vertical, fitted into reveal .
- Shutter curtain: Galvanized steel .
 - Finish as delivered: BSC Colorcoat HP200 high performance Plastisol coating on one side and a painted finish on the other .
- Frame/ Guides: Galvanized steel channels .
 - Finish as delivered: _____ .
- Operation: Manual hand chain hoist operation / Electrical drive unit .
- Ironmongery: As recommended by manufacturer for selected shutter .
- Other requirements: Fully enclosed roller casing and motor cover. Escape doors where indicated on drawings .

INSTALLATION

710 PROTECTION OF COMPONENTS

- General: Do not deliver to site components that cannot be installed immediately or placed in clean, dry, floored and covered storage.
- Stored components: Stacked on level bearers, separated with spacers to prevent damage by and to projecting ironmongery, beads, etc.

730 PRIMING/ SEALING

- Wood surfaces inaccessible after installation: Primed or sealed as specified before fixing components.

740 CORROSION PROTECTION

- Surfaces to be protected: _____ .
- Protective coating: Two coats of bitumen solution to BS 6949 or an approved mastic impregnated tape.
 - Timing of application: Before fixing components.

750 FIXING DOORSETS

- Timing: After associated rooms have been made weathertight and the work of wet trades is finished and dried out.

760 BUILDING IN

- General: Not permitted unless indicated on drawings.

770 DAMP PROOF COURSES ASSOCIATED WITH BUILT IN WOOD FRAMES

- Method of fixing: To backs of frames using galvanized clout nails.

780 DAMP PROOF COURSES IN PREPARED OPENINGS

- Location: Correctly positioned in relation to door frames. Do not displace during fixing operations.

800 FIXING OF LOOSE THRESHOLDS

- Spacing of fixings: Maximum 150 mm from each end and at 600 mm maximum centres.

809 FIRE RESISTING AND/ OR SMOKE CONTROL DOORS/ DOORSETS

- Installation: By a firm currently registered under a third party accredited fire door installer scheme in accordance with instructions supplied with the product conformity certificate, test report or engineering assessment.

820 SEALANT JOINTS

- Sealant:
 - Manufacturer: Submit proposals . Product reference: _____ .
 - Colour: From selected manufacturers range .
 - Application: As section Z22 to prepared joints. Triangular fillets finished to a flat or slightly convex profile.

830 FIXING IRONMONGERY GENERALLY

- Fasteners: Supplied by ironmongery manufacturer.
 - Finish/ Corrosion resistance: To match ironmongery.
- Holes for components: No larger than required for satisfactory fit/ operation.
- Adjacent surfaces: Undamaged.
- Moving parts: Adjusted, lubricated and functioning correctly at completion.

840 FIXING IRONMONGERY TO FIRE RESISTING DOOR ASSEMBLIES

- General: All items fixed in accordance with door leaf manufacturer's recommendations ensuring that integrity of the assembly, as established by testing, is not compromised.
- Holes for through fixings and components: Accurately cut.
 - Clearances: Not more than 8 mm unless protected by intumescent paste or similar.
- Lock/ Latch cases for FD 60 doors: Coated with intumescent paint or paste before installation.

850 LOCATION OF HINGES

- Primary hinges: Where not specified otherwise, positioned with centre lines 250 mm from top and bottom of door leaf.
- Third hinge: Where specified, positioned with centre line 250 mm below centre line of top hinge .
- Hinges for fire resisting doors: Positioned in accordance with door leaf manufacturer's recommendations.

860 INSTALLATION OF EMERGENCY EXIT DEVICES

- Standard: Unless specified otherwise, install panic bolts/ latches in accordance with BS EN 1125.

L35 FIXED UTILITARIAN ACCESS SYSTEMS

To be read with Preliminaries/ General conditions.

GENERAL

140 HIGH LEVEL WALKWAY SYSTEMS _____.

- Type: Fabricated.
- Positions and dimensions: As drawing ???.
- Components generally: Galvanised mild steel.
- Decking: Chequer plate.
- Stringers: Channels.
- Posts and rails: Tubes.
- Guarding system: Required.
- Connections: Weld.
- Fixings and supports: Weld.
- Accessories: 100 mm high toe plate.

150 STAIR SYSTEMS _____.

- Type: Fabricated.
- Positions and dimensions: As drawing ???.
- Components generally: Galvanised mild steel.
- Treads: Chequer plate.
- Stringers: Channels.
- Landings: Chequer plate.
- Posts and rails: Tubes.
- Guarding system: Required.
- Connections: Weld.
- Fixings and supports: Weld.
- Accessories: Contrast nosings to treads.

170 GUARDING SYSTEMS _____.

- Type: Fabricated.
- Positions and dimensions: As drawing ???.
- Components generally: Galvanised mild steel.
- Standards: Tubes.
- Handrails: Tubes.
- Infill panels: Not required.
- Connections: Weld.
- Fixings and supports: Weld.
- Accessories: None required.

SYSTEM PERFORMANCE

210 DESIGN

- Design: Complete the design of the fixed utilitarian access systems.
- Design standards:
 - Companionways: To BS 5395-3.

- Ladders for access to tall structures: To BS 4211.
- Straight stairs and winders: In accordance with BS 5395-1.
- Helical and spiral stairs: In accordance with BS 5395-2.
- Industrial stairs, permanent ladders and walkways: In accordance with BS 5395-3.
- Access to machinery: To BS EN ISO 14122-1, -2, and -3.
- Industrial type flooring, walkways and stair treads: To BS 4592.
- Proposals: Submit drawings, technical information, calculations and manufacturers' literature.

240 STRUCTURAL DESIGN WITH CHARACTERISTIC LOADS

- Design standard: To BS 5950-3.
 - Deflections (maximum): 6.5mm.
 - Characteristic loads: _____.

PRODUCTS

340 HIGH LEVEL WALKWAYS

- Standard: To BS 5395-3.
- Manufacturer: Submit proposals.
 - Product reference: _____.

350 STAIRCASES

- Standard:
 - Dog-leg and straight: To BS 5395-1.
 - Spiral: To BS 5395-2.
- Type: Straight .
- Manufacturer: Submit proposals.
 - Product reference: _____.

370 GUARDINGS

- Standard: To BS EN 14122-3.
- Manufacturer: Submit proposals.
 - Product reference: _____.

420 CARBON STEEL COMPONENTS

- Carbon steel, hot rolled and plate: To BS 10025.
- Solid plates: To BS 4592-5.
 - Thickness: 6 mm.
- Expanded metal grating panels: To BS 4592-2.
- Open bar gratings: To BS 4592-1.
 - Depth: _____.
- Fixings: As section Z20.
- Finish: Hot dip galvanized.
 - Colour: Self finished.

450 ANCHORS

- Manufacturer: Submit proposals.
 - Product reference: _____.

- Material and finish: Submit proposals.
- Size: Submit proposals.

FABRICATION

510 FABRICATION GENERALLY

- Shop drawings: Submit.
- Frameworks: Assemble and brace, including temporary members required for installation.
- Contact between dissimilar metals: Avoid.
- Fixings: Fully bolt together. Tighten bolts.
- Finish: Hot-dip galvanizing .
- Temporary transport support: Do not subject members to non-design loadings.

EXECUTION

620 EXECUTION GENERALLY

- Structural members: Do not modify, cut, notch or make holes in structural members.
- Frameworks: Assemble and brace, including temporary members required for installation.
 - Temporary support: Do not use access systems as temporary support or strutting for other work.
- External durability of fastenings: Corrosion resistant material or with a corrosion resistant finish.
- Bolted joints:
 - Contact between dissimilar metals: Avoid.
 - Bolts and washers: Select types, sizes and quantities of fasteners or packings and spacings to retain supported components without distortion or loss of support.
 - Fixed with isolating washers/ sleeves to avoid bimetallic corrosion.
 - Fully bolt together. Tighten bolts.
- Welded joints:
 - Standard: For carbon steel use metal arc welding to BS EN 1011-1 and -2.
 - Clean: Surfaces to be jointed.
 - Tack welds: Use only for temporary attachment.
 - Traces of flux residue, slag and weld spatter: Remove.
 - Surface of welds: Grind smooth.
 - Joints: Fully bonded with no holes or cracks.
- Finished components:
 - Free: From distortion, cracks, burrs and sharp arrises.
 - Corner junctions of identical sections: Mitre.
 - Handrails: Smooth and continuous, with no sharp edges.

660 ANCHORING

- Fixing positions: Coordinate location of holding down bolts and wall fixings with services fixing positions.

COMPLETION

910 CLEANING

- Cleaning: Clean off splashes and wipe down finishes.
- Notice for inspection (minimum): 5 days.

920 INSPECTION 930 DOCUMENTATION

- Operation and maintenance instructions: Submit.
- Record drawings: Submit.



M12 RESIN FLOORING

To be read with Preliminaries/ General conditions

TYPES OF FLOORING

110 RESIN FLOORING _____

- Substrate: Power floated concrete or cement:sand screed.
 - Preparation: All loose substrate material, laitence and contamination to be carefully removed. The surface must be prepared by grinding, vacuum-blasting, carborundum grinding, diamond grinding or machine sanding with coarse sandpaper to achieve minimum surface tensile strength of 1.5Mpa. Substrate to be carefully vacuum-cleaned after treatment. The substrate must not be washed after treatment.
- Resin flooring system:
 - Manufacturer: Flowcrete UK Limited, The Flooring Technology Centre, Booth Lane, Sandbach, Cheshire, CW11 3QF. Tel: 01270 753000 Fax: 01270 753333 Email: uk@flowcrete.com Web: www.flowcrete.com.
 - Primer reference: Not required.
 - Resin flooring reference: Flowseal EPW.
 - Nominal thickness: 2 coats, roller or rubber spatula applied to minimum thickness of 60µm for each coat. Drying time between coats to be minimum 1 hour or until surface feels dry and hard.
 - Surface finish/ treatment: None.
- Flatness/ Surface regularity:
 - Sudden irregularities: Not permitted.
 - Classification of surface regularity to BS 8204-6: SR2.
- Other requirements: Skirtings/upstands & sealant movement joints.

PREPARATION OF SUBSTRATES

210 MOISTURE CONTENT OF SUBSTRATES

- Drying aids: Before testing, turn off and wait at least four days.
- Test for moisture content: To BS 8203, Annex A using an accurately calibrated hygrometer. Take readings in corners, along edges, and at various points over the area being tested.
- Acceptability: Do not lay resin flooring until readings show 75% relative humidity or less.

220 SURFACE HARDNESS OF SUBSTRATES

- General: Substrates must restrain stresses that occur during setting and hardening of resin.
- Test for surface hardness: To BS EN 12504-2 using a rebound hammer.
- Report: Submit:
 - Details of areas where substrate hardness values do not comply with the resin flooring manufacturer's requirements.
 - Proposals for repair or cut out and replacement.

230 PREPARATION OF SUBSTRATES GENERALLY

- Chases/ Saw cuts: Cut/ break out at skirtings, free edges, movement joints, etc. for termination of resin flooring.
- Blow holes, cavities, cracks, etc: Fill with repair product recommended by resin flooring manufacturer.

- Cleanliness: Remove surface contaminants, debris, dirt and dust.
- Texture of surface: Suitable to accept resin flooring and achieve a full bond over the complete area.

260 METAL SUBSTRATES

- Cleaning: Shot blast to BS EN ISO 8501-1, grade Sa2.5.

LAYING FLOORING

310 WORKMANSHIP

- Operatives: Trained/ experienced in the application of resin floorings. Submit evidence of training/ experience on request.
- Fillers and incorporated aggregates: Thoroughly mix in to ensure wetting. Avoid over-vigorous mixing resulting in excessive air entrainment.
- Scattered aggregates: Broadcast onto wet surface of resin to give consistent appearance.
- Curing: Allow appropriate periods between coats, before surface treatments, and before trafficking/ use.

330 PRIMING

- Application: Spread primer evenly to completely wet, penetrate and seal substrates.

340 REINFORCEMENT

- Fabric: Lay glass fibre or similar as local reinforcement to resin flooring.
- Overlap at joints: Not less than 50 mm.
- Bedding: Roll into preliminary thin layer of resin flooring.
-

350 RESIN FLOORING

- Application: Spread coats evenly with uniform thickness, surface finish and colour.

400 ADHESION OF RESIN FLOORING

- Bond strength between substrate and fully cured resin flooring: In accordance with manufacturer's performance data.
- Test for bond strength: To BS 8204-6, clause 11.4 and BS EN 1542.

410 RESIN SKIRTINGS/ UPSTANDS

- General: Use a vertical grade of resin for skirting/ upstand. Form radius cove. Tuck top edge of resin into a chase in the skirting/ upstand or retain with a stop bead.
- Transition with resin flooring: Strip movement joints, abutments at chases in slab & sealant movement joints.

420 FREE EDGES OF RESIN FLOORING

- General: Form a straight, smooth transition to abutting floor finishes.
- Retention of resin edges: Not required

430 SEALANT MOVEMENT JOINTS _____

- General: Position over movement joints in substrate.
- Preparation and application: As section Z22.
- Retention of resin edges: Not required.
- Joint width: _____.

- Sealant: _____.
 - Colour: _____.

440 STRIP MOVEMENT JOINTS _____

- General: Position over movement joints in substrate.
- Retention of resin edges: Not required.
- Joint width: _____.
- Strip:
 - Manufacturer: _____.
 - Product reference: _____.
 - Insert type: _____.
 - Colour: _____.
 - Fixing: To exact finished level of resin flooring: _____.



M40 STONE/ CONCRETE/ QUARRY/ CERAMIC TILING/ MOSAIC

To be read with Preliminaries/ General conditions.

TYPES OF TILING/ MOSAIC

110 TILING TO SHOWERS & SPLASHBACKS

- Tiles: _____ .
 - Manufacturer/ Supplier: Submit proposals .
Product reference: _____ .
 - Colour: From selected manufacturers standard range .
 - Finish: Glazed .
 - Size: Modular 200 x 100 mm .
 - Thickness: 6 mm .
- Background/ Base: Metal stud partition as K10/127 .
 - Preparation: Stippling as clause 431 .
- Intermediate substrate: As clause 470 to shower areas .
- Bedding: Thin bed adhesive - ribbed, as clause 650 or cement:sand - as clause 690 .
 - Reinforcement: Not applicable .
 - Adhesive: Not applicable .
- Joint width: As spacer lugs .
- Grout: As clause 875 .
 - Type/ classification: Not applicable .
- Movement joints: As clause 815 at all perimeters including door thresholds and around columns etc .
- Accessories: Tile edge trim .

GENERALLY

210 SUITABILITY OF BACKGROUNDS/ BASES

- Background/ base tolerances: To permit specified flatness/ regularity of finished surfaces given the permissible minimum and maximum thickness of bedding.
- New background drying times (minimum):
 - Concrete walls: 6 weeks.
 - Brick/ block walls: 6 weeks.
 - Rendering: 2 weeks.
 - Gypsum plaster: 4 weeks.
- New base drying times (minimum):
 - Concrete slabs: 6 weeks.
 - Cement:sand screeds: 3 weeks.

250 SAMPLES

- General: Submit representative samples of the following: Each type of tile.

PREPARATION

390 PLASTERBOARD BACKGROUNDS

- Boards: Dry, securely fixed and rigid with no protruding fixings and face to receive decorative finish exposed.

431 STIPLING FOR KEY

- Stipple mix:
 - Cement: Portland to BS EN 197-1 type CEM I/42.5.
 - Sand: Clean coarse.
 - Proportions (cement:sand): 1-1.5:2.
 - Admixture: in accordance with tiling manufacturer's recommendations.
- Application/ finish: Brush applied to a deep close texture.
- Curing/ drying: Keep damp until hardened. Dry out to provide securely bonded finish.



470 INTERMEDIATE SUBSTRATE

- Substrate: _____ .
 - Manufacturer: RIW Limited (or equal approved), Arc House, Terrace Road South, Binfield, Bracknell, Berkshire RG42 4PZ. Tel: 01344 397 777. Fax: 01344 862010. E-mail: technical@riw.co.uk Web: www.riw.co.uk . Product reference: RIW Tilesafe .
 - Thickness: 2 mm .
- Fixing system: Self-adhesive .
 - Manufacturer: Not applicable . Product reference: Not applicable .
 - Application: Preparation: All surfaces to be clean, dry, free from dust, grease and oil etc. Surfaces must be smooth, sound and flat, with all indentations filled with a suitable filler and all protrusions be struck off. Primer: All surfaces to be primed with RIW Tilesafe Primer .
 - Substrate surface: Secure, true and even.
- Joints: Close butt.
 - Treatment: All butt joints in the membrane, wall to wall and wall to floor interfaces and at pipe entries to be sealed with RIW Tilesafe Jointing Compound, gun applied. All small gaps or voids around pipe entries etc to be filled with RIW Tilesafe Putty prior to application of jointing compound .
- . Penetrations: Seal.
- Accessories: RIW Tilesafe Reinforcing Tape to reinforce all wall junctions, butt joints and other detail work in membrane .

FIXING

510 FIXING GENERALLY

- Colour/shade: Unintended variations within tiles for use in each area/room are not permitted.
 - Variegated tiles: Mix thoroughly.
- Adhesive: Compatible with background/base. Prime if recommended by adhesive manufacturer.
- Cut tiles: Neat and accurate.
- Fixing: Provide adhesion over entire background/base and tile backs.
- Final appearance: Before bedding material sets, make adjustments necessary to give true, regular appearance to tiles and joints when viewed under final lighting conditions.
- Surplus bedding material: Clean from joints and face of tiles without disturbing tiles.

530 SETTING OUT

- Joints: True to line, continuous and without steps.
 - Joints on walls: Horizontal, vertical and aligned round corners.
 - Joints in floors: Parallel to the main axis of the space or specified features.
- Cut tiles: Minimise number, maximise size and locate unobtrusively.
- Joints in adjoining floors and walls: Align
- Joints in adjoining floors and skirtings: Align.
- Movement joints: If locations are not indicated, submit proposals.
- Setting out of Openings: Drawing references: See drawing schedule.
- Setting out of Fittings: Submit proposals.



550 FLATNESS/ REGULARITY OF TILING

- Sudden irregularities: Not permitted.
- Deviation of surface: Measure from underside of a 2 m straightedge placed anywhere on surface. The straightedge should not be obstructed by the tiles and no gap should be greater than 3 mm.

560 LEVEL OF TILING ACROSS JOINTS

- Deviation (maximum) between tile surfaces either side of any type of joint:
 - 1 mm for joints less than 6 mm wide.
 - 2 mm for joints 6 mm or greater in width.
 -

570 MORTAR BEDDING

- Bedding mix:
 - Cement: Portland to BS EN 197-1 type CEM I/42.5.
 - Sand for walls: To BS 1199, type A.
 - Sand for floors: To BS 882.
 - Grading limit: To BS 8204-1, table 1.
- Batching: Select from:
 - Batch by weight.
 - Batch by volume: Permitted on the basis of previously established weight:volume relationships of the particular materials. Use accurate gauge boxes. Allow for bulking of damp sand.
- Mixing: Mix materials thoroughly to uniform consistence. Use a suitable forced action mechanical mixer. Do not use a free fall type mixer.
- Application: At normal temperatures use within two hours. Do not use after initial set. Do not retemper.

580 POROUS TILES

- Tiles to be bedded in cement:sand: Soak in clean water for at least 30 minutes. Fix as soon as surface water has drained.

650 THIN BED ADHESIVE - RIBBED (WALLS)

- Application: Apply 3 mm floated coat of adhesive to dry background in areas of approximately 1 m². Trowel to ribbed profile.
- Tiling: Press tiles firmly onto float coat.

690 CEMENT|SAND - (WALLS)

- Preparation: Dampen background.
- Application: Apply floated coat: 1:3 - 4 cement:sand mortar bedding.

- Thickness (maximum): 10 mm.
- Finish: Equivalent to wood float. Before tiling allow to stiffen slightly.
- Tiling: Without delay, apply 2 mm thick coat of 1:2 cement:fine sand mortar to backs of tiles, filling keys. Press tiles firmly onto float coat. Tap firmly into position.

MOVEMENT JOINTS/ GROUTING/ COMPLETION

815 SEALANT MOVEMENT JOINTS IN TILING TO WALLS

- Joints: Extend through tiles and bedding to base/ background. Centre over joints in base/ background.
 - Width: To match that of structural movement joint .
- Sealant: Submit proposals .
 - Colour: To match grout .
 - Preparation and application: As section Z22.

875 GROUTING

- Sequence: Grout when bed/adhesive has set sufficient to prevent disturbance of tiles.
- Joints: 6 mm deep (or depth of tile if less). Free from dust and debris.
- Grouting: Fill joints completely, tool to profile, clean off surface. Leave free from blemishes.
 - Profile: Flush.
- Polishing: When grout is hard, polish tiling with a dry cloth.

885 COLOURED GROUT

- Staining of tiles: Not permitted
- Evaluating risk of staining: Apply grout to a few tiles in a small trial area. If discoloration occurs apply a protective sealer to tiles and repeat trial.



M50 RUBBER/ PLASTICS/ CORK/ LINO/ CARPET TILING/ SHEETING

To be read with Preliminaries/ General conditions.

TYPES OF COVERING

138 NEEDED CARPET TILING

- Location: Refer to drawings & schedules.
- Base: Power floated concrete or cement:sand screed.
 - Preparation: As clause 410, 430, 440 & 460.
- Fabricated underlay: Not applicable.
- Carpet tiles: _____.
 - Manufacturer: Interface Europe Ltd. (or equal approved), Shelf Mills, Shelf, Halifax, West Yorkshire HX3 7PA Tel: 08705 304030, Fax: 01274 698300, Email: enquiries@eu.interfaceinc.com Web: www.interfaceeurope.com. Product reference: Superflor Heavy Contract Carpet Tile.
 - Type: Needle punch, synthetic hair.
 - BS EN 1470 classification: Level of use class: 4.
 - Size: 500 x 500 mm.
 - Colour/ pattern: Refer to drawings & schedules.
- Method of laying: Spray adhesive as manufacturers recommendations.
- Accessories: Edging strip at thresholds as clause 740.
- Other requirements: _____.

146 NEEDED PILE CARPET TILING

- Location: _____.
- Base: _____.
 - Preparation: _____.
 - Fabricated underlay: _____.
- Carpet tiles: _____.
 - Manufacturer: _____. Product reference: _____.
 - Type: _____.
 - BS EN 13297 classification: _____.
 - Level of use class: _____.
 - Size: _____.
 - Colour/ pattern: _____.
- Method of laying: _____.
- Accessories: _____.
- Other requirements: _____.

150 SHEETING PVC

- Location: Refer to drawings & schedules.
- Base: Power floated concrete or cement:sand screed.
 - Preparation: As clause 410, 430, 440 & 460.
- Fabricated underlay: Not applicable.
- Flooring roll: Heterogeneous PVC to BS EN 649.
 - Manufacturer: Armstrong Floor Products UK Ltd (or equal approved), Hitching Court, Abingdon Business Park, Abingdon, Oxon. OX14 1RB Tel: 01235 554848 Fax: 01235 553583 Email: uk-info@armstrong.com Web: www.armstrong-europe.com. Product reference: Contract Interior.

- BS EN 685 class: 42.
- Width: 2000 mm.
- Thickness: 2.0 mm.
- Colour/ pattern: Refer to drawings & schedules.
- Adhesive (and primer if recommended by manufacturer): As clause 640.
- Seam welding: Hot welding with complimentary coloured rod.
- Accessories: Edging trim for thresholds as clause 740.
- Finishing: As clause 820 & 830.
- Other requirements: _____.

155 PVC SHEET FLOORING IN WET AREAS

- Refer to drawings & schedules.
- Base: Power floated concrete or cement:sand screed.
 - Preparation: As clause 410, 430, 440 & 460.
- Fabricated underlay: _____.
- Flooring roll: PVC to BS EN 13553.
 - Manufacturer: Armstrong Floor Products UK Ltd 9or equal approved), Hitching Court, Abingdon Business Park, Abingdon, Oxon. OX14 1RB Tel: 01235 554848 Fax: 01235 553583 Email: uk-info@armstrong.com Web: www.armstrong-europe.com. Product reference: Protech.
 - Identity code: W1.
 - BS EN 685 class: 42.
 - Width: 2000 mm.
 - Thickness: 2.0 mm.
 - Colour/ pattern: Refer to drawings & schedules.
- Adhesive (and primer if recommended by manufacturer): As clause 640.
- Seam welding: Hot welding with complimentary coloured rod.
- Accessories: PVC skirtings as clause 770.
- Finishing: As clause 820 & 830.
- Other requirements: _____.

GENERAL REQUIREMENTS

210 WORKMANSHIP GENERALLY

- Base condition after preparation: Rigid, dry, sound, smooth and free from grease, dirt and other contaminants.
- Finished coverings: Accurately fitted, tightly jointed, securely bonded, smooth and free from air bubbles, rippling, adhesive marks and stains.

220 SAMPLES

- Covering samples: Before placing orders, submit representative sample of each type.

250 LAYOUT - ROLL MATERIALS

- Setting out of seams: Before placing orders agree setting out for sheeting types M50/ _____.

251 LAYOUT - SEAMS IN ROLL MATERIALS

- Setting out: Minimise occurrences of seams and cross seams.
- Cross seams: Not permitted in following locations: _____.

330 COMMENCEMENT

- Required condition of works prior to laying materials:
 - Building is weathertight and well dried out.
 - Wet trades have finished work.
 - Paintwork is finished and dry.
 - Conflicting overhead work is complete.
 - Floor service outlets, duct covers and other fixtures around which materials are to be cut are fixed.
- Notification: Submit not less than 48 hours before commencing laying.

340 CONDITIONING

- Prior to laying: Condition materials by unpacking and separating in spaces where they are to be laid. Maintain resilient flooring rolls in an upright position. Unroll carpet and keep flat on a supporting surface.
- . Conditioning time and temperature (minimum): As recommended by manufacturer with time extended by a factor of two for materials stored or transported at a temperature of less than 10 °C immediately prior to laying.

350 ENVIRONMENT

- Temperature and humidity: Before, during and after laying, maintain approximately at levels which will prevail after building is occupied.
- Ventilation: Before during and after laying, maintain adequate provision.

PREPARING BASES

410 NEW BASES

- Suitability of bases and conditions within any area: Commencement of laying of coverings will be taken as acceptance of suitability.

430 NEW WET LAID BASES

- Base drying aids: Not used for at least four days prior to moisture content testing.
- Base moisture content test: Carry out in accordance with BS 5325, Annexe A or BS 8203, Annexe A.
 - Locations for readings: In all corners, along edges, and at various points over area being tested.
- Commencement of laying coverings: Not until all readings show 75% relative humidity or less.

440 SUBSTRATES TO RECEIVE THIN COVERINGS

- Trowelled finishes: Uniform, smooth surface free from trowel marks and other blemishes. Abrade suitably to receive specified floor covering material.

460 SMOOTHING/ LEVELLING UNDERLAYMENT COMPOUND

- Type: Latex cement.
- Selection: As recommended by covering manufacturer.

LAYING COVERINGS

610 SETTING OUT TILES

- Method: Set out from centre of area/ room, so that wherever possible:

- Tiles along opposite edges are of equal size.
- Edge tiles are more than 50% of full tile width.

620 COLOUR CONSISTENCY

- Finished work in any one area/ room: Free from banding or patchiness.

640 ADHESIVE FIXING GENERALLY

- Adhesive: Type to be as specified, recommended by covering/ underlay manufacturer or as approved.
- Primer: Use and type as recommended by adhesive manufacturer.
- Application: As necessary to achieve good bond.
- Finished surface irregularities: Trowel ridges and high spots caused by particles on the substrate not acceptable.

650 SEAMS

- Patterns: Matched.
- Joints: Tight without gaps.

680 SEAM WELDING COVERINGS

- Commencement: Not before a period of 24 hours after laying, or until adhesive has set.
- Joints: Neat, smooth, strongly bonded, flush with finished surface.

700 LOOSE LAID CARPET TILES

- Areas of adhered tiles: Secure using double sided tape or peelable adhesive.
- Joints: Tightly butted.
 - Perimeter joints: Accurately cut to match abutment and prevent movement.

720 DOORWAYS

- Joint location: On centre line of door leaf.

740 EDGINGS/ COVER STRIPS

- Manufacturer: Submit proposals .
 - Product reference: _____ .
- Material/ finish: _____ .
- Fixing: Secure (using matching fasteners where exposed to view) with edge of covering gripped.

770 SKIRTINGS

- Types: PVC.
- Manufacturer: Submit proposals.
 - Product reference: _____.
- Fixing: Secure with top edge straight and parallel with floor.
 - Corners: Mitre joints.

780 TRAFFICKING AFTER LAYING

- Covering types: _____.
- Traffic free period: 7 hours or until adhesive is set.

COMPLETION

820 FINISHING PLASTICS FLOORING

- Cleaning operations:
 - Wash floor with water containing neutral (pH 6-9) detergent. If necessary, lightly scrub heavily soiled areas.
 - Rinse with clean water, removing surplus to prevent damage to adhesive. Allow to dry.
- Emulsion polish: Two coats of a type recommended by covering manufacturer.

830 FINISHING RUBBER FLOORING

- Cleaning operations:
 - Wash floor with a cleaner recommended by covering manufacturer.
 - Wet vacuum or mop up residue.
 - Rinse with clean water. Wet vacuum or mop up and allow to dry.
- Final treatment: Follow recommendations of covering manufacture and spray buff with wetting agent or dry burnish.

880 WASTE

- Spare covering material: Retain suitable material for patching. On completion submit pieces for selection. Hand over selected pieces to Employer.



M60 PAINTING/CLEAR FINISHING

To be read with Preliminaries/General conditions.

COATING SYSTEMS

110 EMULSION PAINT TO INTERNAL PLASTERED SURFACES

- Manufacturer: Contractor's choice.
 - Product reference: _____.
- Surfaces: Uncoated.
 - Preparation: Tape and fill joints and Wash down surfaces .
- Initial coats:
 - Type: 10% thinned coat .
 - Number of coats: 1 coat .
- Finishing coats:
 - Type: Matt vinyl .
 - Number of coats: 2 coats .

130 GLOSS PAINT TO INTERNAL EXPOSED SOFTWOOD

- Manufacturer: Contractor's choice .
 - Product reference: _____ .
- Surfaces: Preprimed and sealed .
 - Preparation: Degrease and abrade to provide key .
- Initial coats:
 - Type: Acrylic gloss .
 - Number of coats: 1 coat .
- Finishing coats:
 - Type: Acrylic gloss .
 - Number of coats: 2 coats .

150 EGGSHELL/ SATIN PAINT TO INTERNAL PLASTERED SURFACES TO KITCHENETTES & CHANGING AREAS .

- Manufacturer: Contractor's choice .
 - Product reference: _____ .
- Surfaces: Uncoated .
 - Preparation: Tape and fill joints and Wash down surfaces .
- Initial coats:
 - Type: Quick Drying Eggshell .
 - Number of coats: 1 coat .
- Finishing coats:
 - Type: Quick Drying Eggshell .
 - Number of coats: 2 coats .

GENERALLY

215 HANDLING AND STORAGE

- Coating materials: Deliver in sealed containers, labelled clearly with brand name, type of material and manufacturer's batch number.
- Materials from more than one batch: Store separately. Allocate to distinct parts or areas of the work.

220 COMPATIBILITY

- Coating materials selected by contractor:
 - Recommended by their manufacturers for the particular surface and conditions of exposure.
 - Compatible with each other.
 - Compatible with and not inhibiting performance of preservative/fire retardant pretreatments.



240 SURFACES NOT TO BE COATED

- Radiator valves and stop valves .

280 PROTECTION

- 'Wet paint' signs and barriers: Provide where necessary to protect other operatives and general public, and to prevent damage to freshly applied coatings.

320 INSPECTION BY COATING MANUFACTURERS

- General: Permit manufacturers to inspect work in progress and take samples of their materials from site if requested.

PREPARATION

400 PREPARATION GENERALLY

- Standard: To BS 6150, Section 4.
- Preparation materials: Types recommended by their manufacturers and the coating manufacturer for the situation and surfaces being prepared.
- Substrates: Sufficiently dry in depth to suit coating.
- Efflorescence salts: Remove.
- Dirt, grease and oil: Remove.
 - Give notice if contamination of surfaces/ substrates has occurred.
- Joints, cracks, holes and other depressions:
 - Fill with stoppers/ fillers. Work well in and finish off flush with surface.
 - Abrade to a smooth finish.
- Water based stoppers and fillers:
 - Apply before priming unless recommended otherwise by manufacturer.
 - If applied after priming: Patch prime.
- Oil based stoppers and fillers: Apply after priming.
- Surface irregularities: Abrade to a smooth finish.
- Dust, particles and residues from abrasion: Remove.
- Doors, opening windows and other moving parts:
 - Ease before coating.
 - Prime resulting bare areas.

420 FIXTURES AND FITTINGS

- Before commencing work: Remove the following: Coverplates, grilles, wall clocks, and other surface mounted fixtures.

- On completion of coating work: Refix.

425 IRONMONGERY

- Before commencing work: Remove ironmongery from surfaces to be coated.
- Hinges: Do not remove.
- On completion of coating work: Refix.

471 PREPRIMED TIMBER

- Defective primer: Abrade back to bare timber.
- Bare areas: Reprime.

481 UNCOATED TIMBER

- General: Abrade to a smooth, even finish with arrises and moulding edges lightly rounded or eased.
- Heads of fasteners: Countersink sufficient to hold stoppers/fillers.
- Resinous areas and knots: Apply two coats of knotting.

580 UNCOATED PLASTER

- Nibs, trowel marks and plaster splashes: Scrape off.
- Overtrowelled 'polished' areas: Abrade lightly.

590 UNCOATED PLASTERBOARD

- Depressions around fixings: Fill with stoppers/ fillers

622 ORGANIC GROWTHS

- Loose growths and infected coatings: Scrape off and remove.
- Treatment biocide: Apply appropriate solution to growth areas and surrounding surfaces.
- Dead growth: Scrape off and remove.
- Residual effect biocide: Apply appropriate solution to inhibit re-establishment of growths.

645 POINTING TO INTERNAL MOVEMENT JOINTS

- General: Apply to junctions of walls and ceilings with architraves, skirtings and other trims.
- Preparation and application: As section Z22.
- Sealant: Water based acrylic.
 - Manufacturer: Contractors choice. Product reference: _____.

APPLICATION

711 COATING GENERALLY

- Application standard: To BS 6150, Section 5.
- Conditions: Maintain suitable temperature, humidity and air quality during application and drying.
- Surfaces: Clean and dry at time of application.
- Thinning and intermixing of coatings: Not permitted unless recommended by manufacturer.
- Overpainting: Do not paint over intumescent strips or silicone mastics.
 - Priming coats:
 - Thickness: To suit surface porosity.

- Application: As soon as possible on same day as preparation is completed.
- Finish:
 - Even, smooth and of uniform colour.
 - Free from brush marks, sags, runs and other defects.
 - Cut in neatly.

720 PRIMING JOINERY

- Preservative treated timber: Retreat cut surfaces before priming.
- End grain: Liberally coat, allow to soak in and then recoat.
- General: Apply coatings to all surfaces of components.

730 WORKSHOP COATING OF CONCEALED JOINERY SURFACES 731 SITE COATING OF CONCEALED JOINERY SURFACES

- General: After priming, apply additional coatings to surfaces that will be concealed when fixed in place.
 - Additional coatings: One undercoat.
- Components: _____.

810 WATER REPELLENT

- Application: Liberally flood surface, giving complete and even coverage.

M61 INTUMESCENT COATINGS FOR FIRE PROTECTION OF STEELWORK

To be read with Preliminaries/General conditions.

PROTECTIVE COATING SYSTEMS

150 OFF SITE COATING TO UNPRIMED STEEL INTERNALLY

- Use/ Location: Exposed internal faces of columns, beams and purlins, but excluding cladding rails.
- Preparation: Blast clean as clause 315.
- Primer:
 - Type: Zinc phosphate as recommended by intumescent coating manufacturer .
 - Manufacturer: Submit proposals. Product reference: _____.
 - Dry film thickness: Manufacturers recommendation .
- Intumescent coat:
 - Manufacturer: As primer. Product reference: _____.
 - Fire resistance to BS 476-21: 60 min. generally, 120 min. to compartments in accordance with fire safety report.
 - Finish: Visible areas: Normal decorative as clause 450.
- Top sealer coat:
 - Manufacturer: As primer. Product reference: _____.
 - Dry film thickness: As clause 490.
 - Colour: Blue, BS 4800 18 E 51.
 - Bolt head/ nut protection: As main steelwork .

GENERAL REQUIREMENTS

205 VALIDATION OF MATERIALS

- Project specific evaluation of intumescent coating materials:
 - Standard: To BS 8202-2, clause 4.
 - Submit on request.

210 WORKING PROCEDURES

- Standard: To BS 8202-2.
- Give notice: Before commencing surface preparation and coating application.
- Quality control: Record project specific procedures for surface preparation and coating application.

215 WORKING CONDITIONS

- General: Maintain suitable temperature, humidity and air quality during coating application and drying.
- Surfaces: Clean and dry at time of application.

220 APPLICATOR'S PERSONNEL

- Operatives: Trained/ experienced in anticorrosive and intumescent coatings.
- Evidence of training/ experience: Submit on request.

270 INSPECTION

- Permit intumescent coating manufacturer to:
 - Inspect work in progress.

- Inspect quality control records.
- Take dry film thickness and other measurements.
- Take samples of coating products.
- Intumescent coating manufacturer's inspection reports: Submit without delay.

280 OFF SITE COATED STEEL

- Masking: Apply to areas of steel not requiring application of intumescent coatings.
- Handling and erection: Use methods and devices designed to minimise damage to intumescent coatings.

PREPARATION OF SURFACES

315 NEW STEEL - BLAST CLEANING

- Preparation: Remove oil and grease.
- Blast cleaning:
 - Atmospheric condition: Dry.
 - Abrasive: Suitable type and size, free from fines, moisture and oil.
 - Finish: To BS EN ISO 8501-1, preparation grade SA2½, with an average profile of approximately 75 micrometres.
 - Abrasive residues and moisture: Remove.
- Primer: Apply as soon as possible after cleaning and before gingering or blackening appears.

APPLICATION OF COATINGS

410 INTUMESCENT DRY FILM THICKNESS (DFT)

- Required dft: Determine for every steel member to give specified period of fire resistance. Use intumescent coating manufacturer's current published loading tables.
 - Special sections and partial fire exposure conditions: Obtain required dft in writing from manufacturer.
- Schedule and drawings: Submit at least two weeks before starting work.
 - Schedule content: Member sizes, weights/thicknesses, loading conditions, etc. showing, for each variant, the exposed perimeter/ sectional area (Hp/A) ratio and required dft.
 - Drawing content: Steelwork drawings marked in colour to show required dft for each member.

420 MEASUREMENT OF INTUMESCENT DFT

- Primer dft: Determine average dft for deduction from total dft after application of intumescent.
- Intumescent dft: Determine at:
 - 500 mm centres along each coated plane of sections and rectangular hollow sections (i.e. 8 planes for a universal column/ beam).
 - 125 mm centres along coated circular hollow sections, spread evenly around circumference.
- Acceptance standard:
 - Average intumescent dft: Not less than required dft (exclusive of primer and top sealer).
- Local intumescent dft: Not less than 80% of required dft. Areas greater than 100 mm equivalent diameter of less than 80% of required dft must be brought up to thickness.

440 BASIC FINISH

- Definition: Reasonably smooth and even. Orange peel, other texture, minor runs and similar minor defects are acceptable.

450 NORMAL DECORATIVE FINISH

- Definition: Good standard of cosmetic finish generally, when viewed from a distance of 5 m or more. Minor orange peel or other texture is acceptable.

490 TOP SEALER COAT

- Application: To achieve dft recommended by manufacturer and to give an even, solid, opaque appearance, free from runs, sags and other visual defects.

520 COMPLETION OF OFF SITE COATED STEEL

- Exposed unprotected areas, including fixings: Following erection of steelwork, locally apply intumescent coatings.
- Unscheduled additional connections to erected steelwork: Locally remove and reinstate intumescent coatings.

530 RECORDS OF COATED STEEL

- On completion of intumescent coating work, submit:
 - Accurate surface preparation and coating application records.
 - Fire resistance certificates.
 - Intumescent coating manufacturer's recommendations for maintenance and overcoating.



N10 GENERAL FIXTURES/ FURNISHINGS/ EQUIPMENT

To be read with Preliminaries/General conditions.

PRODUCTS

145 VANITY AND WASHROOM FURNITURE

- Manufacturer: Armitage Venesta Washroom Systems (or equal approved), Chartwell Court, West Mill, Imperial Business Park, Gravesend, Kent, DA11 0DL Tel: 01474 353333 Fax: 01474 533558 Web: www.armitage-venesta.co.uk.
 - Product reference: C1 Profile fully inset vanity unit.
- Material: 13mm thick compact laminate.
- Dimensions: Top: 13mm thick compact laminate. Top comprises 555mm overall width bed with 100mm front downstand, 800mm overall height from FFL to top of bed. Underpanels: 13mm thick compact laminate. All edges radiused and polished.
- Supports:
 - Material: _____.
 - Finish/ Colour: _____.
 - Edges: _____.
- Doors/ Drawers/ Shelves:
 - Material: _____.
 - Finish/ Colour: _____.
 - Edges: _____.
- Worktops:
 - Material: _____.
 - Finish/ Colour: _____.
 - Exposed edges: _____.
- Other requirements: _____.

150 FITTED KITCHEN UNITS

- Standard: To BS 6222 and BS EN 1153.
- Manufacturer: Submit proposals.
 - Product reference: _____.
- Structural performance: To BS 6222-2, grade G.
- Dimensions: To BS EN 1116.
- Surface finishes: To BS 6222-3.
- Doors/ Drawer fronts:
 - Material: Plastics laminate.
 - Finish/ Colour: Refer to drawings/schedules.
 - Edges: _____.
- Side panels/ Plinths/ Shelves:
 - Material: Plastics laminate.
 - Finish/ Colour: Refer to drawings/schedules.
 - Edges: _____.
- Worktops:
 - Classification: To BS 6222-3, type: 1..
 - Material: Plastics laminate.
 - Finish/ Colour: Refer to drawings/schedules.
 - Exposed edges: _____.

- Accessories: Worktop trim.
- Other requirements: _____.

165 OFFICE SHELVING SYSTEM GENERAL OFFICES

- Standard: To BS 5459-3.
- Manufacturer: Submit proposals.
 - Product reference: _____.
- Test level: To BS 5459-3: G.
- Shelves:
 - Material: Plastics laminate.
 - Finish/ Colour: Refer to drawings/schedules.
- Other components: Support brackets.

200 WORKTOPS GENERALLY

- Manufacturer: Submit proposals.
 - Product reference: _____.
- Standard: None applicable.
- Material: Laminate covered particleboard.
- Dimensions: refer to drawings.
- Exposed edges: Laminate.
- Support: Battens off walls & legs.
- Other requirements: _____.

220 LOCKERS CHANGING ROOMS

- Manufacturer: Helmsman (or equal approved), Northern Way, Bury St Edmunds, Suffolk, IP32 6NH. Tel: 01284 727 600. Fax: 01284 727 666. Service Department Tel: 01284 727 633. Fax: 01284 727 601. Sales Email: sales@helmsman.co.uk Service Email: service@helmsman.co.uk Web: www.helmsman.co.uk.
 - Product reference: System 1301 Vedette Range.
- Standard: To BS 4680.
 - Test level: _____.
- Material: Mild steel-stove enamelled, Five stage automatic degreasing and phosphating process finished with a demineralising rinse to remove final surface contamination. This base gives improved adhesive and humidity resistance. The paint is applied by high-speed electrostatic rotating bells, which are computer controlled to ensure the correct thickness of paint is applied. The component parts are then stove enamelled at the correct temperature to ensure a hard paint finish is achieved.
 - Finish/ Colour: From; Locker frame: Ice / Stone. Locker doors: Ice / Stone / Rockpool / Smoke / Snow / Skye / Kingfisher / Royale / Forest / Cherry / Metallic Blue / Metallic Green / Metallic Grey / Metallic Pink.
- Fittings: Standard steel doors. End panels. Locker numbering: Number labels / Number plates / Number discs. Locking: Standard 10 disc tumbler cam lock in mastered suite.
- Other requirements: Standard locks fitted with espagnolette bolts. 100 mm high aluminium plinth.

240 BLINDS

- Manufacturer: _____.
 - Product reference: _____.
- Type: _____.
- Material: _____.

- Finish/ Colour: _____.
- Operation: _____.
- Accessories: _____.
- Other requirements: _____.

245 VENETIAN BLINDS

- Manufacturer: Submit proposals.
 - Product reference: _____.
- Standard: To BS 3415.
- Material: From selected manufacturers standard range.
 - Finish/ Colour: From selected manufacturers standard range.
- Accessories: From selected manufacturers standard range.
- Other requirements: _____.

270 MIRRORS CHANGING ROOMS

- Material: Class C safety glass to BS 6206 .
- Quality: Free from tarnishing, discoloration, scratches and other defects visible in the designed viewing conditions. Reflection undistorted.
- Size: Refer to drawings.
- Backing: Aluminium foil.
- Edges: Smooth.
- Fixing: CP dome top wood screws with polyethylene sleeves and washers.
- Installation: Accurately with sides vertical.

290 MATWELL FRAMES

- Manufacturer: Gradus Limited (or equal approved), Park Green, Macclesfield, Cheshire SK11 7LZ, Tel: 01625 428922, Fax: 01625 433949 Email: sales@gradusworld.com Web: <http://www.gradusworld.com>.
 - Product reference: FRA031 Matwell Frame.
- Material: Mill finished aluminium angles.
 - Finish: _____.
- Size: Refer to drawings.
- Angles: _____.
 - Corners: Mitred and welded.
 - Angle size: _____.

300 ENTRANCE MATTING

- Manufacturer: Gradus Limited (or equal approved), Park Green, Macclesfield, Cheshire SK11 7LZ, Tel: 01625 428922, Fax: 01625 433949 Email: sales@gradusworld.com Web: <http://www.gradusworld.com> .
 - Product reference: Access 4000 – 17mm thick .
- Colour: Dark grey or black. Insert colour from manufacturers standard range .
- Size: Refer to drawings .

EXECUTION

710 MOISTURE CONTENT OF WOOD AND WOOD BASED BOARDS

- Temperature and humidity: During delivery, storage, fixing and to handover maintain conditions to suit specified moisture contents of timber components.

- Testing: When instructed, test components with approved moisture meter to manufacturer's recommendations.

720 INSTALLATION GENERALLY

- General: _____.
- Fixing and fasteners: As section Z20.
- Services: As Engineering Services specification.

740 TAPS

- Fixing: Secure, watertight seal with the appliance.
- Positioning: Hot tap to left of cold tap as viewed by user of appliance.

750 WASTES AND OVERFLOWS

- Bedding: Waterproof jointing compound.
- Fixing: With resilient washer between appliance and backnut.

760 SEALANT POINTING

- Manufacturer: Contractor's choice.
 - Product reference: _____.
- Material: Silicone based to BS 5889, type B with fungicide.
- Colour: _____.
- Application: As section Z22.

770 TRIMS

- Lengths: Wherever possible, unjointed between angles or ends of runs.
- Running joints: Where unavoidable, obtain approval of location and method of jointing.
- Angle joints: Mitred.

COMPLETION

910 GENERAL

- Doors and drawers: Accurately aligned, not binding. Adjusted to ensure smooth operation.
- Ironmongery: Checked, adjusted and lubricated to ensure correct functioning.

920 APPLIANCES

- Test: Ensure that all functions and features work correctly.
- Documentation: Submit guarantees, instruction manuals, etc.

N13 SANITARY APPLIANCES AND FITTINGS

To be read with Preliminaries/ General conditions.

PRODUCTS

300 WCS AND CISTERNS _____

- WC standard: To DEFRA WC suite performance specification or approved by relevant water company.
- Type: Floor-mounted pan with exposed low-level cistern.
- Pan:
 - Standards: Magnia WC pan S3510 (149701A).
 - Manufacturer: Armitage Shanks (or equal approved), Armitage, Rugeley, Staffordshire WS15 4BT. Tel: 01543 490253, Fax: 01543 491677. Web: www.thebluebook.co.uk. Product reference: _____.
 - Material: _____.
- Seat and cover:
 - Standard: _____.
 - Manufacturer: _____. Product reference: Gemini plastics seat and cover S4055 (68940B1).
 - Material: _____.
- Pan connector:
 - Standard: To BS 5627.
 - Manufacturer: _____. Product reference: Panekta pan connector to P trap, finned S4305 (9013000) .
 - Colour: To match pan.
- Cistern:
 - Standard: _____.
 - Manufacturer: _____. Product reference: Compact low-level cistern for bottom supply and internal, not reversible, chromium plated lever S3902 (17730BA).
 - Material: _____.
 - Colour and finish (exposed cisterns): To match pan.
- Flushing arrangement: _____.
 - Manufacturer: _____. Product reference: _____.
 - Operating control: _____.
 - Water supply connection: _____.
 - Flush volume: 6 litres.
- Flush pipe: _____.
 - Manufacturer: _____. Product reference: _____.
 - Material: _____.
- Accessories: Domex screws (four) S9102 (7550BFR).

315 URINALS AND AUTO FLUSHING CISTERNS _____

- Urinals:
 - Standard: _____.
 - Manufacturer: Manufacturer: Armitage Shanks (or equal approved), Armitage, Rugeley, Staffordshire WS15 4BT. Tel: 01543 490253, Fax: 01543 491677. Web: www.thebluebook.co.uk. Product reference: Contour wall urinal S6110 (261229E), Urinal division with hanger and screw S6120 (2606000).
 - Material: Vitreous china, white.



- Wastes: 1½in plastics domed strainer waste, 50mm unslotted tail S885067 (90568N0 SC).
 - Standards: To BS EN 274-1, -2 and -3.
 - Manufacturer: _____. Product reference: _____.
 - Size: _____.
 - Material: _____.
 - Tail: Unslotted.
- Traps: 1½in plastics P trap with 75mm seal S8970 (71428Q4).
 - Standards: To BS EN 274-1, -2 and -3.
 - Manufacturer: _____. Product reference: _____.
 - Size: _____.
 - Material: _____.
 - Depth of seal (minimum): 75 mm.
- Cistern, complete with automatic siphon, lid, supports and fixings: Conceala automatic cistern 4.5 litres S6215 (4225100).
 - Standard: To BS 1876.
 - Manufacturer: _____. Product reference: _____.
 - Material: _____.
 - Flush volume: _____.
 - Operating control: _____.
 - Flush pipe: _____. Product reference: _____.
 - Material: _____.
- Accessories: _____.

331 SINKS _____

- Standard: _____.
- Manufacturer: Armitage Shanks (or equal approved), Armitage, Rugeley, Staffordshire WS15 4BT. Tel: 01543 490253, Fax: 01543 491677. Web: www.thebluebook.co.uk.
 - Product reference: Doon single sink bowl, two tapholes, no overflow, 60x60cm S5860MY (533013UPO).
- Size: _____.
- Material: _____.
- Tap/ Chainstay/ Overflow holes: _____.
- Taps: Nimbus ½" bib taps, (pairs)dual indices, chromium plated S7206AA (661041GCP) TB3 75mm extension pieces, (pairs)chromium plated S8346AA (8174BPRCP).
 - Manufacturer: _____. Product reference: _____.
 - Wastes: 1 ½" chain waste, unslotted 50mm tail with bolt stay, chromium plated
 - Standards: To BS EN 274-1, -2 and -3.
 - Manufacturer: _____. Product reference: _____.
 - Size: _____.
 - Material: _____.
 - Tail: _____.
- Traps: Plastic 1 ½" resealing bottle trap, 75mm seal S892567 (7360700SC) TRR2-P.
 - Standards: To BS EN 274-1, -2 and -3.
 - Manufacturer: _____. Product reference: _____.
 - Size: _____.
 - Material: _____.
 - Depth of seal (minimum): 75 mm.
- Accessories: Stainless steel legs to floor (2) S9272MY (S09TQ05PO).

335 WASH BASINS INSET

- Manufacturer: Armitage Shanks (or equal approved), Armitage, Rugeley, Staffordshire WS15 4BT. Tel: 01543 490253, Fax : 01543 491677. Web: www.thebluebook.co.uk.
 - Product reference: Sandringham 50cm countertop washbasin, two tapholes, chainstay hole and overflow S2508.
- Size: _____.
- Material: _____.
- Tap/ Chainstay/ Overflow holes: Sandringham basin pillar taps (pair), chrome plated finish S7031AA.
- Water supply fittings: _____.
 - Manufacturer: _____. Product reference: _____.
 - Operation: _____.
- Wastes: Plastic 1 ¼" waste with bead chain and metal plug, 80mm slotted tail, bolt stay, chromium plated S8805AA Brass 1 ¼" waste, swivel plug, slotted tail, chromium plated S8733.
 - Standards: To BS EN 274-1, -2 and -3.
 - Manufacturer: _____. Product reference: _____.
 - Size: _____.
 - Material: _____.
 - Tail: _____.
- Traps: Plastic 1 ¼" bottle trap, 75mm seal S891067 (70237Q4SC).
 - Standards: To BS EN 274-1, -2 and -3.
 - Manufacturer: _____. Product reference: _____.
 - Size: _____.
 - Material: _____.
 - Depth of seal (minimum): 75 mm.
- Accessories: Isovalve servicing valves S900067 (9060400SC).

375 SHOWER UNITS _____

- Tray: _____.
 - Manufacturer: Armitage Shanks (or equal approved), Armitage, Rugeley, Staffordshire WS15 4BT. Tel: 01543 490253, Fax: 01543 491677. Web: www.thebluebook.co.uk. Product reference: L5154 Lincoln shower tray 75cm x 75cm.
 - Size: _____.
 - Material: _____.
- Shower fittings: Shower handset (white), hose and adjustable bracket, for surface-mounted shower valves S9400 (7892400).
 - Manufacturer: _____. Product reference: _____.
 - Finish: _____.
 - Operating control: _____.
- Wastes: 1½ in flat grid strainer waste, chromium plated S8726AA (70038P3CP), HTM reference WT2 .
 - Standards: To BS EN 274-1, -2 and -3.
 - Manufacturer: _____. Product reference: _____.
 - Size: DN 40.
 - Material: _____.
 - Tail: Unslotted.
- Traps: Top access combined waste and trap.

- Standards: To BS EN 274-1, -2 and -3.
- Manufacturer: _____. Product reference: _____.
- Size: DN 40.
- Material: _____.
- Depth of seal (minimum): 50 mm.
- Enclosure: From selected manufacturers range.
 - Manufacturer: _____. Product reference: _____.
- Accessories: _____.

580 SEALANT FOR POINTING

- Standard: To BS EN ISO 11600.
 - Class: _____.
- Type: _____.
 - Manufacturer: Contractor's choice. Product reference: _____.
 - Colour: _____.

EXECUTION

610 INSTALLATION GENERALLY

- Assembly and fixing: Surfaces designed to falls to drain as intended.
- Fasteners: Nonferrous or stainless steel.
- Supply and discharge pipework: Fix before appliances.
- Fixing: Fix appliances securely to structure. Do not support on pipework.
- Jointing and bedding compounds: Recommended by manufacturers of appliances, accessories and pipes being jointed or bedded.
- Appliances: Do not use. Do not stand on appliances.
- On completion: Components and accessories working correctly with no leaks.
- Labels and stickers: Remove.

613 COMPATIBILITY OF COMPONENTS

- General: Each sanitary assembly must consist of functionally compatible components obtained from a single manufacturer.
 - Exceptions: Wastes and traps.

620 NOGGINGS AND BEARERS

- Noggings, bearers, etc. to support sanitary appliances and fittings: Position accurately. Fix securely.

630 TILED BACKGROUNDS OTHER THAN SPLASHBACKS

- Timing: Complete before fixing appliances.
- Fixing appliances: Do not overstress tiles.

650 INSTALLING WC PANS

- Floor mounted pans: Screw fix and fit cover caps over screw heads. Do not use mortar or other beddings.
- Seat and cover: Stable when raised.

670 INSTALLING CISTERNS

- Cistern operating components: Obtain from cistern manufacturer.

- Float operated valve: Matched to pressure of water supply.
- Overflow pipe: Fixed to falls and located to give visible warning of discharge.
 - Location: Agreed, where not shown on drawings.

710 INSTALLING TAPS

- Fixing: Secure against twisting.
- Seal with appliance: Watertight.
- Positioning: Hot tap to left of cold tap as viewed by user of appliance.

720 INSTALLING WASTES AND OVERFLOWS

- Bedding: Waterproof jointing compound.
- Fixing: With resilient washer between appliance and backnut.

755 SEALANT BEDDING AND POINTING

- Bedding: Bed and point basins to underside of vanity units. .
- Pointing: Joints between appliances and splashbacks. Joints between appliances and walls. Joints between appliances and floors. .



N25 PERMANENT ACCESS AND SAFETY EQUIPMENT

To be read with Preliminaries/ General conditions.

TYPES OF SYSTEM/ EQUIPMENT

210 GUIDED TYPE FALL ARREST SYSTEM FOR ROOF MAINTENANCE ACCESS

- Manufacturer: Latchways plc, Hopton Park, Devizes Wiltshire SN10 2JP Tel: +44 (0)1380 732700 Fax: +44 (0)1380 732701 E-mail: spec@latchways.com Web: www.latchways.com.
 - System reference: 210 Guided Type Fall Arrest System; Latchways ManSafe® for Standing Seam Roofing.
- Anchorage device: Latchways Constant Force™ Post for bulb type standing seam roofing. Finish: Polyester powder coated. Colour: Neutral Silver Attachment: Mechanical attachment using 4no. Latchways split clamps per post which locate over the standing seam.
- Overall system length: As shown on drawings.
- Intermediate support spacing: Between 6 and 10m and as designed by the Latchways approved installer to meet the requirements of the Latchways ManSafe design software.
- Accessories/ Other requirements: System access: To be from && Consult with Latchways for recommendations and complete accordingly. Completion: On completion of the installation, system to be inspected and fully tested and a test certificate covering a period of 12 months to be issued. Notice showing date and period of validity of the test certificate to be attached to the system at each access point. && no. of complete sets of Personal Protective Equipment PPE to be supplied with the system. Set to comprise - && no. of attachment devices x Removable Transfastener – Part number 85085. All to be supplied in a suitable PPE storage holdall.
- Installation: To BS 7883 by the system manufacturer or a contractor approved by the system manufacturer.
- Structural anchors: Type recommended by the system manufacturer to suit the structure/ fabric into which they will be fixed.

220 SINGLE POINT ANCHORAGE DEVICES FOR PERSONAL PROTECTIVE EQUIPMENT

- Standard: To BS EN 795.
- Manufacturer: Latchways plc, Hopton Park, Devizes Wiltshire SN10 2JP Tel: +44 (0)1380 732700 Fax: +44 (0)1380 732701 E-mail: info@latchways.com Web: http://www.latchways.com.
 - Product reference: Latchways PushLock System.
- Type: 200 mm x M12 eyebolt, 19 mm I/D - TBC.
- Material/ Finish: stainless steel.
- Locations: As shown on drawings.
- Installation: To BS 7883.
- Other requirements: Provide with each anchor:
 - A backing disc giving the manufacturer's name and telephone number and the date of installation.
 - A certificate of compliance with testing and examination requirements of BS EN 365.

DESIGN/ PERFORMANCE REQUIREMENTS

410 WIND LOADING

- General: Design access/ safety system to withstand calculated wind loads with equipment in position of maximum exposure and in parked position.

- Wind loads: Calculate in accordance with BS 6399-2, Standard Method:
 - Basic wind speed (V_b): by reference to BS 6399-2.
 - Altitude factor (S_a): by reference to BS 6399-2.
 - Direction factor (S_d): 1.
 - Seasonal factor (S_s): 1.
 - Probability factor (S_p): 1.
 - Terrain and building factor (S_b), determined from BS 6399-2, table 4 _____.

430 SAFETY

- General: The equipment as installed must have no irregularities/ projections capable of inflicting personal injury.
- Finished surfaces and edges of all accessible parts: Regular and smooth.

440 DESIGN LIFE/ MAINTENANCE PROGRAMME

- Design life of access/ safety system: Not less than Confirm with manufacturer years.
- Schedule for maintenance and for replacement of components: Submit.

460 ASSESSMENT/ TESTING OF FIXING POINTS FOR ANCHOR DEVICES

- Design and installation of fixings in steelwork or timber: Verified by calculation to be capable of sustaining the relevant static and dynamic test forces specified in BS EN 795, clause 4.3.
- Fixings in other materials: Verify suitability by carrying out a test in a sample of the material. The sample must be capable of sustaining the relevant static and dynamic test forces specified in BS EN 795, clause 4.3. Thereafter, each structural anchor installed in that material must be subjected to an axial pull out force of 5 kN to confirm the soundness of the fixing. The structural anchor must sustain the force for a minimum of 15 seconds.

FABRICATION, ASSEMBLY AND INSTALLATION

510 FABRICATION AND ASSEMBLY GENERALLY

- Machine cutting, drilling and assembly: Carry out as much as possible in the workshop. Obtain approval for any reassembly on site.
- Dissimilar metal surfaces of assembly components/ supports/ fixings: Isolate to prevent electrolytic corrosion.

520 PROTECTION

- General: Do not deliver to site any components or assemblies that cannot be installed immediately or unloaded into a suitable well protected storage area.

530 SUITABILITY OF STRUCTURE/ FABRIC

- Visual and geometric survey of supporting structure and fabric: Carry out before commencing installation of access/ safety system. Report immediately if structure/ fabric will not allow required accuracy/ security of erection/ fixing.

540 MECHANICAL FIXINGS

- Materials: Unless otherwise recommended by equipment manufacturer:
 - Connecting bolts and other fixings fully accessible for inspection: Mild steel hot dip galvanized to BS 7371-6. Nuts: Tapped after galvanizing.
 - Cast-in anchors and other fixings not accessible for routine inspection: Austenitic stainless steel, grade 1.4401 (316) to BS EN 10088-1.

550 FASTENERS, INSERTS AND BOLTS FOR BUILDING IN

- Supplier: Equipment manufacturer/ supplier.

560 FIXINGS FOR SECURING EQUIPMENT

- Adjustment capability: Adequate three dimensional adjustment to accommodate building structure/ fabric irregularities.

570 FIXING ANCHOR INSTALLATION

- Site drilling or cutting into structure/ fabric: Permitted only in approved locations.
- Distance between all fixing devices and edges of supporting material: Not less than recommended by fixing manufacturer.

640 MARKING OF ANCHOR DEVICES

- Provision: Provide on or near each anchor device a label or other clear marking giving:
 - Manufacturer's name and telephone number.
 - Serial number and year of manufacture of device.
 - Maximum number of personnel that may be attached to the device at any one time.
 - Requirements for energy absorbers, ground clearance, etc.
- Anchor devices intended solely for use with personal protective equipment: Indicate restriction of use by pictogram or other suitable marking on or near the device.

810 SERVICE/ MAINTENANCE GUIDED TYPE FALL ARREST SYSTEM

- General: Following acceptance of the completed installation, service and maintain the equipment for the period stated below as and at intervals recommended by the manufacturer. Such maintenance to include a 'call-out' service during normal working hours to maintain the equipment in an acceptable and safe condition.
- Service/ Maintenance period: TBC.

820 OPERATING INSTRUCTIONS

- Equipment and accessories: Where appropriate, mark in such a way that it is possible to identify the correct mode of operation for their safe use.

830 OPERATING AND MAINTENANCE MANUAL

- General: Provide, for inclusion in the Building Manual, printed instructions and recommended procedures to be established by the Employer for operating and routinely maintaining the equipment. Provide diagrams where appropriate.
- Content:
 - Instructions for assembling/ erecting equipment for use.
 - Comprehensive operating instructions, including safety and emergency procedures, for all motions including upward, downward and lateral travel, and slew.
 - Servicing and planned maintenance procedures, including assembly instructions where maintenance necessitates dismantling of machinery parts.
 - List of replacement parts, with references.
 - Recommended procedures for testing equipment.

840 AS INSTALLED DRAWINGS

- General: After commissioning/ testing of the equipment provide as installed drawings for inclusion in the Building Manual.
 - Number of sets: TBC.
- Drawing content:

- Contractor's name and contract number.
- Location and date of installation.
- Manufacturer's name, model and type numbers.
- General arrangement of the complete installation.
- Electrical circuit wiring diagrams complete with details and ratings of all items of equipment.



P10 SUNDRY INSULATION/ PROOFING WORK

SUNDRY INSULATION/ PROOFING WORK

To be read with Preliminaries/ General conditions.

TYPES OF INSULATION

410 FLEXIBLE CAVITY BARRIERS

- . Fire resistance: 60 minutes integrity; 30 minutes insulation .
- Material: Wired mineral wool with foil face .
- Manufacturer: Rockwool Limited (or equal approved), Pencoed, Bridgend, CF35 6NY.
Tel: 01656 862 621. Fax: 0871 222 1784. Email: info@rockwool.co.uk Web:
www.rockwool.co.uk .
 - Product reference: Rockwool Cavity Barrier. Single layer 60 mm, foil faced Fire Barrier with overlapped joints .
- Installation requirements:
 - Spacing: Installed in the void so the maximum dimension in any direction is 20 m.
- Fixing: Secure at perimeters and joints with no gaps, to provide a complete barrier to smoke and flame.

432 MINERAL WOOL SLAB CAVITY BARRIERS

- Cavity width: Refer to drawings.
- Fire resistance: 60 minutes .
- Manufacturer: Rockwool Limited (or equal approved), Pencoed, Bridgend, CF35 6NY.
Tel: 01656 862 621. Fax: 0871 222 1784. Email: info@rockwool.co.uk Web:
www.rockwool.co.uk .
 - Product reference: Rockwool Firepro SP 60 Firestop Slab .
- Thickness: 75mm.
- Installation requirements:
 - Installed: Continuous, with minimum joints.
 - Fasteners: As selected manufacturers recommendations .
- Other requirements: _____ .

P12 FIRE STOPPING SYSTEMS

To be read with Preliminaries/ General conditions.

GENERAL

130 FIRE STOPPING SYSTEM TO INDIVIDUAL SERVICES PENETRATIONS TO DUCTS TO FIRE RESISTING WALLS

- Joint filler: Pipe collars - concealed intumescent.
 - Size or thickness: 110 mm.
 - Pipe outside diameter: Varies.
- Sealant: Fire resisting silicone.
 - Colour: Grey.

140 FIRE STOPPING SYSTEM TO MULTIPLE SERVICES PENETRATIONS TO PIPES TO FIRE RESISTING FLOORS

- Panel material: Mineral wool rigid batts.
 - Thickness: 50 mm.
 - Number of layers: Two.
 - Framing: 75 x 38 mm softwood arranged and trimmed around services.
- Finish: Intumescent mortars.
- Sealant: One-part fire resisting acrylic.
 - Colour: Grey.

160 MOVEMENT JOINTS STRUCTURAL IN FIRE RESISTING WALLS

- Soffit profiles: As drawings.
- Joint width or height (minimum): 25 mm.
- Joint filler: Sealant backing material.
- Sealant: Two-part fire resisting polysulfide.
 - Colour: Grey.

170 EXTERNAL WALLS TO PLANT ROOM

- Panel material or joint filler: Boards - calcium silicate filled and fibre.
- Sealant: One-part fire resisting acrylic.
 - Colour: Grey.

190 TEMPORARY FIRE STOPPING _____

- Panel material or joint filler: Intumescent pillows.
 - Panel or board thickness: Not applicable.
 - Intumescent pillow size: 300 x 100 x 30 mm.
- Sealant: Fire resisting silicone.

SYSTEM PERFORMANCE

210 DESIGN

- Design: Complete the design of the fire stopping system.
- Proposals: Submit drawings, technical information, calculations and manufacturers' literature.

240 FIRE RESISTANCE OF COMPARTMENT WALLS

- Fire resistance:
 - Rating to BS 476-20: 60/60.
 - Rating to BS EN 13501-2: EI 60.
- Surface spread of flame to BS 476-7: Class 0.
- Smoke resistance:
 - Air leakage rate (maximum): 3 m³/m²/hr.

PRODUCTS

305 PRODUCT CERTIFICATION

- Certification: For products specified generically, submit evidence of compliance with the specification.
- Acceptable evidence: Fire test certificate from NAMAS laboratory or Agrément certificate.

310 BOARDS - CALCIUM SILICATE FIBRE REINFORCED

- Manufacturer: Contractor's choice .
Product reference: _____.

315 BOARDS - CALCIUM SILICATE FILLED AND FIBRE REINFORCED

- Manufacturer: Contractor's choice .
Product reference: _____.

320 BOARDS - EXFOLIATED VERMICULITE

- Manufacturer: Contractor's choice .
Product reference: _____.

325 BOARDS - MINERAL BOUND LIGHTWEIGHT

- Manufacturer: Contractor's choice .
Product reference: _____.

330 FIRE STOP LAMINATES

- Manufacturer: Contractor's choice .
Product reference: _____.

335 INTUMESCENT FOAMS

- Manufacturer: Contractor's choice .
Product reference: _____.

340 INTUMESCENT MORTARS

- Manufacturer: Contractor's choice .
Product reference: _____.

345 INTUMESCENT PILLOWS

- Composition: Sealed polyethylene bags containing graphite and vermiculite granules treated with fire activated chemicals.
- Integral reinforced eyelets: Required.
- Linking cable: Non-corrosive cotton-coated wire.

- Length: The cable should be long enough to stitch the pillows together and to allow them to be unpacked, stored and refitted whilst work on the services is carried out.
- Manufacturer: Contractor's choice .
 - Product reference: _____.

350 INTUMESCENT PUTTIES

- Manufacturer: Contractor's choice.
Product reference: _____.

360 MINERAL WOOL RIGID BATTS

- Standard: To BS 3958-5.
- Manufacturer: Contractor's choice .
Product reference: _____.

365 MINERAL WOOL INTUMESCENT COATED RIGID BATTS

- Uncoated batts: To BS 3958-5.
- Manufacturer: Contractor's choice .
Product reference: _____.

370 PIPE COLLARS - CONCEALED INTUMESCENT

- Manufacturer: Contractor's choice .
Product reference: _____.

375 PIPE COLLARS - INSULATED WRAP

- Manufacturer: Contractor's choice .
Product reference: _____.

380 PIPE COLLARS - SURFACE MOUNTED INTUMESCENT

- Manufacturer: Contractor's choice .
Product reference: _____.

385 SEALANT BACKING MATERIAL

- Manufacturer: Contractor's choice .
Product reference: _____.

390 SEALANTS - FIRE RESISTING SILICONE

- Standard: To BS 5889 Type A.
- Manufacturer: Contractor's choice.
Product reference: _____.

395 SEALANTS - ONE-PART FIRE RESISTING ACRYLIC

- Manufacturer: Contractor's choice .
Product reference: _____.

410 SEALANTS - TWO-PART FIRE RESISTING POLYSULFIDE

- Manufacturer: Contractor's choice.
Product reference: _____.

EXECUTION

620 WORKMANSHIP GENERALLY

- Gaps: Seal gaps between building elements and services, to provide fire resistance and resist the passage of smoke.
- Adjacent surfaces: Prevent overrun of sealant or mortar on to finished surfaces.

640 INSTALLING BOARDING

- Position of boarding: Across face of opening.
- Framing: Provide framing to all edges of boarding.
- Bedding: Bed boarding on two-part fire resisting polysulfide.
- Double layers of board: Staggered butt joints.
 - Joints: Seal with vermiculite adhesive.
- Fixing: Plugged and screwed with 38 mm x No. 10 wood screws.

650 INSTALLING FIRE STOP LAMINATE

- Strip width: Wider than joint width.
- Fitting of strips: Compress strips and fit into joint, so that as they decompress the strips wedge themselves in the void.
- Shrink wrapping: Do not remove.
- Joints:
 - Ends of strips: Fit intumescent 'end piece' at the end of joints.
 - Joints in strips: Fit two intumescent 'end pieces' at each butt joint.

660 INSTALLING INTUMESCENT FOAM

- New joints: Remove builder's debris, mortar droppings, grease, and the like.
- Old joints: Clean and remove existing sealant from the joint.
- Priming: Lightly moisten substrate with water.
- Application: Fill joint to approximately half its depth, allowing foam to expand to face of joint.
- Trimming: Do not trim or cut the face of the cured foam.

670 APPLYING INTUMESCENT MORTAR

- Sequence: Install mortar after services are permanently installed.
- Loose dust and combustible materials: Remove from the opening.
- Shuttering: Install suitable shuttering panels to the faces of the opening.
- Temperature: Do not apply mortar when it could be damaged by frost.
- Powder:water ratio: 1 kg:1.2-1.3 L.
- Mortar cure: Do not disturb mortar before final set has taken place.
- Shuttering: Remove after mortar has cured.

680 INSTALLING INTUMESCENT PILLOWS

- Number of pillows (per m² of opening): Number necessary to achieve fire resistance.
- Orientation of bags: Lay parallel to plane of element containing opening.

690 APPLYING INTUMESCENT PUTTY

- Sequence: Install putty after services are permanently installed.
- Loose dust and combustible materials: Remove from the opening.

710 INSTALLING BATTS

- Installing batts: Fit tight into void between the floor or wall and the penetrating services.

- Face of batts: Flush with the surface of wall, floor or soffit.
- Joints: Closed butt joints..
- Gaps between services and bulkhead: Seal with fire resisting sealant.

730 FIXING PIPE COLLARS

- Collar fixing: Plugged and screwed with 38 mm x No. 10 wood screws.
- Gap around collar: Seal with intumescent foam.
- Length of wraps: Project 50 mm from each side of the element.

740 INSERTING SEALANT BACKING MATERIAL

- New joints: To full depth of joint leaving sufficient depth to apply sealant.
- Old joints: Not applicable.

750 APPLYING SEALANT TO JOINTS

- De-greasing: Cleaner by sealant manufacturer.
- Priming: Primer by sealant manufacturer.
- Depth of sealant: Check with manufacturer.
- Temperature: Do not apply water based sealants when they could be damaged by frost.

COMPLETION

910 CLEANING

- Masking tapes: Remove.
- Cleaning: Clean off splashes and droppings. Wipe down finishes.

920 INSPECTION

- Notice for inspection (minimum): 5 days.

P20 UNFRAMED ISOLATED TRIMS/ SKIRTINGS/ SUNDRY ITEMS

To be read with Preliminaries/ General conditions

110 SOFTWOOD SKIRTINGS GENERALLY

- Quality of wood and fixing: To BS 1186-3.
 - Species: Contractor's choice.
 - Class: 2.
- Moisture content at time of fixing: 9-13%.
- Preservative treatment: Not required.
- Fire rating: Not applicable.
- Profile: As drawing.
 - Finished size: As drawing.
- Finish as delivered: Sanded.
- Fixing: Plugged, screwed and pelleted at 450mm centres.



200 MEDIUM DENSITY FIBREBOARD WINDOW SILL BOARDS

- Manufacturer: Contractor's choice.
 - Product reference: _____.
- Standard: To BS EN 622-5.
 - Type: MDF.
 - Formaldehyde class: To BS EN 622-1, Class E1.
- Fire rating: To BS 476-7, Class 1.
- Edges: Quarter rounded.
- Thickness: 20 mm.
- Finish: Prepared and primed as section M60 .
- Support/ Fixing: Fix to softwood grounds with lost head nails at 600 mm centres..

510 INSTALLATION GENERALLY

- Joinery workmanship: As section Z10.
- Metal workmanship: As section Z11.
- Methods of fixing and fasteners: As section Z20 where not specified.
- Straight runs: To be in one piece, or in long lengths with as few joints as possible.
- Running joints: Location and method of forming to be agreed where not detailed.
- Joints at angles: Mitre unless shown otherwise.
- Position and level: To be agreed where not detailed.

P21 DOOR/ WINDOW IRONMONGERY

To be read with Preliminaries/ General conditions.

PRE-TENDER

10 QUANTITIES AND LOCATIONS

- Quantities and locations of ironmongery are given in the schedules .
- Fixing: As sections L10 and L20.

GENERALLY

121 IRONMONGERY FROM SINGLE PROPRIETARY RANGE

- Manufacturer: Submit proposals .
 - Product reference: _____ .
- Principal material/ finish: Satin stainless steel, grade 1.4301 (304) .
- Items unavailable within selected range: Submit proposals.

140 SAMPLES

- General: Before placing orders with suppliers submit labelled samples of the following: Hinges, closers, handles & locks .
 - Conformity: Retain samples on site for the duration of the contract. Ensure conformity of ironmongery as delivered with labelled samples.

141 SAMPLE BOARDS

- General: Before placing orders with suppliers submit a sample board, containing labelled samples of ironmongery and showing methods of fixing.
- Range: Include As Clause 140 .
 - Conformity: Retain board on site in an approved location for the duration of the contract. Ensure conformity of ironmongery as delivered with labelled samples

170 IRONMONGERY FOR FIRE DOORS

- Relevant products: Ironmongery fixed to, or morticed into, the component parts of a fire resisting door assembly.
- Compliance: Ironmongery included in successful tests to BS 476-22 or BS EN 1634-1 on door assemblies similar to those proposed.
 - Certification: Submit CERTIFIRE certificates or Submit evidence of successful testing by UKAS accredited laboratory .
- Melting point of components (except decorative non functional parts): 800°C minimum.

180 CATEGORY OF DUTY FOR DOOR IRONMONGERY

- Standard: To DD 171.
 - Category of duty of doors: Medium duty .
- General: Durability of ironmongery components to be compatible with stated category of duty of each door leaf.
 - Exclusions: Ironmongery with specific duty or 'category of use' defined elsewhere. Documentation: Before placing orders with suppliers submit documentation showing product compliance with stated category of duty.

DOOR HANGING DEVICES

320 DOOR HINGES GENERALLY

- Manufacturer: Submit proposals .
-Product reference: _____ .
- Type: As schedule .
- Size: As schedule .
- Material/ finish: Satin stainless steel, grade 1.4301 (304) .
- Other requirements: _____ .

WINDOW HANGING DEVICES

370 WINDOW HINGES GENERALLY

- Manufacturer: As clause 121 .
- Product reference: _____ .
- Type: As drawings/schedule .
- Size: As drawings/schedule .
- Material/ finish: Powder coated steel, colour as windows .

380 SLIDING FRICTION STAY HINGES GENERALLY

- Manufacturer: As clause 121 .
- Product reference: _____ .
- Type: As drawings/schedule .
- Size: As drawings/schedule .
- Material/ finish: Austenitic stainless steel .

385 PIVOT HINGES GENERALLY

- Manufacturer: As clause 121 .
- Product reference: _____ .
- Type: As drawings/schedule .
- Material/ finish: Extruded aluminium, powder coated, colour as windows .

DOOR OPERATING DEVICES

410 OVERHEAD DOOR CLOSERS GENERALLY

- Standard: To BS EN 1154.
 - Door closing devices to fire/ smoke control doors: CE marked.
 - Manufacturer: Submit proposals .
 - Product reference: _____ .
- Type: As schedule .
- Power size: As schedule .
- Other functions: As schedule .
- Casing finish: As schedule .
- Operational adjustment:
 - Variable power: Matched to size, weight and location of doors.
 - Latched doors: Override latches and/ or door seals when fitted.
 - Unlatched doors: Hold shut under normal working conditions.
 - Closing against smoke seals of fire doors: Positive. No gaps.

450 FLOOR SPRINGS GENERALLY

- Standard: To BS EN 1154.
 - Door closing devices to fire/ smoke control doors: CE marked.
- Manufacturer: Submit proposals .
 - Product reference: _____ .
- Power size: As schedule .
- Other functions: As schedule .
- Material/ finish: Satin stainless steel .
- Operational adjustment:
 - Variable power: Matched to size, weight and location of doors.
 - Latched doors: Override latches and/ or door seals when fitted.
 - Unlatched doors: Hold shut under normal working conditions.
 - Closing against smoke seals of fire doors: Positive. No gaps.



470 ELECTROMAGNETIC HOLD OPEN DEVICES (MAINS VOLTAGE) GENERALLY

- Manufacturer: Submit proposals .
 - Product reference: _____ .
- Type: As schedule .
- Material/ finish: Satin stainless steel .
- Means of release: Alarm system and/ or failure of power supply.
- Test switch: Located in a convenient position adjacent to door.

471 ELECTROMAGNETIC HOLD OPEN/ SWING-FREE DEVICES (24 V) GENERALLY

- Standard: To BS EN 1155.
 - Electromagnetic devices to fire/ smoke control doors: CE marked.
- Manufacturer: Submit proposals .
 - Product reference: _____ .
- Type: As schedule .
- Material/ finish: Satin stainless steel .
- Means of release: Alarm system and/ or failure of power supply.
- Test switch: Located in a convenient position adjacent to door. Operational adjustment of integral closer:
 - Variable power: Matched to size, weight and location of doors.
 - Latched doors: Override latches and/ or door seals when fitted.
 - Unlatched doors: Hold shut under normal working conditions.

481 DOOR CO-ORDINATORS GENERALLY

- Standard: To BS EN 1158.
 - Door co-ordinators to fire/ smoke control doors: CE marked.
- Manufacturer: Submit proposals .
 - Product reference: _____ . Material/ finish: Satin stainless steel .
- Application: To all single swing double doors with rebated meeting stiles and fitted with self closers.

490 UNCONTROLLED DOOR CLOSERS GENERALLY

- Manufacturer: Submit proposals .
 - Product reference: _____ .
- Type: As schedule .

- Material/ finish: As schedule .
- Action: As schedule .
- Operation:
 - Power: To suit the size and weight of doors to which they are fitted.
 - Unlatched doors: Hold closed under normal conditions.

DOOR SECURING DEVICES

515 DOOR LOCKS _____

- Standard: To BS EN 12209.
- Manufacturer: Submit proposals .
 - Product reference: _____ .
- Type: As schedule .
- Backset: As schedule .
- Material/ finish: As schedule .
- Keying: As schedule .

530 SPECIAL FUNCTION DOOR LOCKS _____

- Manufacturer: Submit proposals.
 - Product reference: _____.
- Type: As schedule.
- Backset: As schedule.
- Material/ finish: As schedule.
- Keying: As schedule.

540 DOOR LATCHES GENERALLY

- Standard: To BS EN 12209.
- Manufacturer: Submit proposals .
 - Product reference: _____ .
- Type: As schedule .
- Backset: As schedule .
- Material/ finish: As schedule .
- Latch spring strength: Select to prevent unsprung lever handles drooping.

571 EMERGENCY EXIT DEVICES GENERALLY

- Standard: To BS EN 179.
 - Emergency exit devices for locked doors on escape routes: CE marked.
- Manufacturer: As clause 121 .
 - Product reference: _____ .
- Type: As schedule .
- Material/ finish: Satin stainless steel .
- Additional requirements: As schedule .

577 PANIC EXIT DEVICES _____

- Standard: To BS EN 1125.
 - Panic exit devices for locked doors on escape routes: CE marked.
- Manufacturer: Submit proposals .
 - Product reference: _____ .

- Type: As schedule .
- Material/ finish: Satin stainless steel .
- Additional requirements: As schedule .

582 DOOR BOLTS GENERALLY

- Standard: To BS EN 12051.
- Manufacturer: Submit proposals .
 - Product reference: _____ .
- Type: As schedule .
- Size: As schedule .
- Material/ finish: Satin stainless steel, grade 1.4301 (304) .
- Additional requirements: None .

586 PRIVACY INDICATOR BOLTS _____

- Manufacturer: As clause 121 .
 - Product reference: _____ .
- Type: As schedule .
- Material/ finish: Satin stainless steel, grade 1.4401 (316) .
- Emergency release facility: Required.

WINDOW SECURING DEVICES

590 WINDOW ESPAGNOLETTE BOLTS GENERALLY

- Manufacturer: Submit proposals .
 - Product reference: _____ .
- Type: As drawings/schedule .
- Size: As drawings/schedule .
- Backset: As drawings/schedule .
- Material/ finish: Electro-galvanized steel .
- Additional requirements: Window lever handles as clause 910 .

DOOR FURNITURE

610 LEVER HANDLES GENERALLY

- Standard: To BS EN 1906.
- Manufacturer: As clause 121 .
 - Product reference: _____ .
- Style: As schedule .
- Size: As schedule .
- Material/ finish: Satin stainless steel, grade 1.4401 (316) As schedule .
- Mounting: As schedule .
- Additional requirements: None .

620 DOOR KNOBS GENERALLY

- Standard: To BS EN 1906.
- Manufacturer: As clause 121 .
 - Product reference: _____ .
- Style: As schedule .

- Size: As schedule .
- Material/ finish: Satin stainless steel, grade 1.4301 (304) .
- Mounting: As schedule .
- . Additional requirements: None .

641 PULL HANDLES GENERALLY

- Standard: To BS 8424.
- Manufacturer: As clause 121 .
 - Product reference: _____ .
 - Shape: As schedule .
- Diameter: As schedule .
- Distance between centres: As schedule .
- Material/ finish: _____ .
- Mounting: _____ .
- Additional requirements: _____ .

670 PUSH PLATES GENERALLY

- Manufacturer: Submit proposals .
 - Product reference: _____ .
- Size: As schedule .
- Material/ finish: Satin stainless steel, grade 1.4401 (316) .
- Mounting: As schedule .
- Additional requirements: Rounded edges and radiused corners .

680 MIDRAIL PLATES GENERALLY

- Manufacturer: As clause 121 .
 - Product reference: _____ .
- Size: As schedule .
- Material/ finish: Satin stainless steel, grade 1.4401 (316) .
- Mounting: As schedule .
- Additional requirements: Rounded edges and radiused corners .

690 KICK PLATES GENERALLY

- Manufacturer: Submit proposals .
 - Product reference: _____ .
- Material/ finish: Satin stainless steel, grade 1.4401 (316) .
- Mounting: As schedule .
- Additional requirements: Rounded edges and radiused corners .

710 ESCUTCHEONS GENERALLY

- Manufacturer: As clause 121 .
 - Product reference: _____ .
- Material/ finish: Satin stainless steel, grade 1.4401 (316) .
- Keyhole type: As schedule .
- Usage: As schedule .

720 DOOR STOPS

- Manufacturer: Submit proposals .

- Product reference: _____ .
- Type: As schedule .
- Usage: As schedule .

750 LIMITING DOOR STAYS GENERALLY

- Manufacturer: As clause 121 .
 - Product reference: _____ .
- Type: As schedule .
- Size: To suit door width .
- Material/ finish: Satin stainless steel, grade 1.4401 (316) .
- Features: _____ .



760 DOOR HOLDERS GENERALLY

- Manufacturer: Submit proposals .
 - Product reference: _____ .
- Type: As schedule .
- Size: As schedule .
- Material/ finish: Satin stainless steel, grade 1.4301 (304) .

895 DOOR MOUNTED AIR TRANSFER GRILLES GENERALLY

- Manufacturer: Submit proposals .
 - Product reference: _____ .
- Type: As schedule .
- Size: As schedule .
- Material/ finish: As schedule .

896 DOOR MOUNTED FIRE RESISTING AIR TRANSFER GRILLES

- Manufacturer: Submit proposals .
 - Product reference: _____ .
- Type: As schedule .
- Size: As schedule .
- Fire resistance: As schedule .
- Material/ finish: As schedule .

897 DOOR MOUNTED FIRE RESISTING AIR TRANSFER GRILLES WITH SMOKE SHUTTER GENERALLY .

- Manufacturer: Submit proposals .
 - Product reference: _____ .
- Type: As schedule .
- Size: As schedule .
- Material/ finish: As schedule .
- Activator: Smoke detection and/ or fire alarm system.

WINDOW FURNITURE

900 CASEMENT HANDLES GENERALLY

- Manufacturer: Submit proposals .
 - Product reference: _____ .

- Type: As drawings/schedule .
- Material/ finish: Powder coated steel, colour as windows .
- Features: None .

905 CASEMENT STAYS GENERALLY

- Manufacturer: Submit proposals .
 - Product reference: _____ .
- Type: As drawings/schedule .
- Size: As drawings/schedule .
- Material/ finish: Powder coated steel, colour as windows .
- Features: None .



910 WINDOW LEVER HANDLES GENERALLY

- Manufacturer: Submit proposals .
 - Product reference: _____ .
- Type: As drawings/schedules .
- Material/ finish: Powder coated steel, colour as windows .
- Features: None .

925 OVER CENTRE CASEMENT STAYS GENERALLY

- Manufacturer: Submit proposals .
 - Product reference: _____ .
- Type: As drawings/schedules .
- Size: As drawings/schedules .
- Material/ finish: Powder coated steel, colour as windows .
- Features: None .

930 FRICTION RESTRICTOR CASEMENT STAYS GENERALLY

- Manufacturer: Submit proposals .
 - Product reference: _____ .
- Type: As drawings/schedules .
- Size: As drawings/schedules .
- Material/ finish: Electroplated steel .
- Features: None .

935 REMOTE WINDOW OPENERS GENERALLY

- Manufacturer: Submit proposals .
 - Product reference: _____ .
- Type: Manufacturer's standard .
- Material/ finish: Manufacturer's standard .
- Features: None .

960 WEATHERSTRIP TO WINDOWS _____

- Manufacturer: Submit proposals .
 - Product reference: _____ .
- Type: Elastomeric compression strip in metal carrier .
- Size: As manufacturer's recommendations .
- Material/ finish: Colour as window furniture .

R10 RAINWATER DRAINAGE SYSTEMS

To be read with Preliminaries/ General conditions.

GENERAL

110 GRAVITY RAINWATER DRAINAGE SYSTEM

- Rainwater outlets: None.
- Gutters: Aluminium.
- Pipework: Aluminium.
- Below ground drainage: As section R12.
- Disposal: To soakaway as section R12.
- Controls: Not applicable.
- Accessories: Sealant for gutters.

121A SIPHONIC RAINWATER DRAINAGE SYSTEM

- Manufacturer / System designer: Fullflow Group Ltd. Fullflow House, Holbrook Avenue, Holbrook, Sheffield S20 3FF. Tel: 0114 247 3655. Fax: 0114 247 7805. Email: info@uk.fullflow.com Web: www.fullflow.com
- Product reference: Fullflow Self Priming Syphonic Roof Drainage System
- System design: To operate on syphonic principles so as to be regularly self cleansing in normal use.
- Pipework: Airtight at maximum and minimum operating pressures. To accommodate thermal movement using flexible joints or other suitable means without damage to fixings and joints or excess stress, abrasion or noise.
- Roof/ gutter construction and finish: Polyester powder coated aluminium as roof cladding.
- Design rate of rainfall: As BS EN 12056-3, National Annex NB.2, Category 1, 2, 3 or 4.
- Design life of building: 25 years.
- Available capacity of below ground drainage (maximum): TBC litres/ second.
- Design water depth after design rainfall of: mm Insert requirements minutes duration (maximum): TBC.
- Overflow arrangements: Wier overflows in ends of gutters. Height to be above the maximum depth of water produced by the syphonic outlets at design intensity, or Secondary Systems (in Valley Gutters) or Secondary Spigots (in Eaves Gutters)
- Siphonic flow velocity (minimum): 1.0 m/second Recommended.
- Pressure imbalance (maximum): 8.0 m/second Recommended.
- Operational pressure (maximum): Consult Fullflow Group Ltd .
- Rainwater pipework: High density polyethylene as clause 415. Pipework pressures must be designed within the range of +5.00 to -0.80 bar gauge pressure.
- Other requirements: Consult Fullflow Group Ltd .

SYSTEM PERFORMANCE

210 DESIGN

- Design: Complete the design of the rainwater drainage system.
- Standard: To BS EN 12056-3, clauses 3-7 and National Annexes.
- Proposals: Submit drawings, technical information, calculations and manufacturers' literature.

221 COLLECTION AND DISTRIBUTION OF RAINWATER

- General: Complete, and without leakage or noise nuisance.

PRODUCTS

311 ALUMINIUM GUTTERS

- Standard: To BS 2997.
- Manufacturer: Submit proposals.
 - Product reference: _____.
- Profile: As drawings.
- Type/ Thickness: Minimum 0.9 mm thickness.
- Nominal size: As drawings.
- Finish: Polyester powder coating.
- Colour: As roof cladding.
- Brackets: Strap type.
 - Fixings: Bolted to galvanized steel support angle. Size: _____.
- Accessories: Spacer rail & ST clips.

360 SEALANT FOR GUTTERS

- Type: Low modulus silicone sealant.

365 PROPRIETARY RAINWATER OUTLETS

- Manufacturer: Fullflow Group Ltd. Fullflow House, Holbrook Avenue, Holbrook, Sheffield S20 3FF. Tel: 0114 247 3655. Fax: 0114 247 7805. Email: info@uk.fullflow.com Web: www.fullflow.com.
 - Product reference: Primaflow Self Priming Patented Outlets
 - Type: Syphonic.
- Roof construction: Refer to drawings.
 - Roof insulation thickness: Refer to drawings.
- Type of grate/ Fittings: Manufacturers recommendation.
- Outlet: Type and direction to suit pipework with suitable adaptors and connections.
- Accessories: _____.

370 ALUMINIUM PIPEWORK

- Standard: To BS 2997.
- Manufacturer: Submit proposals.
 - Product reference: _____.
- Type/ Thickness : Minimum 0.7 mm.
- Section: Round.
- Nominal size: 102 mm.
- Finish Polyester powder coating.
- Colour: TBC.
- Brackets: Extruded aluminium pipe clips coated as pipes.
 - Fixings: Stainless steel screws. Size: _____.
- Accessories: Access fittings.

415A HPPE PIPEWORK FOR INTERNAL USE

- Standard: To BS EN 1519-1 or Agreement certified.

- Manufacturer: Fullflow Group Ltd. Fullflow House, Holbrook Avenue, Holbrook, Sheffield S20 3FF. Tel: 0114 247 3655. Fax: 0114 247 7805. Email: info@uk.fullflow.com Web: www.fullflow.com
- Product reference: Fullflow HPPE pipework
- Nominal size: Consult Fullflow Group Ltd. HPPE pipework suitable for use with diameters of 200mm and above. Brackets: Consult Fullflow Group Ltd. Jointing: Butt fusion at fabrication stage in controlled, factory conditions, additionally butt fusion and electro-fusion sleeves to be utilised for on site jointing where applicable. Hot air welding of polyethylene pipework is not allowed on a Fullflow Rainwater Management system. All joints must be either capable of resisting longitudinal forces without sliding apart or restrained to prevent relative movement. Accessories: Consult Fullflow Group Ltd.

450 INSULATION TO INTERNAL PIPELINES

- Material: Preformed flexible mineral fibre split tube.
- Thermal conductivity (maximum): 0.045 W/m•K.
- Manufacturer: Submit proposals.
 - Product reference: _____.
- Thickness: 25 mm.
- Fire performance: Class 1 spread of flame when tested to BS 476-7 or Class C to BS EN 13501.

455 MASKING PLATES

- Manufacturer: Submit proposals.
 - Product reference: _____.
- Material and finish: _____.
- Fixing: _____.

EXECUTION

600 PREPARATION

- Work to be completed before commencing work specified in this section:
 - Below ground drainage. Alternatively, make temporary arrangements for dispersal of rainwater without damage or disfigurement of the building fabric and surroundings.
 - Painting of surfaces which will be concealed or inaccessible.

605A INSTALLATION GENERALLY

- Discharge of rainwater: Complete, and without leakage or noise nuisance. All Components must be procured within an ISO 9001 accredited approved suppliers procedure.
- Electrolytic corrosion: Avoid contact between dissimilar metals where corrosion may occur.
- Plastics and galvanized steel pipes: In accordance with current and relevant British and European standards.
- Protection: Fit purpose made temporary caps to prevent ingress of debris. Fit access covers, cleaning eyes and blanking plates as the work proceeds.
- Fixings/ Fasteners: As section R10/610

610 FIXING AND JOINTING GUTTERS

- Joints: Watertight.
- Brackets: Securely fixed.

- Fixings: Supported off straps. Fixing centres: 600 mm.
- Additional brackets: Where necessary to maintain support and stability, provide at joints in gutters and near angles and outlets.
- . Roofing underlay: Dressed into gutter.

615 SETTING OUT EAVES GUTTERS - TO FALLS

- Setting out: To true line and even gradient to prevent ponding or backfall. Position high points of gutters as close as practical to the roof and low points not more than 50 mm below the roof.
- Outlets: Align with connections to below ground drainage.

616 SETTING OUT EAVES GUTTERS - LEVEL

- Setting out: Level and as close as practical to the roof.
- Outlets: Aligned with connections to below ground drainage.

630A INSTALLING RAINWATER OUTLETS

- Fixing: Outlets to be issued to nominated gutter manufacturer/roofing contractor for installation into roof/gutter via approved fixing method. Outlets can be fixed to gutters by hot air welding, mechanical fixing or metallic welding. Outlets can be clamp ring, PVC Coated or standard flanged. Junctions between outlets and pipework: Where necessary, to accommodate movement in structure and pipework.

635 FIXING PIPEWORK

- Pipework: Fix securely, plumb and/ or true to line.
- Branches and low gradient sections: Fix with uniform and adequate falls to drain efficiently.
- Externally socketed pipes and fittings: Fix with sockets facing upstream.
- Additional supports: Provide as necessary to support junctions and changes in direction.
- Vertical pipes:
 - Provide a loadbearing support at least at every storey level.
 - Tighten fixings as work proceeds so that every storey is self supporting.
 - Wedge joints in unsealed metal pipes to prevent rattling.
- Wall and floor penetrations: Isolate pipework from structure.
 - Pipe sleeves: As section P31.
 - Masking plates: Fix at penetrations if visible in the finished work.
- Expansion joint pipe sockets: Fix rigidly to buildings. Elsewhere, provide brackets and fixings that allow pipes to slide.

650 JOINTING PIPEWORK AND GUTTERS

- General: Joint with materials and fittings that will make effective and durable connections.
- Jointing differing pipework and gutter systems: Use adaptors intended for the purpose.
- Cut ends of pipes and gutters: Clean and square. Remove burrs and swarf. Chamfer pipe ends before inserting into ring seal sockets.
- Jointing or mating surfaces: Clean and, where necessary, lubricate immediately before assembly.
- Junctions: Form with fittings intended for the purpose.
- Jointing material: Strike off flush. Do not allow it to project into bore of pipes and fittings.
- Surplus flux, solvent jointing materials and cement: Remove.



670A INSTALLING SIPHONIC DRAINAGE PIPEWORK

- Pipework support: Fullflow Suspension Rail System comprising a steel channel section to BS 2994 securely anchored to the main building below the level of the roof with support from the secondary structure, as detailed in the accompanying drawing. Pipes and fittings shall be securely suspended from the rail by means of purpose made clamps, zinc plated to BS 3382 and fixed to the pipe at fabrication stage in controlled, factory conditions with impact torque wrenches. All in accordance with enclosed detailed drawings SD079 & SD080.
- Pipe clamp location: Fixed at each side of branch connections and at not more than ten times the nominal pipe diameter on the horizontal collector pipe. Vertical pipework shall be rigidly fixed with pipe clamps to vertical structural steelwork or suitable faces at not more than ten times the nominal pipe diameter. When designing the building structural frame, due note should be taken of pipe loadings as detailed in the enclosed standard drawings.
- Installation: The Fullflow roof drainage system shall be installed only by trained and qualified operatives or an approved installation company. Installation companies will:- Install pipework system to ensure the complete discharge of rainwater from the building in compliance with design and performance requirements and without leaking. Obtain all components for all pipework systems and rainwater outlets from Fullflow Group Limited. Shall be supplied within an ISO 9001 environment. Where not specified plated, sherardized, galvanised or non-ferrous fastenings must be used, these must be fit for purpose, and compatible with the material being fixed or fixed to. Before installing ensure that any specified painting of surfaces which will be concealed or otherwise inaccessible is completed. Internal Pipework Test will be carried out as clause 560.

675 CUTTING COATED PIPEWORK AND GUTTERS

- Cutting: Recoat bare metal.

680 FIXING INSULATION TO INTERNAL PIPELINES AND GUTTERS

- Fixing: Secure and neat. Provide continuity at supports and leave no gaps. Fix split pipe insulation with the split on 'blind' side of pipeline.
 - Method: Contractor's choice.
- Timing: Do not fit insulation until completion of pipe airtightness or leakage testing.

685 IDENTIFICATION OF INTERNAL RAINWATER PIPEWORK

- Markings: To BS 1710.
 - Type: Black, with arrows to indicate direction of flow.
- Wording: White lettering 'RAINWATER DRAINAGE' on a black background.
- Type: Permanent; integral or painted pipe colour, self-adhesive bands or identification clips.
- Locations: Junctions, both sides of slabs, bulkheads and wall penetrations.

690 ELECTRICAL CONTINUITY - PIPEWORK

- Joints in metal pipes with flexible couplings: Clips (or suitable standard pipe couplings) supplied for earth bonding by pipework manufacturer to ensure electrical continuity.

695 ELECTRICAL CONTINUITY - GUTTERS

- Joints in metal gutters: Purpose made links supplied by the gutter manufacturer to ensure electrical continuity.

700 ACCESS FOR TESTING AND MAINTENANCE

- General: Install pipework and gutters with adequate clearance to permit testing, cleaning and maintenance, including painting where necessary.
- Access fittings and rodding eyes: Position so that they are not obstructed.

COMPLETION

900 TESTING GENERALLY

- Dates for testing: Give notice.
 - Period of notice (minimum): 7 days.
- Preparation:
 - Pipework: Complete, securely fixed, free from defects, obstruction and debris before testing.
- Testing:
 - Supply clean water, assistance and apparatus.
 - Do not use smoke to trace leaks.
- . Records: Submit a record of tests.

910 GUTTER TEST

- Preparation: Temporarily block all outlets.
- Testing: Fill gutters to overflow level and after 5 minutes closely inspect for leakage.

915 MAINTENANCE INSTRUCTIONS

- General: At completion, submit printed instructions recommending procedures for maintenance of the rainwater installation, including full details of recommended inspection, cleaning and repair procedures.

920 IMMEDIATELY BEFORE HANDOVER

- Construction rubbish, debris, swarf, temporary caps and fine dust which may enter the rainwater system: Remove. Do not sweep or flush into the rainwater system.
- Access covers, rodding eyes, outlet gratings and the like: Secure complete with fixings.

Z10 PURPOSE MADE JOINERY

To be read with Preliminaries/ General conditions.

110 FABRICATION

- Standard: To BS 1186-2.
- Sections: Accurate in profile and length, and free from twist and bowing. Formed out of solid unless shown otherwise.
 - Machined surfaces: Smooth and free from tearing, wooliness, chip bruising and other machining defects.
- Joints: Tight and close fitting.
- Assembled components: Rigid. Free from distortion.
- Screws: Provide pilot holes.
 - Screws of 8 gauge (4 mm diameter) or more and screws into hardwood: Provide clearance holes.
 - Countersink screws: Heads sunk at least 2 mm below surfaces visible in completed work.
- Adhesives: Compatible with wood preservatives applied and end uses of timber.

120 CROSS SECTION DIMENSIONS OF TIMBER

- General: Dimensions on drawings are finished sizes.
- Maximum permitted deviations from finished sizes:
 - Softwood sections: To BS EN 1313-1:-Clause 6 for sawn sections. Clause NA.2 for further processed sections.
 - Hardwood sections: To BS EN 1313-2:-Clause 6 for sawn sections. Clause NA.3 for further processed sections.

130 PRESERVATIVE TREATED WOOD

- Cutting and machining: Completed as far as possible before treatment.
- Extensively processed timber: Retreat timber sawn lengthways, thickened, planed, ploughed, etc.
- Surfaces exposed by minor cutting and/ or drilling: Treat with two flood coats of a solution recommended by main treatment solution manufacturer.

140 MOISTURE CONTENT

- Wood and wood based products: Maintained within range specified for the component during manufacture and storage.

210 LAMINATED PLASTICS VENEERED BOARDS/ PANELS

- Fabrication: To British Laminated Plastics Fabricators Association Ltd (BLF) fabricating standards.
- Balancing veneer: From decorative veneer manufacturer and of similar composition. Applied to reverse side of core.
- Finished components: Free from defects, including bow, twist, scratches, chipping, cracks, pimpling, indentations, glue marks, staining and variations in colour and pattern.
- Joints visible in completed work: Tight butted, true and flush.

220 WOOD VENEERED BOARDS/ PANELS

- Core material and veneers: Conditioned before bonding.
- Setting out: Veneer features and grain pattern aligned regularly and symmetrically unless instructed otherwise.
- Balancing veneer: Applied to reverse side of core material.

- Moisture and temperature movement characteristics: As facing veneer.
- Veneer edges: Tight butted and flush, with no gaps.
- Tolerance of veneer thickness (maximum): ± 0.5 mm.
- Finished components: Free from defects, including bow, twist, scratches, chipping, splits, blebs, indentations, glue marks and staining.
- Surface finish: Fine, smooth, free from sanding marks.

250 FINISHING

- Surfaces: Smooth, even and suitable to receive finishes.
 - Arrises: Eased unless shown otherwise on drawings.
- End grain in external components: Sealed with primer or sealer as section M60 and allowed to dry before assembly.



Z11 PURPOSE MADE METALWORK

To be read with Preliminaries/ General conditions.

310 METAL PRODUCTS

- Standards: Generally, as specified in the following clauses.
- Fasteners: Generally, same metal as component, with matching coating and finish.

320 STEEL LONG AND FLAT PRODUCTS

- Hot rolled structural steels (excluding structural hollow sections and tubes): To BS EN 10025-1.
- Fine grain steels, including special steels: To BS EN 10025-3 and -4.
- Improved atmospheric corrosion resistance: To BS EN 10025-5.

330 STEEL PLATE, SHEET AND STRIP

- Plates and wide flats, high yield strength steel: To BS EN 10025-6.

340 HOT ROLLED STEEL PLATE, SHEET AND STRIP

- Flat products, high yield strength for cold forming: To BS EN 10149-1, -2 and -3.
- Low carbon steel sheet and strip for cold forming: To BS EN 10111.
- Narrow strip, formable and general engineering purposes: To BS 1449-1.8 and BS 1449-1.14.

350 COLD ROLLED STEEL PLATE, SHEET AND STRIP

- Steel sections: To BS EN 10162.
- Flat products, high yield strength micro-alloyed steels for cold forming: To BS EN 10268.
- Low carbon steel flat products for cold forming: To BS EN 10130 and BS EN 10131.
- Uncoated mild steel narrow strip for cold forming: To BS EN 10139 and BS EN 10140.
- Narrow strip, general engineering purposes: To BS EN 10132-1, -2, and -3.
- Low carbon steel flat products for vitreous enamelling: To BS EN 10209.

360 STEEL COATED FLAT PRODUCTS

- Hot dip zinc coated low carbon steel sheet and strip for cold forming: To BS EN 10327 and BS EN 10143.
- Hot dip zinc coated structural steel sheet and strip: To BS EN 10143 and BS EN 10326.
- Hot dip zinc-aluminium (za) coated sheet and strip: To BS EN 10326 and 10327.
- Hot dip aluminium-zinc (az) coated sheet and strip: To BS EN 10327.
- Organic coated flat products: To BS EN 10169-1.

370 STEEL STRUCTURAL HOLLOW SECTIONS (SHS)

- Non alloy and fine grain steels, hot finished: To BS EN 10210-1 and -2.
- Non-alloy and fine grain steels, cold formed welded: To BS EN 10219-2.
- Weather resistant steels, hot finished: To BS 7668.

380 OTHER STEEL SECTIONS

- Equal flange tees: To BS EN 10055.
- Equal and unequal angles: To BS EN 10056-1 and -2.
- Wire, mild steel for general engineering purposes: To BS 1052.
- Wire and wire products, general: To BS EN 10218-2.
- Tubes:

- Cold finished electric resistance welded: To BS 6323-6.
- Cold finished seamless: To BS 6323-4.
- Seamless and welded generally: To BS 6323-1.
- Seamless circular: To BS EN 10297-1.
- Seamless cold drawn: To BS EN 10305-1.
- Welded and cold sized square and rectangular: To BS EN 10305-5.
- Welded circular: To BS EN 10296-1.
- Welded cold drawn: To BS EN 10305-2.
- Welded cold sized: To BS EN 10305-3.

400 STAINLESS STEEL PRODUCTS

- Chemical composition and physical properties: To BS EN 10088-1.
- Sheet, strip and plate: To BS EN 10088-2.
- Semi-finished products bars, rods and sections: To BS EN 10088-3.
- Wire: To BS 1554.
- Tubes, welded: To BS 6323-8.

410 ALUMINIUM ALLOY PRODUCTS

- Designations:
 - Designation system, chemical composition and forms: To BS EN 573-1 to -4.
 - Temper designations: To BS EN 515.
- Sheet, strip and plate: To BS EN 485-1 to -4.
- Cold drawn rods, bars and tubes: To BS EN 754-1 and -2.
- Extruded rods, bars, tubes and profiles: To BS EN 755-1 and -2.
- Drawn wire: To BS EN 1301-1, -2 and -3.
- Rivet, bolt and screw stock: To BS 1473.
- Structural sections: To BS 1161.

420 COPPER ALLOY PRODUCTS

- Sheet, strip, plate and circles for general purposes: To BS EN 1652.
- Sheet and strip for building purposes: To BS EN 1172.
- Rods: To BS EN 12163.
- Profiles and rectangular bars: To BS EN 12167.
- Wire: To BS EN 12166.
- Tubes: To BS EN 12449.

510 PREPARATION FOR APPLICATION OF COATINGS

- General: Complete fabrication, and drill fixing holes before applying coatings.
- Paint, grease, flux, rust, burrs and sharp arrises: Remove.

515 FABRICATION GENERALLY

- Contact between dissimilar metals in components: Avoid.
- Finished components: Rigid and free from distortion, cracks, burrs and sharp arrises.
 - Moving parts: Free moving without binding.
- Corner junctions of identical sections: Mitre.
- Prefinished metals: Do not damage or alter appearance of finish.

520 COLD FORMED WORK

- Profiles: Accurate, with straight arrises.

525 ADHESIVE BONDING

- Preparation of surfaces of metals to receive adhesives:
 - Degrease.
 - Abrade mechanically or chemically etch.
 - Prime: To suit adhesive.
- Adhesive bond: Form under pressure.

530 THERMAL CUTTING OF STAINLESS STEEL

- Carbonation in the heat affected zone: Remove, after cutting.

535 WELDING/ BRAZING GENERALLY

- Surfaces to be joined: Clean thoroughly.
- Tack welds: Use only for temporary attachment.
- Joints: Fully bond parent and filler metal throughout with no inclusions, holes, porosity or cracks.
- Surfaces of materials that will be self-finished and visible in completed work: Protect from weld spatter.
- Flux residue, slag and weld spatter: Remove.

540 WELDING OF STEEL

- Method: Metal arc welding to BS EN 1011-1 and -2.

545 WELDING OF STAINLESS STEEL

- Method: TIG welding to BS EN 1011-3.
- Butt welds: Double bevel.

550 WELDING OF ALUMINIUM ALLOYS

- Method: TIG or MIG welding to BS EN 1011-4.

555 BRAZING

- Standard: To BS EN 14324.

560 BRONZE WELDING

- Standard: To BS 1724.

565 FINISHING WELDED AND BRAZED JOINTS VISIBLE IN COMPLETE WORK

- Butt joints: Smooth, and flush with adjacent surfaces.
- Fillet joints: Neat.
- Grinding: Grind smooth where indicated on drawings.

570 LIQUID ORGANIC COATING FOR ALUMINIUM ALLOY COMPONENTS

- Standard: To BS 4842.

575 ZINC AND CADMIUM PLATING OF IRON AND STEEL SURFACES

- Zinc plating: To BS EN 12329.
- Cadmium plating: To BS EN 12330.

- Standard: To BS EN 12540.

580 CHROMIUM PLATING 585 GALVANIZING

- Standard: To BS EN ISO 1461.
- Vent and drain holes:
 - Location: Submit proposals..
 - Sealing after galvanizing: Required. Submit proposals.

590 VITREOUS ENAMELLING

- Standard: To BS 3830.



Z12 PRESERVATIVE/ FIRE RETARDANT TREATMENT

To be read with Preliminaries/ General conditions.

110 TREATMENT APPLICATION

- Timing: After cutting and machining timber, and before assembling components.
- Processor: Licensed by manufacturer of specified treatment solution.
- Certification: For each batch of timber provide a certificate of assurance that treatment has been carried out as specified.

120 COMMODITY SPECIFICATIONS

- Standard: Current edition of the British Wood Preserving and Damp-proofing Association (BWPDA) Manual.

130 PRESERVATIVE TREATMENT SOLUTION STRENGTHS/ TREATMENT CYCLES

- General: Select to achieve specified service life and to suit treatability of specified wood species.

140 COPPER-ORGANIC PRESERVATIVE TREATMENT

- Solution:
 - Manufacturer: Contractor's choice. Product reference: _____.
- Application: High pressure impregnation.
- Moisture content of wood at time of treatment: Not more than 28%. After treatment, allow timber to dry before using.

160 ORGANIC SOLVENT PRESERVATIVE TREATMENT

- Solution:
 - Manufacturer: Contractor's choice . Product reference: _____ .
 - Application: Double vacuum + low pressure impregnation, or immersion.
- Moisture content of wood at time of treatment: As specified for the timber/ component at time of fixing. After treatment, timber to be surface dry before use.

165 WATER BASED MICROEMULSION PRESERVATIVE TREATMENT

- Solution:
 - Manufacturer: Contractor's choice . Product reference: _____ .
 - Application: Double vacuum + low pressure impregnation.
- Moisture content of wood at time of treatment: As specified for the timber/ component at time of fixing. After treatment, timber to be surface dry before use.

167 BORON COMPOUND PRESERVATIVE TREATMENT

- Solution:
 - Manufacturer: Contractor's choice. Product manager: _____.
- Application: High pressure impregnation.
- Moisture content of wood at time of treatment: Not more than 28%. After treatment allow timber to dry before using.

210 FIRE RETARDANT TREATMENT

- Solution type: Interior.
 - Manufacturer: Contractor's choice. Product reference: _____.
 - Application: Vacuum + pressure impregnation.

- Moisture content of wood at time of treatment: As specified for the timber/ component at time of fixing. After treatment, timber to be redried slowly at temperatures not exceeding 65°C to minimize distortion and degradation.

220 LEACH RESISTANT FIRE RETARDANT TREATMENT

- Solution:
 - Manufacturer: Contractor's choice. Product reference: _____.
 - Application: Vacuum + pressure impregnation.
 - Moisture content of wood at time of treatment: As specified for the timber/ component at time of fixing.



Z20 FIXINGS/ ADHESIVES

To be read with Preliminaries/ General conditions.

110 FIXINGS GENERALLY

- Integrity of supported components: Types, sizes and quantities of fasteners/ packings and spacings of fixings selected to retain supported components without distortion or loss of support.
- Components/ substrates/ fasteners of dissimilar metals: Fixed with isolating washers/ sleeves to avoid bimetallic corrosion.
- General usage: To recommendations of fastener manufacturers and/ or manufacturers of components, products or materials fixed and fixed to.
- Appearance: As approved samples.

130 FASTENER DURABILITY

- Fasteners in external construction:
 - Fasteners not directly exposed to weather: Of corrosion resistant material or with a corrosion resistant finish.
 - Fasteners directly exposed to weather: Of corrosion resistant material.

140 FIXINGS THROUGH FINISHES

- Penetration of fasteners/ plugs into substrate: To achieve a secure fixing.

150 PACKINGS

- Function: To take up tolerances and prevent distortion of materials/ components.
- Materials: Noncompressible, noncorrodible, rot proof.
- Locations: Not within zones to be filled with sealant.

170 NAILED TIMBER FIXINGS

- Nails:
 - Steel: To BS 1202-1 or BS EN 10230-1
 - Copper: To BS EN 1202-2.
 - Aluminium: To BS 1202-3.
- Penetration: Fully driven in without splitting or crushing timber.
- Surfaces visible in completed work: Nail heads punched below surfaces.
- Nailed timber joints: Two nails per joint (minimum), opposed skew driven.

180 FIXINGS TO MASONRY

- Fasteners:
 - Light duty: Plugs and screws.
 - Heavy duty: Expansion anchors or chemical anchors.

210 PLUGS

- Type: Proprietary types suited to background, loads to be supported and conditions expected in use.

220 SCREW FIXINGS

- Screws: To BS 1210.
- Finished level of countersunk screw heads:
 - Exposed: Flush with timber surface.
 - Concealed (holes filled/ stopped): Sunk minimum 2 mm below surface.

- Washers and screw cups: Of same material as screw.

230 PELLETED COUNTERSUNK FIXINGS

- Finished level of countersunk screw heads: Minimum 6 mm below timber surface.
- Cut from matching timber, grain matched and glued in to full depth of hole.
- Finished level of pellets: Flush with surface.

240 PLUGGED COUNTERSUNK FIXINGS

- Finished level of countersunk screw heads: Minimum 6 mm below timber surface.
- Plugs: Glued in to full depth of hole.
- Finished level of plugs: Projecting above surface.



250 POWDER ACTUATED FIXING SYSTEMS

- Powder actuated fixing tools: To BS 4078-2 and Kitemark certified. Operatives trained and certified as competent by tool manufacturer.
- Types of fastener, accessories and consumables: As recommended by tool manufacturer.
- Protective coating to exposed fasteners used externally or in other locations subject to dampness: Zinc rich primer to fastener heads.

510 ADHESIVES

- Storage/ Usage: In accordance with manufacturer's and statutory requirements.
- Surfaces: Clean. Regularity and texture adjusted to suit bonding and gap filling characteristics of adhesive.
- Finished adhesive joints: Fully bonded. Free of surplus adhesive.

Z22 SEALANTS

To be read with Preliminaries/General conditions.

PRODUCTS

310 JOINTS GENERALLY

- Primer, backing strip, bond breaker: Types recommended by sealant manufacturer.

EXECUTION

610 SUITABILITY OF JOINTS

- Presealing checks:
 - Joint dimensions: Within limits specified for the sealant.
 - Substrate quality: Surfaces regular, undamaged and stable.
- Joints not fit to receive sealant: Submit proposals for rectification.

620 PREPARING JOINTS

- Surfaces to which sealant must adhere:
 - Remove temporary coatings, tapes, loosely adhering material, dust, oil, grease, surface water and contaminants that may affect bond.
 - Clean using materials and methods recommended by sealant manufacturer.
- Vulnerable surfaces adjacent to joints: Mask to prevent staining or smearing with primer or sealant.
- Backing strip and/ or bond breaker installation: Insert into joint to correct depth, without stretching or twisting, leaving no gaps.
- Protection: Keep joints clean and protect from damage until sealant is applied.

630 APPLYING SEALANTS

- Substrate: Dry (unless recommended otherwise) and unaffected by frost, ice or snow.
- Environmental conditions: Do not dry or raise temperature of joints by heating.
- Sealant application: Fill joints completely and neatly, ensuring firm adhesion to substrates.
- Sealant profiles:
 - Butt and lap joints: Slightly concave.
 - Fillet joints: Flat or slightly convex.
- Protection: Protect finished joints from contamination or damage until sealant has cured.



Z31 POWDER COATINGS

To be read with Preliminaries/ General conditions.

120 POWDER COATING MATERIALS

- Manufacturer: Obtain from one only of the following: Submit proposals .
- Selected manufacturer: Submit details before commencement of powder coating.

210 WORKING PROCEDURES

- Requirement: Comply with:
 - BS 6496 for aluminium alloy backgrounds.
 - BS 6497 for galvanized steel backgrounds.
 - British Coatings Federation: Code of safe practice - Application of thermosetting powder coatings by electrostatic spraying.
 - Powder coating manufacturer's guarantee.

220 POWDER COATING APPLICATORS

- Applicator requirements:
 - Approved by powder coating manufacturer.
 - Currently certified to BS EN ISO 9001.
 - Comply with quality procedures, guarantee conditions, standards and tests required by powder coating manufacturer.
 - Each applicator to use only one plant.
- Selected applicator: Submit details before commencement of powder coating.

225 GUARANTEES

- Powder coating manufacturer and applicator guarantees:
 - Submit sample copies before commencement of powder coating.
 - Submit signed project specific copies on completion of work.

230 CONTROL SAMPLES

- Sequence: Prior to ordering materials for the works, obtain approval of appearance for:
 - Powder coated samples: Of various grades and forms of background metal to be used, showing any colour, texture and gloss variation.
 - Fabrication samples: Showing joint assembly, how powder coating is affected and how any cut metal edges are protected.
- Samples to include the following information:
 - Product reference.
 - Colour:
 - Reference number.
 - Name.
 - Gloss level.

235 INDEPENDENT INSPECTION AT PLANT

- Requirement: Contractors/ suppliers of the following designated components must commission an approved Independent Inspection Authority to carry out acceptance inspections to confirm that powder coating application complies with this specification.
 - Designated components: All major external components .
- Acceptance inspections: Carry out for each variation of colour and finish of each component work package at applicator's plant prior to any fabrication of units, in accordance with the following:

- Where three or more production runs are required for application of coatings, not less than three acceptance inspections must be carried out in accordance with BS 6001-1, general inspection level 2, with an acceptance quality limit of 1%.
- Where less than three production runs are required for application of coatings, one acceptance inspection must be carried out in accordance with BS 6001-2, with a limiting quality of 5% where the probability of acceptance is 10%.
- Components failing inspection: Reprocess or replace and reinspect.
- Inspection reports: Independent Inspection Authority must submit copies.

241 INDEPENDENT INSPECTION ON SITE

- Requirement: Where coating specification compliance at plant is unresolved, contractors/ suppliers must commission an approved Independent Inspection Authority to carry out site acceptance inspections.
- Acceptance inspections: Carry out in accordance with BS 6001-2 with a limiting quality of 5% where the probability of acceptance is 10%. Each component of a fabricated unit will be assessed as an individual part for sampling procedures.
- Components failing inspection: Remove and reprocess/ replace.

250 COMPONENT DESIGN

- Condition of components to be powder coated:
 - To comply with relevant recommendations of BS 4479-1, -3, and -4.
 - Of suitable size to fit plant capacity.
 - Of suitable thickness to withstand oven curing.

310 PRETREATMENT

- Condition of components to be powder coated:
 - Free from corrosion and damage.
 - Free from impurities including soil, grease, oil.
 - Suitable for and compatible with the pretreatment and powder coating process.
- Process: Clean, conversion coat, condition, rinse in demineralised water, drain and dry components in accordance with the powder coating manufacturer's requirements and the pretreatment supplier's recommendations.

430 EXTENT OF POWDER COATINGS

- Application: To visible component surfaces, and concealed surfaces requiring protection. Coated surfaces will be deemed 'significant surfaces' for relevant BS 6496/ BS 6497 performance requirements.

435 APPLICATION OF POWDER COATINGS

- Surfaces to receive powder coatings: Free from dust or powder deposits.
- Completion of powder coatings: Within 48 hours of pre-treatment of components.
- Jig points: Not visible on coated components.
- Curing: Controlled to attain metal temperatures and hold periods recommended by powder coating manufacturer.
- Stripping and recoating of components: Only acceptable by prior agreement of powder coating manufacturer. Stripping, pretreatment and powder coating are to be in accordance with manufacturer's requirements.
- Overcoating of components: Not acceptable.

440 PERFORMANCE AND APPEARANCE OF POWDER COATINGS

- Standard: To BS 6496/ BS 6497.

450 ALUMINIUM ALLOY FABRICATIONS

- Units may be assembled:
 - Before powder coating.
 - From components powder coated after cutting to size.
 - Where approved, from components powder coated before cutting to size.
- Exposure of uncoated background metal: Not acceptable.
- Assembly sealants: Compatible with powder coatings. Obtain approval of colour if sealants are visible after fabrication.

460 STEEL FABRICATIONS

- Unit assembly: Wherever practical, before powder coating.
- Exposure of uncoated background metal: Not acceptable.
- Assembly sealants: Compatible with powder coatings. Obtain approval of colour if sealants are visible after fabrication.

470 FIXINGS

- Exposed metal fixings: Powder coat together with components, or coat with matching repair paint system applied in accordance with the powder coating manufacturer's recommendations.

480 FABRICATION DAMAGE REPAIR/ REPLACEMENT

- Inspection: Check all components before delivery to site for damage to powder coatings. Submit proposals for repair or replacement.

510 PROTECTION

- Powder coated surfaces of components: Protect from damage during handling and installation, or by subsequent site operations.
- Protective coverings: Must be:
 - Resistant to weather conditions.
 - Partially removable to suit building in and access to fixing points.
- Protective tapes in contact with powder coatings: Must be:
 - Low tack, self adhesive and light in colour.
 - Applied and removed in accordance with tape and powder coating manufacturers' recommendations. Do not use solvents to remove residues as these are detrimental to the coating.
- Inspection of protection: Carry out monthly. Promptly repair any deterioration or deficiency.

530 SITE DAMAGE REPAIR/ REPLACEMENT

- Damage to powder coatings: Rectify immediately damage caused during handling and installation, or by subsequent site operations. Submit proposals for extensive repair or replacement.

535 DOCUMENTATION

- Submit the following information for each batch of powder coated components:
 - Supplier.
 - Trade name.
 - Colour.
 - Type of powder.
 - Method of application.

- Batch and reference number.
- Statutory requirements.

540 COMPLETION

- Cleaning and maintenance of powder coatings: Carry out in accordance with procedures detailed in powder coating manufacturer and applicator guarantees.





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DUBAI LRT PROJECT

DESIGN CALCULATION FOR BUILDING 2 LIGHT REPAIR WORKSHOP IN RASHIDIYA MAIN DEPOT (VOLUME 1 OF 1)

(Document No. MIPL/CAL/0008)

Prepared by



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Rev	Date	Description	Approved For Issue By



OBAYASHI-KAJIMA-YAPI MERKEZI JT METRO JV

DUBAI LRT PROJECT

DESIGN CALCULATION FOR BUILDING 2 LIGHT REPAIR WORKSHOP IN RASHIDIYA MAIN DEPOT (VOLUME 1 OF 1)

(Document No. MIPL/CAL/0008)

Prepared by

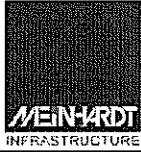


DOCUMENT PREPARATION RECORD

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CONTENTS

1.0	TYPICAL PORTAL FRAME TRANSVERSE ANALYSIS	001
2.0	TYPICAL PORTAL FRAME 3-D ANALYSIS	021
3.0	ROOF BRACING SYSTEM ANALYSIS	069
4.0	FOUNDATION DESIGN	090
5.0	STRUCTURAL ELEMENT DESIGN	112
5.1	ROOF BEAMS & COLUMNS DESIGN	113
5.2	PURLIN & SIDE C CHANNEL DESIGN	122
5.3	GROUND SLAB DESIGN	128
5.4	GROUND BEAM DESIGN	246
5.5	OFFICE & STORES DESIGN	263
5.6	STEEL STAIRCASE & MAINTENANCE PLATFORM DESIGN	283
5.7	BRACING MEMBERS DESIGN	310
5.8	RAIL SUPPORTING COLUMN	327
6.0	CONNECTIONS DESIGN	334
7.0	SETTLEMENT ANALYSIS	346



1.0 TYPICAL PORTAL FRAME TRANSVERSE ANALYSIS

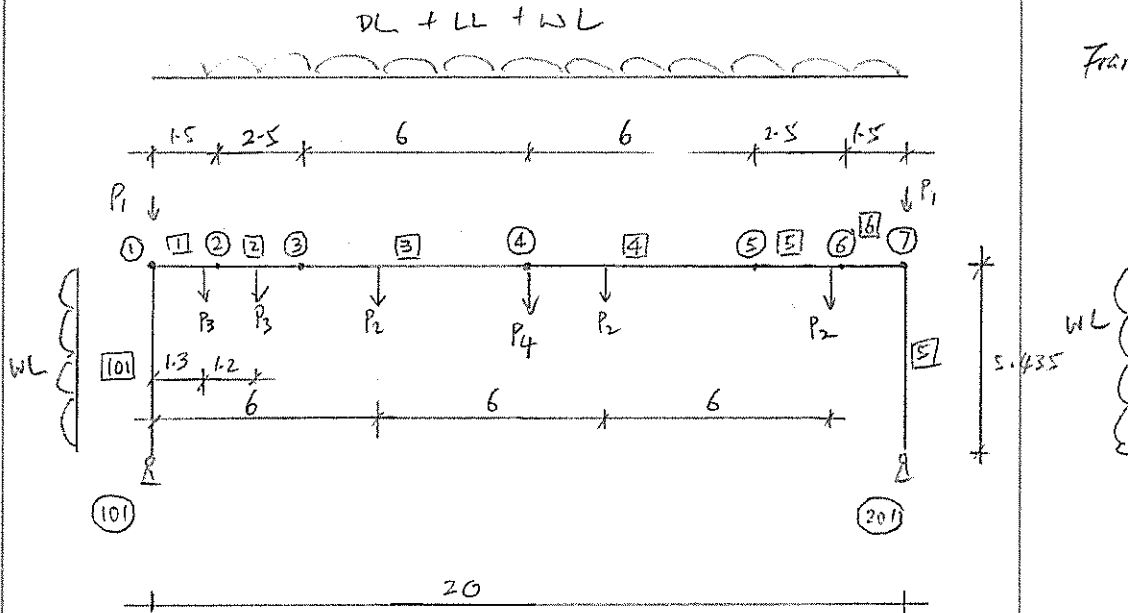
Site: Dubai: Metro Depot - Rashidiya Bldg 10

Date

Job No. 0620/0.01

Designed by GEM

Sheet No.



Beam / Column sizes

Beam 1,2,3,4,5,6 B33 UB 610 x 229 x 139.9

Column 101, 201 C23 UB 610 x 229 x 139.9

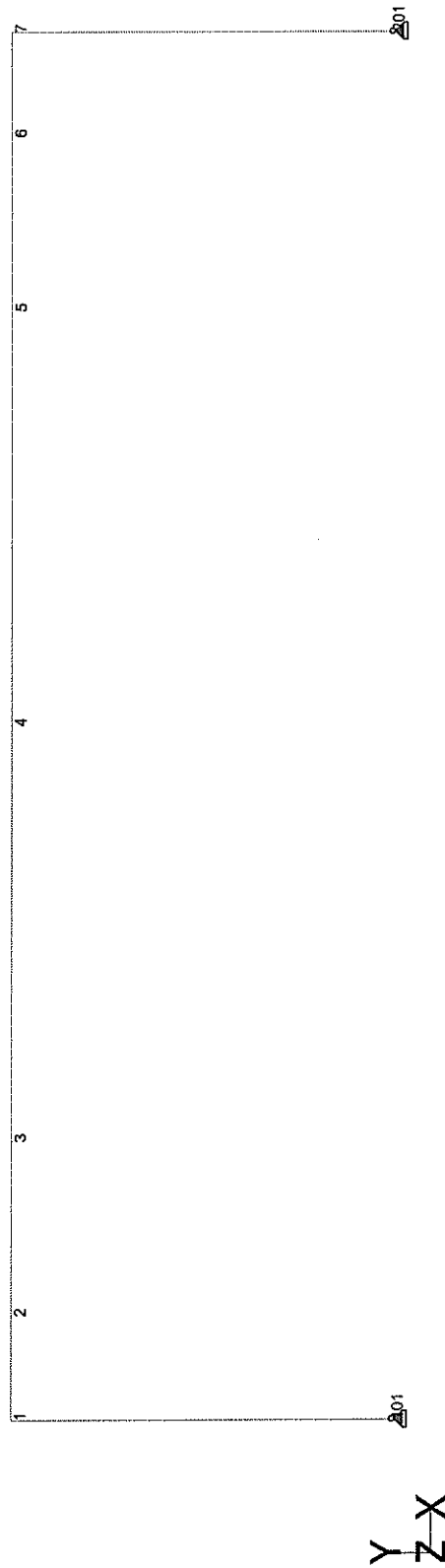


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Job Title - Dubai Metro Project

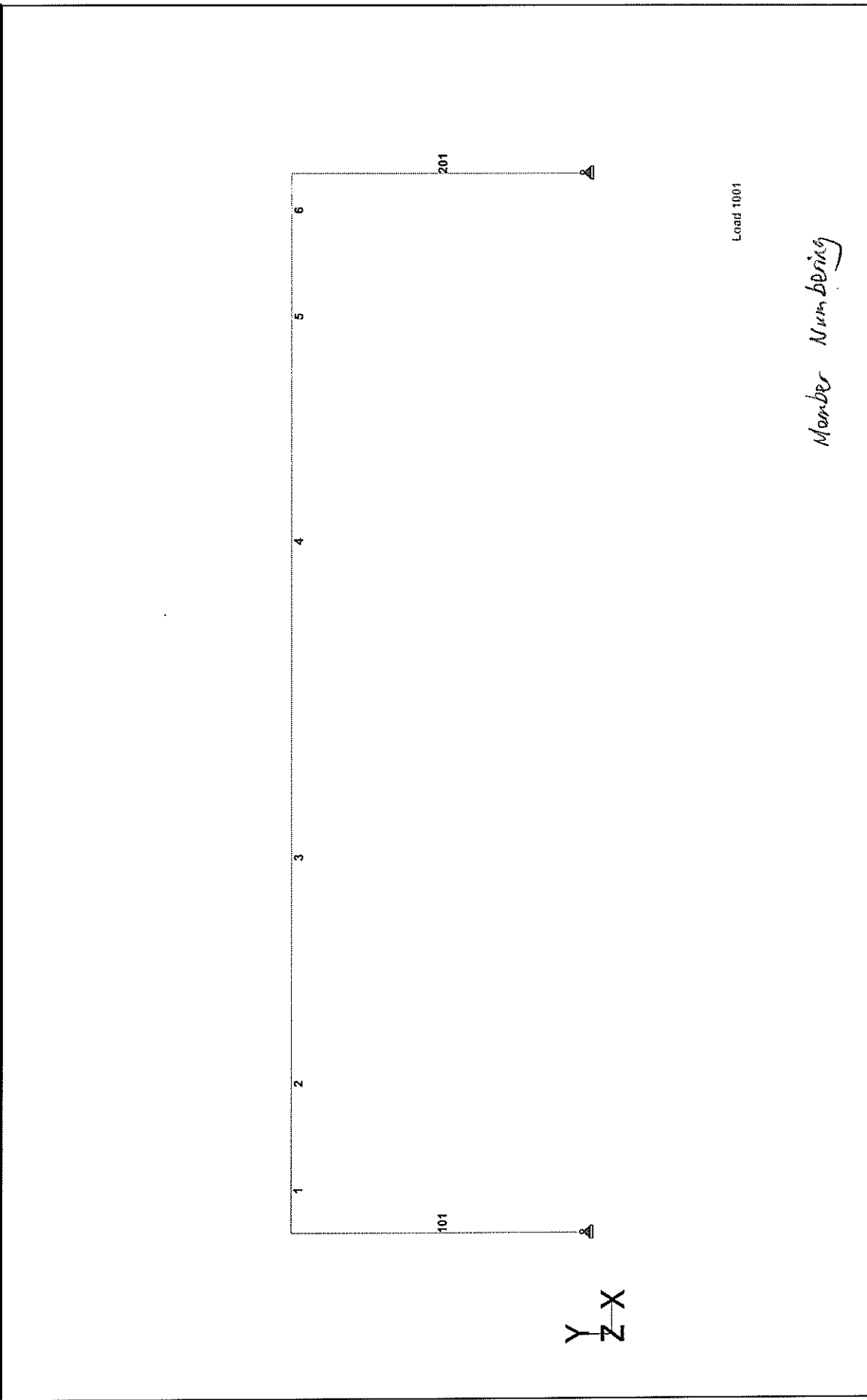
Client - Government of Dubai _ Roads & Transport Authority

Job No	Sheet No	Rev
	1	
Part		
Ref		
By	Date	Chd
22-Jun-06		
File RA_BLD 2_FR1.std		
Date/Time 22-Jun-2006 15:46		



Node Numbering

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	Part	1	
	Ref		
	By	Date 22-Jun-06	Chd
	File RA_BLD 2_FR1.std	Date/Time 22-Jun-2006 15:46	



Member Numbering

Site Dubai Metro Depot - Rashidya Bldg 2

Date

Job No. 0620/10.01

Designed by OFH

Sheet No.

Loadings:

DL

Roof DL

Sheeting	0.1 kPa
Purlins	0.1 kPa
Insulation	0.3 kPa
M&E finishes	0.3 kPa
	$\frac{0.8 \text{ kPa}}{0.8 \text{ kPa}} \times 8 = 6.4 \text{ kN/m}$

"Side cladding + insulation
+ side C-channel + overhung roof"
s/w, P₁, DL

$$= 6 + 18 + 6.4 \times 0.7$$

$$= 28.5 \text{ kN}$$

Frame 1

LL

Roof LL

$$0.6 \text{ kPa} \times 8 = 4.8 \text{ kN/m}$$

P₁, LL

$$4.8 \times 0.7 = 3.4 \text{ kN}$$

$$\text{Stinger system loading, } P_2 = (38.7 + 10.4 + 1.8) \times 0.01 \times 8$$

$$+ (2 \times 50) \times 0.01$$

$$+ 1 \text{ kN (supporting frame s/w)}$$

$$+ 2 \text{ kN (breaking load)}$$

$$= 8 \text{ kN}$$

$$\Rightarrow \text{adopt } 10 \text{ kN}$$

A/C equipment, P₄ 5 kN at mid-span

Site Dubai Metro Depot - Rashidiya Bldg 2

Date

Job No. 068010-01

Designed by OEH

Sheet No.

Maintenance Platform Loading

DL:

Adopt $DL = 2 \text{ kN/m}$

$$\Rightarrow P_{3,DL} = 2 \times 8/2 = 8 \text{ kN (on each beam support)}$$

LL:

5 kPa

$$\Rightarrow P_{3,LL} = 5 \times 1.2 \times 8/2 = 24 \text{ kN (on each beam support)}$$

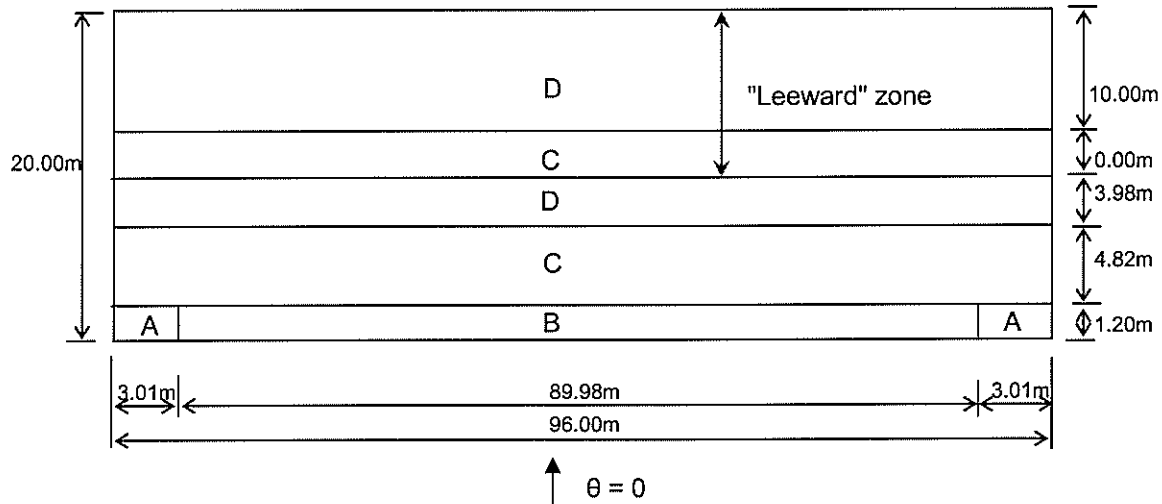
Notional horizontal load, P_h =

From Staad-pro analysis output = (Frame 1, LC 1001)

$$\begin{aligned} W_u &= 1.4DL + 1.6LL = 354 + 284 \\ &= 638 \text{ kN} \end{aligned}$$

$$\begin{aligned} P_h &= 0.570 W_u \\ &= 362 \text{ kN} \end{aligned}$$

Site:	Dubai Metro Project - Rashidiya Depot	Date:	21-Jun-06	Job No.:	68010.01
	Building 2	Frame 1			
	Wind Load Computation	Designed by: OEH		Sheet No.:	

Average Cpe for roof (Wind direction normal to eave)
PLAN


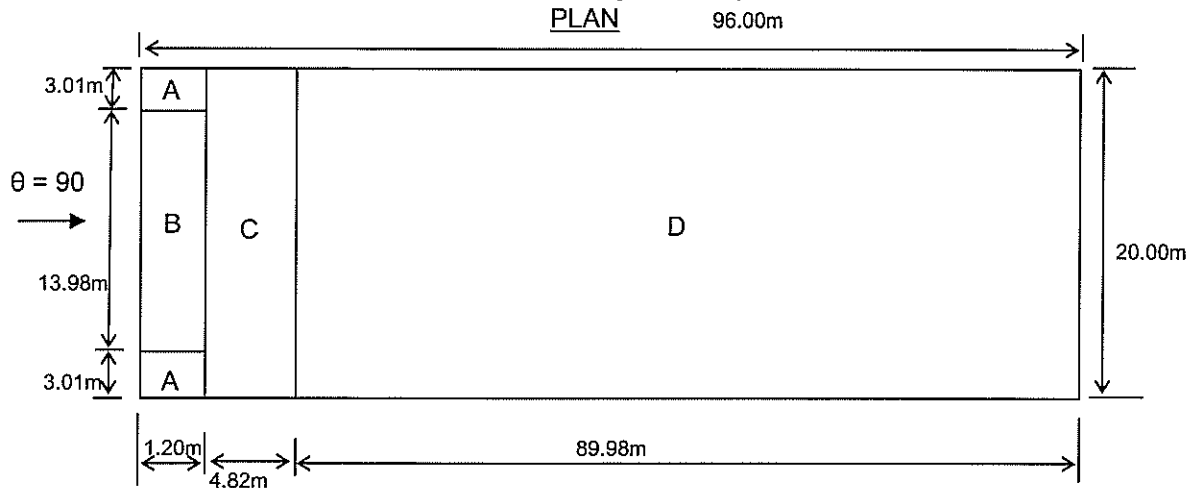
Design as flat roof (with sharp eaves)

$$\begin{aligned}
 B &= 96 \text{ m} \\
 H &= 6.024 \text{ m} \\
 b &= \text{smaller of } (B, 2H) \\
 &= 2 \times 6.024 \\
 &= 12.048 \text{ m}
 \end{aligned}$$

Zone	Windward		Leeward		
	Area (m ²)	Cpe	Area	Cpe1	Cpe2
A	7.3	-2			
B	108.4	-1.4			
C	462.6	-0.7	0.0	-0.7	-0.7
D	381.7	-0.2	960.0	-0.2	0.2
Total	960.0		960.0		
Average		-0.59		-0.20	0.20



Site:	Dubai Metro Project - Rashidiya Depot	Date:	21-Jun-06	Job No.:	68010.01
	Building 2	Frame 1			
	Wind Load Computation	Designed by: OEH		Sheet No.:	

Average Cpe for roof (Wind direction normal to gable end)


Design as flat roof (with sharp eaves)

$$\begin{aligned}
 B &= 20 \text{ m} \\
 H &= 6.024 \text{ m} \\
 b &= \text{smaller of } (B, 2H) \\
 &= 2 \times 6.024 \\
 &= 12.048 \text{ m}
 \end{aligned}$$

For maximum uplift, assume zone C coefficient =>

$$C_{pe} = -0.7$$

For maximum downward force, assume zone D coefficient =>

$$C_{pe} = 0.2$$

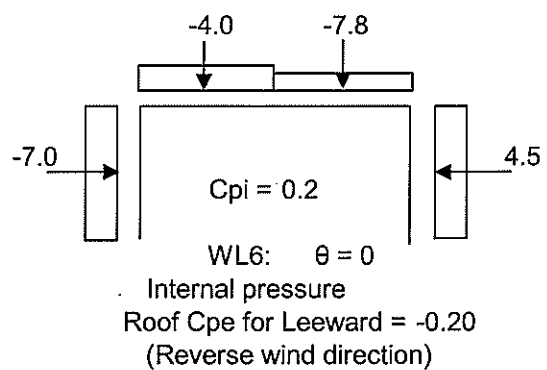
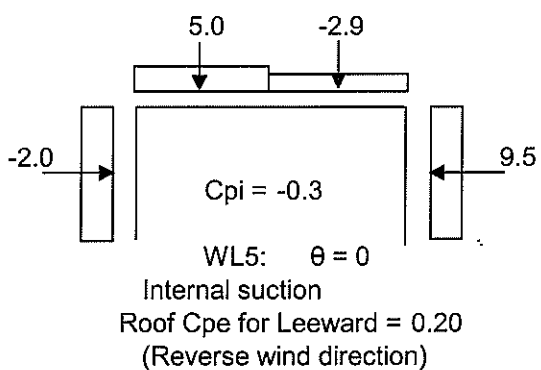
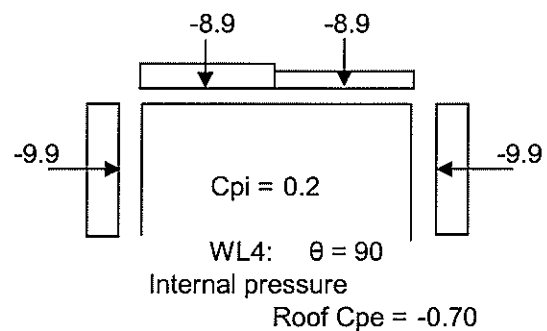
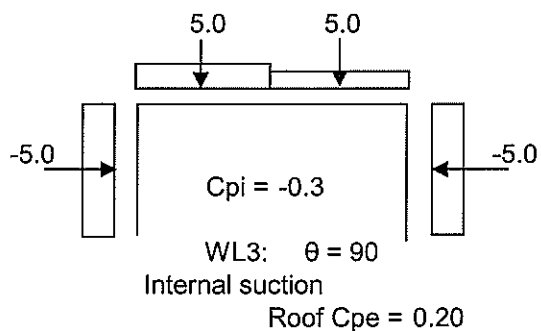
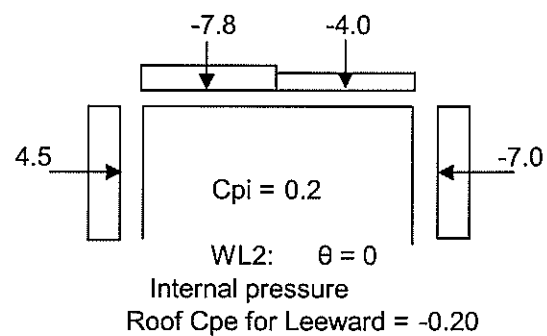
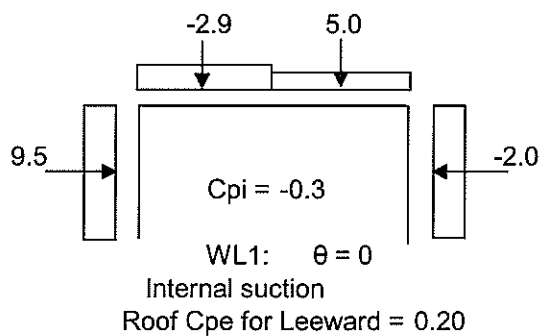
Wind Load Computation

$$\begin{aligned}
 D &= 20.00 \text{ m} \\
 H &= 6.024 \text{ m} \\
 D/H &= 3.32
 \end{aligned}$$

Elements	Effective Wind Speed V_e m/s	Dynamic Wind Pressure q kN/m ²	External Pressure Coefficient, C_{pe}			
			$\theta = 0$ (normal to eave)		$\theta = 90$ (normal to gable end)	
			Windward	Leeward		
Roof	45	1.24	-0.59	-0.20 0.20	-0.7 (Zone C)	0.2 (Zone D)
Wall	45	1.24	0.66	-0.5 -0.5	-0.8 (Zone B)	-0.8 (Zone B)

Site:	Dubai Metro Project - Rashidiya Depot	Date:	21-Jun-06	Job No.:	68010.01
	Building 2		Frame 1		
	Wind Load Computation	Designed by:	OEH	Sheet No.:	

$$\text{Wind Load} = 8 \times q \times (C_{pe} - C_{pi}) \quad \text{kN/m}$$



Site: Dubai Metro Project - Rashidiya Depot
Building 2 Frame 1
Earthquake Loading

Date: 23-Jun-06
Designed by: OEH

Job No.: 68010.01
Sheet No.:

Estimation of Earthquake Loading

Total Dead Load on the Building Column W = Given below (including part of Live Load)
Height of the Column h_n = Given below

Design Information:

- 1) Seismic Zone = 2A (from Design Brief)
2) Occupancy Category = 1 (Tanle- 16-K, Conservatively adopt 1)
3) Lateral Force procedure = **Static Procedure** (Section 1629.8 seismic zone 2A, structures not more than 5 stories)

Earthquake Loads

- 4) **Base Shear** $V_{base-S} = (C_v \times I \times W) / (R \times T)$ (Section 1630.2.1)
Seismic Coefficient $C_v = 0.32$ (Table 16-R assuming soil profile type S_D)
Importance Factor $I = 1.00$ (Table 16-K)
Numeric. Coef. Strength & Ductility $R = 4.50$ (Table 16-N for OMRT steel only)
Elastic Fundamental Period of Vibration $T = C_t (h_n)^{3/4}$ Sec.
Numerical Coefficient $C_t = 0.0853$ (Steel frame t.b.c)

- 5) **Limiting Values of Base shear** $V_{base-S-Max} = (2.5 \times C_a \times I \times W) / R = 22.24$
 $V_{base-S-Min} = 0.11 \times C_a \times I \times W = 4.40$

Design Earthquake Loads

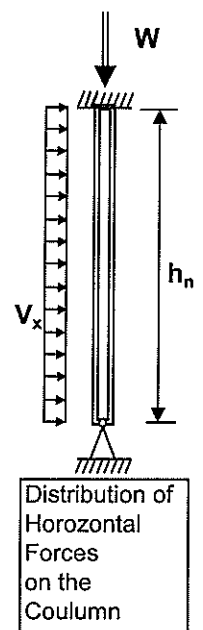
(Section 1630.6)

- $E_x = \rho \times \text{Min. } (E_h \text{ or } V_{base-S-Max}) + E_v$
 $E_{x-max} = \Omega \times \text{Min. } (E_h \text{ or } V_{base-S-Max})$
 $V_{base-S} = E_h =$ Earthquake load in the element due to the base shear, V , as set out above (for non-structural components refer to section 1632)
Vert. component of EQ ground motion $E_v = 0.5 C_a I D$ (Table 16-K)
Seismic Coefficient $C_a = 0.22$ (Table 16-Q, for soil type S_D)
Seismic Force Amplification Factor $\Omega = 2.80$ (Table 16-N)
Dead Load on the element $W = D =$ Given below
 $\rho = 2 - 6.1 / (r_{max} \sqrt{A_B}) = 1.25$ (conservative) ($1.0 \leq \rho \leq 1.5$)
 $r_{max} =$ The maximum element-storey shear ratio
 $A_B =$ Ground floor area of the structure

Distribution of Horizontal Forces:

Distribution of Horizontal Force on the Column $V_x = E_x / h_n$

Col	Gr. Load, W , kN	Ht h_n , m	Time, T sec.	V_{base-S} kN	V_{base-S} kN	E_v kN	E_x kN	E_{x-max} kN	V_x kN/m	V_{x-max} kN/m
C101	182	5.435	0.30	42.6	22.24	20.02	47.8	62.3	8.8	11.5
C201	153	5.435	0.30	35.8	18.70	16.83	40.2	52.4	7.4	9.6
C301	0	5	0.29	0.0	0.00	0	0.0	0.0	0.0	0.0
C401	0	5	0.29	0.0	0.00	0	0.0	0.0	0.0	0.0
C501	0	5	0.29	0.0	0.00	0	0.0	0.0	0.0	0.0
C601	0	5	0.29	0.0	0.00	0	0.0	0.0	0.0	0.0
C701	0	5	0.29	0.0	0.00	0	0.0	0.0	0.0	0.0
C801	0	5	0.29	0.0	0.00	0	0.0	0.0	0.0	0.0
C901	0	5	0.29	0.0	0.00	0	0.0	0.0	0.0	0.0
C1001	0	5	0.29	0.0	0.00	0	0.0	0.0	0.0	0.0





Site:	Dubai Metro Project - Rashidiya Depot	Date:	28-Jun-06	Job No.:	68010.01
	Building 2 Frame 1				
	Effective Length	Designed by:	OEH	Sheet No.:	

Effective Length Computation

Column member 101 (use for 201 as well)

Effective Lengths for Compression Member

Column

Length, L_c = 5.435 m

I_c = 111800 cm⁴

Rafter

Length, L_r = 20 m

I_r = 111800 cm⁴

$$k_1 = (I_c/L_c) / (I_c/L_c + 1.5 \cdot I_r/L_r)$$

$$= 0.71$$

$$k_2 = 1 \text{ (pinned base)}$$

From BS5950-1:2000 Figure E.2:

$$k_c = L_{ex}/L_c = 3.2$$

$$L_{ex} = 18 \text{ m}$$

$$L_{ey} = 3 \text{ m} \quad (\text{assumed})$$

Effective Length for lateral torsional buckling check

$$L_e = 3 \text{ m} \quad (\text{provide lateral restraint at 3m c/c})$$

Rafter member 1, 2, 3, 4, 5, 6

Effective Lengths for Compression Member

$$P_c = 354 \text{ kN} \quad (\text{From LC 1001})$$

$$P_r = 125 \text{ kN} \quad (\text{From LC 1001})$$

$$L_{ex}/L_r = (k_c \cdot L_c/L_r) \cdot [(P_c \cdot I_r)/(P_r \cdot I_c)]^{0.5}$$

$$= 1.46$$

$$L_{ex} = 30 \text{ m}$$

$$L_{ey} = 1.5 \text{ m}$$

Effective Length for lateral torsional buckling check

$$L_e = 3 \text{ m} \quad (\text{For member 1, 2, 5, 6, provide lateral restraint at 3m c/c})$$

$$L_e = 1.5 \text{ m} \quad (\text{For member 3, 4})$$

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*****
*
*          STAAD.Pro
*          Version 2005      Bld 1002.US
*          Proprietary Program of
*          Research Engineers, Intl.
*          Date=      JUN 28, 2006
*          Time=      10:53:31
*
*          USER ID: Meinhardt Singapore Pte Ltd
*****

```

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1. STAAD PLANE      FRAME 1
INPUT FILE: RA BLD 2 FR1.STD
2. START JOB INFORMATION
3. JOB NAME - DUBAI METRO PROJECT
4. JOB CLIENT - GOVERNMENT OF DUBAI  _ ROADS & TRANSPORT AUTHORITY
5. END JOB INFORMATION
6. INPUT WIDTH 79
7. UNIT METER KN
9. JOINT COORDINATES
10. 1 0 5.435
11. 2 1.5 5.435
12. 3 4 5.435
13. 4 10 5.435
14. 5 16 5.435
15. 6 18.5 5.435
16. 7 20 5.435
18. 101 0 0
19. 201 20 0
21. MEMBER INCIDENCES
22. 1 1 2
23. 2 2 3
24. 3 3 4
25. 4 4 5
26. 5 5 6
27. 6 6 7
29. 101 101 1
30. 201 201 7
32. MEMBER PROPERTY BRITISH
33. 1 6      TABLE ST UB610X229X140
34. 2 3 4 5  TABLE ST UB610X229X140
35. 101 201  TABLE ST UB610X229X140
37. DEFINE MATERIAL START
38. ISOTROPIC STEEL
39. E 2.05E+08
40. POISSON 0.3
41. DENSITY 76.8195
42. ALPHA 1.20E-05
43. DAMP 0.03
44. END DEFINE MATERIAL
45. CONSTANTS
46. MATERIAL  STEEL  ALL
    FRAME 1
    2

48. SUPPORTS
49. 101 201 PIN
51. LOAD 1 SELFWEIGHT
52. SELFWEIGHT Y -1
53. LOAD 2 SDL
54. JOINT LOAD
55. 1 7  FY -28.5
56. MEMBER LOAD
57. 1 TO 6 UNI GY -6.4
58. 1  CON GY -8 1.3

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-- PAGE NO.

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59. 2 CON GY -8 1
60. LOAD 3 LL
61. JOINT LOAD
62. 1 7 FY -3.4
63. 4 FY -5
64. MEMBER LOAD
65. 1 TO 6 UNI GY -4.8
66. 3 CON GY -10 2
67. 4 CON GY -10 2
68. 5 CON GY -10 2
69. 1 CON GY -24 1.3
70. 2 CON GY -24 1
71. LOAD 4 PH (NOTIONAL HORIZONTAL LOAD)
72. JOINT LOAD
73. 1 FX 3.2
74. LOAD 11 WL1 (INTERNAL SUCTION, WIND ANGLE 0)
75. MEMBER LOAD
76. 101 UNI Y -9.5
77. 201 UNI Y -2
78. 1 2 3 UNI Y 2.9
79. 4 5 6 UNI Y -5
80. LOAD 12 WL2 (INTERNAL PRESSURE, WIND ANGLE 0)
81. MEMBER LOAD
82. 101 UNI Y -4.5
83. 201 UNI Y -7
84. 1 2 3 UNI Y 7.8
85. 4 5 6 UNI Y 4
86. LOAD 13 WL3 (INTERNAL SUCTION, WIND ANGLE 90)
87. MEMBER LOAD
88. 101 UNI Y 5
89. 201 UNI Y -5
90. 1 2 3 UNI Y -5
91. 4 5 6 UNI Y -5
92. LOAD 14 WL4 (INTERNAL PRESSURE, WIND ANGLE 90)
93. MEMBER LOAD
94. 101 UNI Y 9.9
95. 201 UNI Y -9.9
96. 1 2 3 UNI Y 8.9
97. 4 5 6 UNI Y 8.9
98. LOAD 15 WL5 (INTERNAL SUCTION, WIND ANGLE 0, REVERSE WIND)
99. MEMBER LOAD
100. 101 UNI Y 2
101. 201 UNI Y 9.5
102. 1 2 3 UNI Y -5
103. 4 5 6 UNI Y 2.9
104. LOAD 16 WL6 (INTERNAL PRESSURE, WIND ANGLE 0, REVERSE WIND)
    FRAME 1
    3
    105. MEMBER LOAD
    106. 101 UNI Y 7
    107. 201 UNI Y 4.5
    108. 1 2 3 UNI Y 4
    109. 4 5 6 UNI Y 7.8
    110. LOAD 21 EQ (EARTHQUAKE LOAD)
    111. MEMBER LOAD
    112. 101 UNI Y 11.5
    113. 201 UNI Y 9.6
    115. LOAD COMB 1001 1.4DL + 1.6LL
    116. 1 1.4 2 1.4 3 1.6
    117. LOAD COMB 1002 1.4DL + 1.6LL + 1.0PH
    118. 1 1.4 2 1.4 3 1.6 4 1.0
    119. LOAD COMB 1011 1.2DL + 1.2LL + 1.2WL1
    120. 1 1.2 2 1.2 3 1.2 11 1.2
    121. LOAD COMB 1111 1.4DL + 1.4WL1
    122. 1 1.4 2 1.4 11 1.4
    123. LOAD COMB 1211 1.0DL + 1.4WL1
    124. 1 1.0 2 1.0 11 1.4
    125. LOAD COMB 1012 1.2DL + 1.2LL + 1.2WL2
    126. 1 1.2 2 1.2 3 1.2 12 1.2
    127. LOAD COMB 1112 1.4DL + 1.4WL2
    128. 1 1.4 2 1.4 12 1.4
    129. LOAD COMB 1212 1.0DL + 1.4WL2

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-- PAGE NO.


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130. 1 1.0 2 1.0 12 1.4
131. LOAD COMB 1013 1.2DL + 1.2LL + 1.2WL3
132. 1 1.2 2 1.2 3 1.2 13 1.2
133. LOAD COMB 1113 1.4DL + 1.4WL3
134. 1 1.4 2 1.4 13 1.4
135. LOAD COMB 1213 1.0DL + 1.4WL3
136. 1 1.0 2 1.0 13 1.4
137. LOAD COMB 1014 1.2DL + 1.2LL + 1.2WL4
138. 1 1.2 2 1.2 3 1.2 14 1.2
139. LOAD COMB 1114 1.4DL + 1.4WL4
140. 1 1.4 2 1.4 14 1.4
141. LOAD COMB 1214 1.0DL + 1.4WL4
142. 1 1.0 2 1.0 14 1.4
143. LOAD COMB 1015 1.2DL + 1.2LL + 1.2WL5
144. 1 1.2 2 1.2 3 1.2 15 1.2
145. LOAD COMB 1115 1.4DL + 1.4WL5
146. 1 1.4 2 1.4 15 1.4
147. LOAD COMB 1215 1.0DL + 1.4WL5
148. 1 1.0 2 1.0 15 1.4
149. LOAD COMB 1016 1.2DL + 1.2LL + 1.2WL6
150. 1 1.2 2 1.2 3 1.2 16 1.2
151. LOAD COMB 1116 1.4DL + 1.4WL6
152. 1 1.4 2 1.4 16 1.4
153. LOAD COMB 1216 1.0DL + 1.4WL6
154. 1 1.0 2 1.0 16 1.4
155. LOAD COMB 1021 1.2DL + 0.5LL + 1.0EQ
156. 1 1.2 2 1.2 3 0.5 21 1.0
157. LOAD COMB 1121 1.2DL + 0.5LL - 1.0EQ
158. 1 1.2 2 1.2 3 0.5 21 -1.0
159. LOAD COMB 1221 0.9DL + 1.0EQ
160. 1 0.9 2 0.9 21 1.0
161. LOAD COMB 1321 0.9DL - 1.0EQ

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FRAME 1

-- PAGE NO.

4

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162. 1 0.9 2 0.9 21 -1.0
164. LOAD COMB 2001 1.0DL + 1.0LL
165. 1 1.0 2 1.0 3 1.0
166. LOAD COMB 2002 1.0DL + 0.5LL
167. 1 1.0 2 1.0 3 0.5
168. LOAD COMB 2011 DL + LL + WL1
169. 1 1.0 2 1.0 3 1.0 11 1.0
170. LOAD COMB 2012 DL + LL + WL2
171. 1 1.0 2 1.0 3 1.0 12 1.0
172. LOAD COMB 2013 DL + LL + WL3
173. 1 1.0 2 1.0 3 1.0 13 1.0
174. LOAD COMB 2014 DL + LL + WL4
175. 1 1.0 2 1.0 3 1.0 14 1.0
176. LOAD COMB 2015 DL + LL + WL5
177. 1 1.0 2 1.0 3 1.0 15 1.0
178. LOAD COMB 2016 DL + LL + WL6
179. 1 1.0 2 1.0 3 1.0 16 1.0
181. PERFORM ANALYSIS

```

P R O B L E M S T A T I S T I C S

```

NUMBER OF JOINTS/MEMBER+ELEMENTS/SUPPORTS =      9/      8/      2
ORIGINAL/FINAL BAND-WIDTH=      7/      1/      6 DOF
TOTAL PRIMARY LOAD CASES =      11, TOTAL DEGREES OF FREEDOM =      23
SIZE OF STIFFNESS MATRIX =      1 DOUBLE KILO-WORDS
REQRD/AVAIL. DISK SPACE =      12.0/ 11311.3 MB

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182. PRINT MAXFORCE ENV
MAXFORCE ENV

```

FRAME 1

-- PAGE NO.

5

MEMBER FORCE ENVELOPE

ALL UNITS ARE KN METE

MAX AND MIN FORCE VALUES AMONGST ALL SECTION LOCATIONS

MEMB	FY/ FZ LD	DIST DIST	LD LD	MZ/ MY	DIST DIST	LD LD	FX	DIST
1 MAX	298.52	0.00	1001	709.31	0.00	1013		
	0.00	0.00	1	0.00	0.00	1	126.73 C	0.00
	1002							
MIN	-89.00	0.00	14	-254.62	0.00	14		
	0.00	1.50	2016	0.00	1.50	2016	73.75 T	1.50
	14							
2 MAX	227.69	0.00	1013	294.13	0.00	1013		
	0.00	0.00	1	0.00	0.00	1	126.73 C	0.00
	1002							
MIN	-75.65	0.00	14	-175.54	2.50	1121		
	0.00	2.50	2016	0.00	2.50	2016	73.75 T	2.50
	14							
3 MAX	136.59	0.00	1013	190.38	6.00	14		
	0.00	0.00	1	0.00	0.00	1	126.73 C	0.00
	1002							
MIN	-53.40	0.00	14	-543.70	6.00	1013		
	0.00	6.00	2016	0.00	6.00	2016	73.75 T	6.00
	14							
4 MAX	53.40	6.00	14	190.38	0.00	14		
	0.00	0.00	1	0.00	0.00	1	126.73 C	0.00
	1002							
MIN	-146.38	6.00	1013	-543.70	0.00	1013		
	0.00	6.00	2016	0.00	6.00	2016	73.75 T	6.00
	14							
5 MAX	75.65	2.50	14	368.96	2.50	1013		
	0.00	0.00	1	0.00	0.00	1	126.73 C	0.00
	1002							
MIN	-211.08	2.50	1013	-144.16	2.50	16		
	0.00	2.50	2016	0.00	2.50	2016	73.75 T	2.50
	14							
6 MAX	89.00	1.50	14	709.31	1.50	1013		
	0.00	0.00	1	0.00	0.00	1	126.73 C	0.00
	1002							
MIN	-242.70	1.50	1013	-254.62	1.50	14		
	0.00	1.50	2016	0.00	1.50	2016	73.75 T	1.50
	14							
101 MAX	73.75	5.43	14	709.31	5.43	1013		
	0.00	0.00	1	0.00	0.00	1	354.26 C	0.00
	1001							
MIN	-146.81	0.00	1013	-254.62	5.43	14		
	0.00	5.43	2016	0.00	5.43	2016	89.00 T	5.43
	14							
201 MAX	146.81	0.00	1013	254.62	5.43	14		
	0.00	0.00	1	0.00	0.00	1	289.90 C	0.00
	1013							
MIN	-73.75	5.43	14	-709.31	5.43	1013		
	0.00	5.43	2016	0.00	5.43	2016	89.00 T	5.43
	14							

FRAME 1

6

-- PAGE NO.

***** END OF FORCE ENVELOPE FROM INTERNAL STORAGE *****

183. LOAD LIST 1001 1002 -
 184. 1011 1111 1211 -
 185. 1012 1112 1212 -
 186. 1013 1113 1213 -
 187. 1014 1114 1214 -
 188. 1015 1115 1215 -
 189. 1016 1116 1216 -
 190. 1021 1121 1221 1321
 191. SECTION 0.25 0.5 0.75
 192. PARAMETER
 193. CODE BS5950
 195. UNL 3 MEM 101 201
 196. LZ 18 MEM 101 201
 197. LY 3 MEM 101 201
 199. UNL 3 MEM 1 2 5 6
 200. UNL 1.5 MEM 3 4
 201. LZ 30 MEM 1 2 3 4 5 6
 202. LY 1.5 MEM 1 2 3 4 5 6
 204. CHECK CODE ALL
 STEEL DESIGN

STAAD.Pro CODE CHECKING - (BSI)

PROGRAM CODE REVISION V2.9_5950-1_2000

FRAME 1
 7

-- PAGE NO.

ALL UNITS ARE - KN METE (UNLESS OTHERWISE NOTED)

MEMBER	TABLE	RESULT/ FX	CRITICAL COND/ MY	RATIO/ MZ	LOADING/ LOCATION
1	ST UB610X229X140	PASS	BS-4.3.6	0.741	1013
		114.20 C	0.00	709.31	0.00
2	ST UB610X229X140	PASS	BS-4.3.6	0.307	1013
		114.20 C	0.00	294.13	0.00
3	ST UB610X229X140	PASS	ANNEX I.1	0.538	1013
		114.20 C	0.00	543.70	-
4	ST UB610X229X140	PASS	ANNEX I.1	0.538	1013
		114.20 C	0.00	543.70	-
5	ST UB610X229X140	PASS	BS-4.3.6	0.386	1013
		114.20 C	0.00	368.96	0.00
6	ST UB610X229X140	PASS	BS-4.3.6	0.741	1013
		114.20 C	0.00	709.31	0.00
101	ST UB610X229X140	PASS	BS-4.3.6	0.741	1013
		344.91 C	0.00	709.31	0.00
201	ST UB610X229X140	PASS	BS-4.3.6	0.741	1013
		289.90 C	0.00	709.31	0.00

***** END OF TABULATED RESULT OF DESIGN *****

208. LOAD LIST ALL
 209. PRINT SUPPORT REACTION ALL
 SUPPORT REACTION ALL
 FRAME 1
 16

-- PAGE NO.

SUPPORT REACTIONS -UNIT KN METE STRUCTURE TYPE = PLANE

JOINT	LOAD	FORCE-X	FORCE-Y	FORCE-Z	MOM-X	MOM-Y	MOM Z
101	1	7.05	21.11	0.00	0.00	0.00	0.00
	2	35.09	106.98	0.00	0.00	0.00	0.00
	3	41.34	109.34	0.00	0.00	0.00	0.00
	4	-1.60	-0.87	0.00	0.00	0.00	0.00
	11	-35.66	-17.74	0.00	0.00	0.00	0.00
	12	-58.39	-76.99	0.00	0.00	0.00	0.00
	13	38.87	50.00	0.00	0.00	0.00	0.00
	14	-19.95	-89.00	0.00	0.00	0.00	0.00
	15	26.84	38.74	0.00	0.00	0.00	0.00
	16	4.11	-41.01	0.00	0.00	0.00	0.00
	21	59.83	15.58	0.00	0.00	0.00	0.00
1001		125.14	354.26	0.00	0.00	0.00	0.00
1002		123.53	353.39	0.00	0.00	0.00	0.00
1011		57.38	263.62	0.00	0.00	0.00	0.00
1111		9.07	154.48	0.00	0.00	0.00	0.00
1211		-7.78	103.25	0.00	0.00	0.00	0.00
1012		30.11	192.52	0.00	0.00	0.00	0.00
1112		-22.75	71.53	0.00	0.00	0.00	0.00
1212		-39.60	20.30	0.00	0.00	0.00	0.00
1013		146.81	344.91	0.00	0.00	0.00	0.00
1113		113.41	249.32	0.00	0.00	0.00	0.00
1213		96.55	198.09	0.00	0.00	0.00	0.00
1014		76.24	178.11	0.00	0.00	0.00	0.00
1114		31.07	54.72	0.00	0.00	0.00	0.00
1214		14.22	3.49	0.00	0.00	0.00	0.00
1015		132.38	331.40	0.00	0.00	0.00	0.00
1115		96.58	233.56	0.00	0.00	0.00	0.00
1215		79.72	182.33	0.00	0.00	0.00	0.00
1016		105.11	235.70	0.00	0.00	0.00	0.00
1116		64.76	121.91	0.00	0.00	0.00	0.00
1216		47.90	70.68	0.00	0.00	0.00	0.00
1021		131.06	223.95	0.00	0.00	0.00	0.00
1121		11.41	192.79	0.00	0.00	0.00	0.00
1221		97.75	130.86	0.00	0.00	0.00	0.00
1321		-21.90	99.70	0.00	0.00	0.00	0.00
2001		83.48	237.43	0.00	0.00	0.00	0.00
2002		62.81	182.76	0.00	0.00	0.00	0.00
2011		47.82	219.68	0.00	0.00	0.00	0.00
2012		25.09	160.43	0.00	0.00	0.00	0.00
2013		122.34	287.43	0.00	0.00	0.00	0.00
2014		63.53	148.43	0.00	0.00	0.00	0.00
2015		110.32	276.17	0.00	0.00	0.00	0.00
2016		87.59	196.42	0.00	0.00	0.00	0.00
201	1	-7.05	21.11	0.00	0.00	0.00	0.00
	2	-35.09	94.02	0.00	0.00	0.00	0.00
	3	-41.34	76.46	0.00	0.00	0.00	0.00
	4	-1.60	0.87	0.00	0.00	0.00	0.00
	11	-26.84	38.74	0.00	0.00	0.00	0.00

FRAME 1
17

-- PAGE NO.

SUPPORT REACTIONS -UNIT KN METE STRUCTURE TYPE = PLANE

JOINT	LOAD	FORCE-X	FORCE-Y	FORCE-Z	MOM-X	MOM-Y	MOM Z
	12	-4.11	-41.01	0.00	0.00	0.00	0.00
	13	-38.87	50.00	0.00	0.00	0.00	0.00
	14	19.95	-89.00	0.00	0.00	0.00	0.00
	15	35.66	-17.74	0.00	0.00	0.00	0.00
	16	58.39	-76.99	0.00	0.00	0.00	0.00
	21	54.85	-15.58	0.00	0.00	0.00	0.00

1001	-125.14	283.51	0.00	0.00	0.00	0.00
1002	-126.73	284.38	0.00	0.00	0.00	0.00
1011	-132.38	276.39	0.00	0.00	0.00	0.00
1111	-96.58	215.42	0.00	0.00	0.00	0.00
1211	-79.72	169.37	0.00	0.00	0.00	0.00
1012	-105.11	180.69	0.00	0.00	0.00	0.00
1112	-64.76	103.77	0.00	0.00	0.00	0.00
1212	-47.90	57.72	0.00	0.00	0.00	0.00
1013	-146.81	289.90	0.00	0.00	0.00	0.00
1113	-113.41	231.18	0.00	0.00	0.00	0.00
1213	-96.55	185.13	0.00	0.00	0.00	0.00
1014	-76.24	123.10	0.00	0.00	0.00	0.00
1114	-31.07	36.58	0.00	0.00	0.00	0.00
1214	-14.22	-9.47	0.00	0.00	0.00	0.00
1015	-57.38	208.61	0.00	0.00	0.00	0.00
1115	-9.07	136.34	0.00	0.00	0.00	0.00
1215	7.78	90.29	0.00	0.00	0.00	0.00
1016	-30.11	137.51	0.00	0.00	0.00	0.00
1116	22.75	53.39	0.00	0.00	0.00	0.00
1216	39.60	7.34	0.00	0.00	0.00	0.00
1021	-16.39	160.80	0.00	0.00	0.00	0.00
1121	-126.09	191.96	0.00	0.00	0.00	0.00
1221	16.93	88.03	0.00	0.00	0.00	0.00
1321	-92.78	119.20	0.00	0.00	0.00	0.00
2001	-83.48	191.59	0.00	0.00	0.00	0.00
2002	-62.81	153.36	0.00	0.00	0.00	0.00
2011	-110.32	230.33	0.00	0.00	0.00	0.00
2012	-87.59	150.58	0.00	0.00	0.00	0.00
2013	-122.34	241.59	0.00	0.00	0.00	0.00
2014	-63.53	102.59	0.00	0.00	0.00	0.00
2015	-47.82	173.84	0.00	0.00	0.00	0.00
2016	-25.09	114.59	0.00	0.00	0.00	0.00

***** END OF LATEST ANALYSIS RESULT *****

210. FINISH

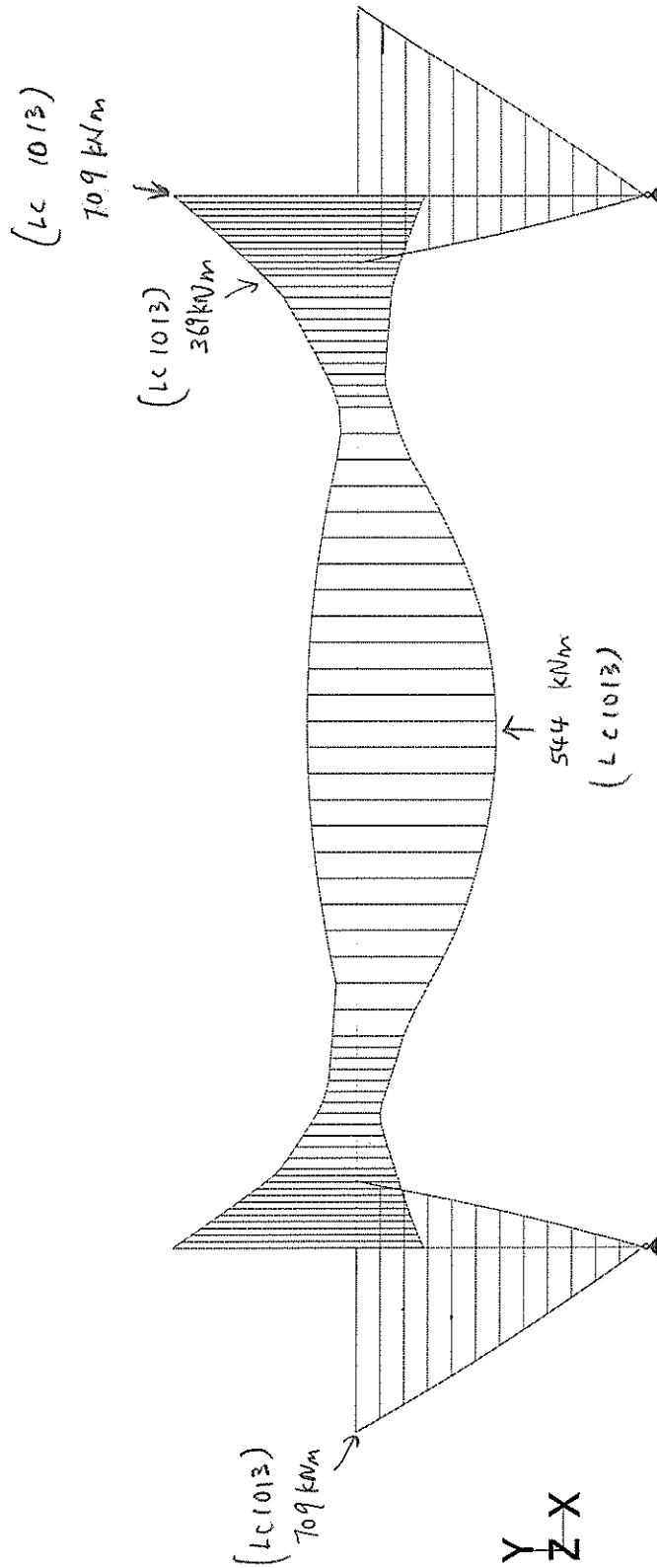


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Job Title - Dubai Metro Project

Client - Government of Dubai - Roads & Transport Authority

Job No	Sheet No	Rev
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Part		
Ref		
By	Date 23-Jun-06	Chd
File RA_BLD 2_FR1.std Date/Time 23-Jun-2006 14:58		



Bending Moment Envelope



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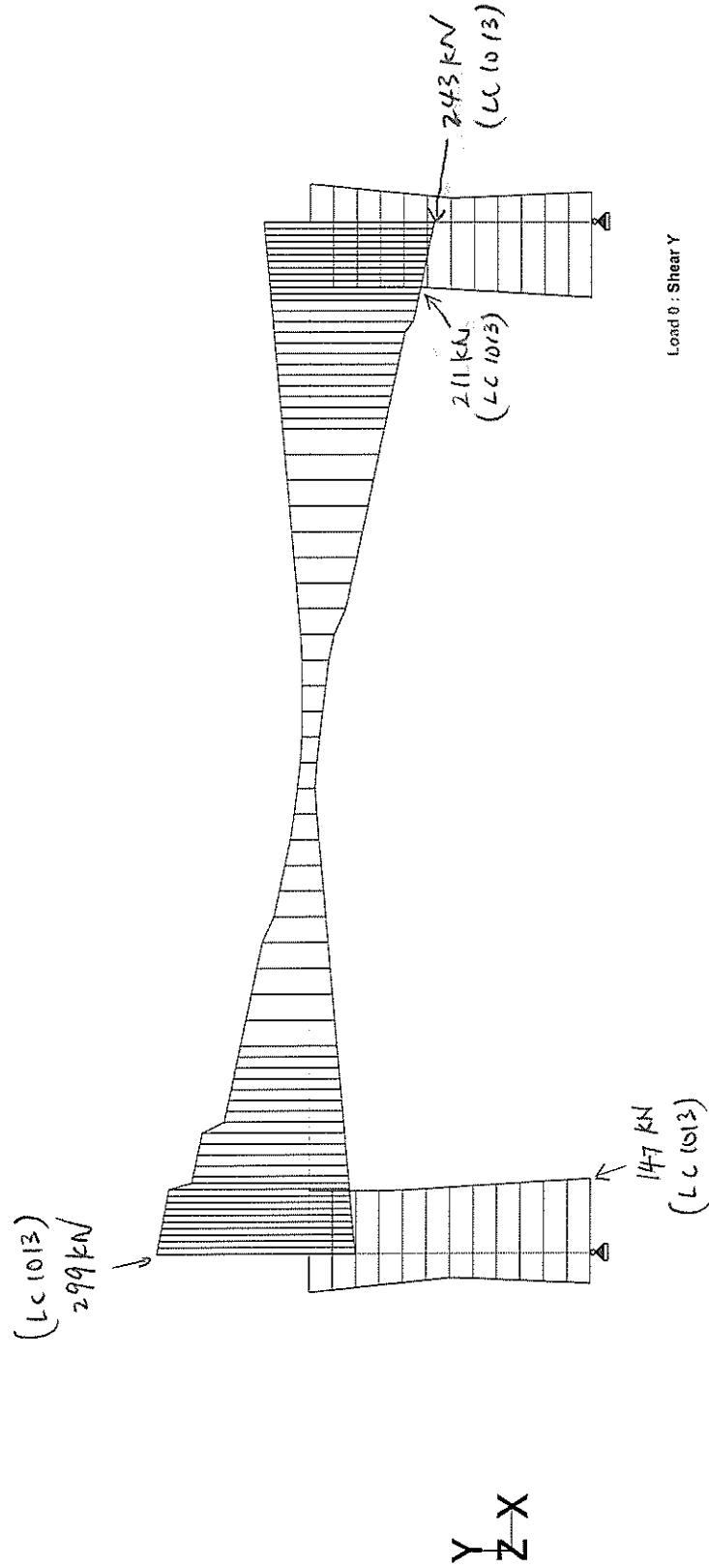
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Job No	Sheet No	Rev
	1	

Part

Ref

By	Date	Chd
	23-Jun-06	



Shear Force Envelope

2.0 TYPICAL PORTAL FRAME 3-D ANALYSIS

Site Dubai Metro Project - Rashidiya Bldg 2

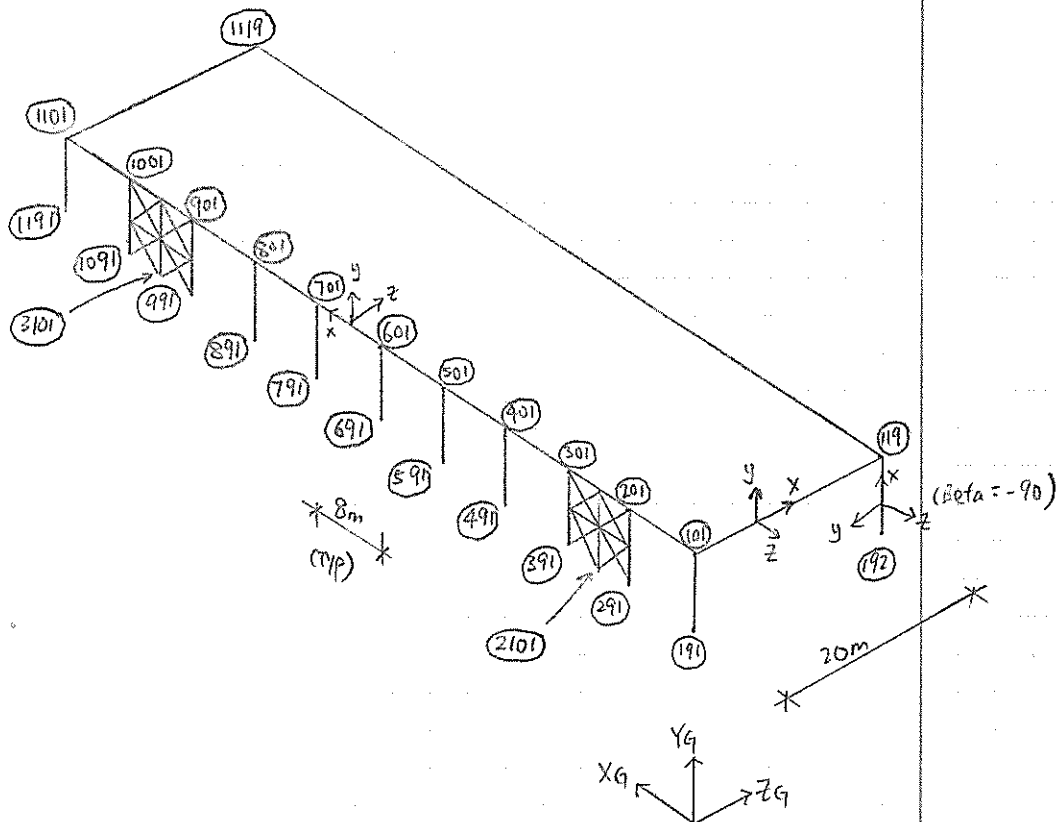
Date

Job No. 068019-01

Designed by OEH

Sheet No.

Frame 2



Roof beam

Column (main)

Column (support bracing)

Horizontal bracing (ll to roof eaves)

Horizontal bracing (near gable ends)

Vertical bracing

B33 UB 610x229x139.9

C23 UB 610x229x139.9

C4 UC 254x254x73.1

P8 $\phi 193.7 \times 5$

P6 $\phi 168.3 \times 5$

P1 $\phi 114.3 \times 4$

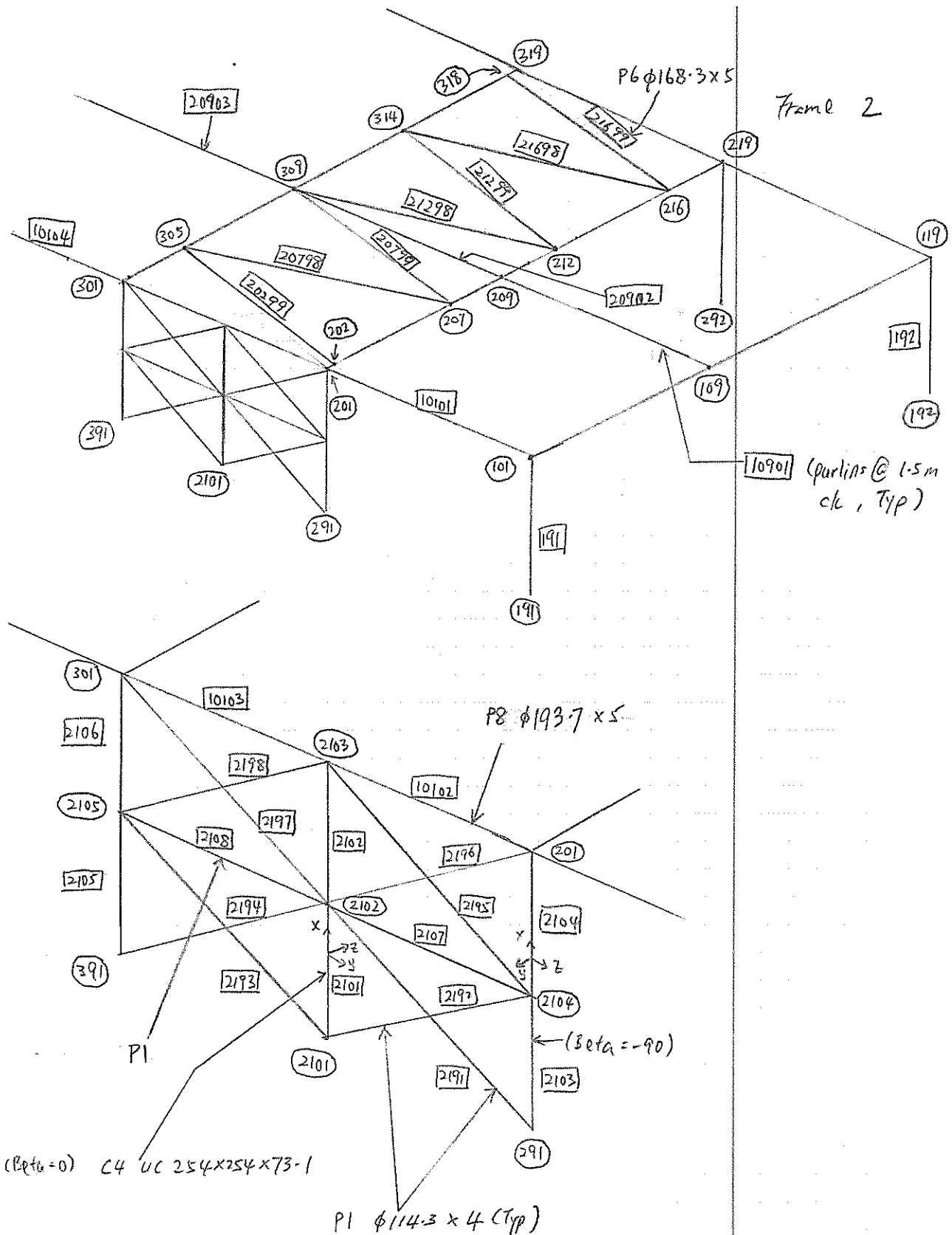
Site Dubai Metro Project - Rashidiya Bldg 2

Date _____

Job No. 062010-01

Designed by 06-11

Sheet No.



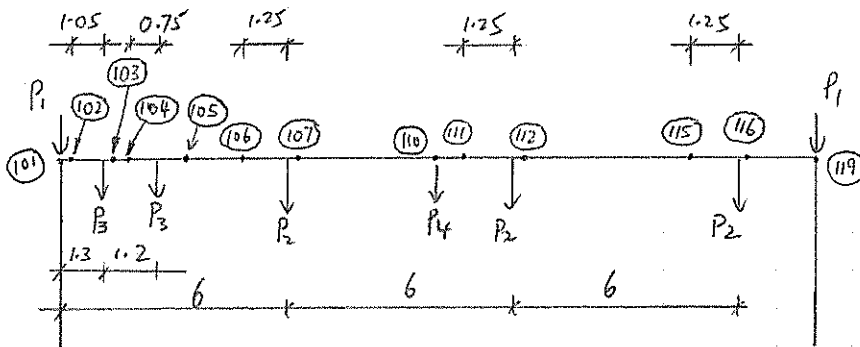
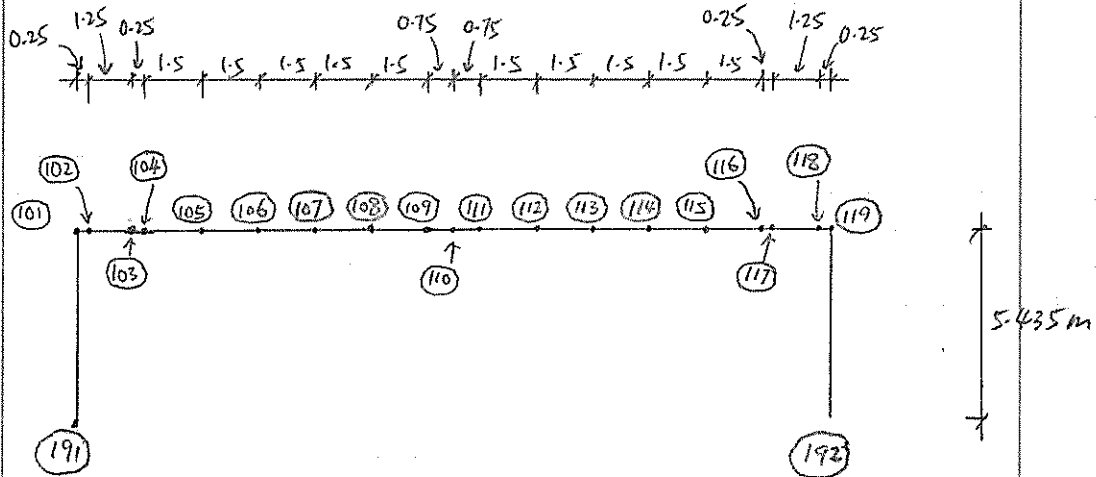
Site Dubai Metro Project - Rashidiya Bldg 2

Date

Job No. 068010.01

Designed by 024

Sheet No.



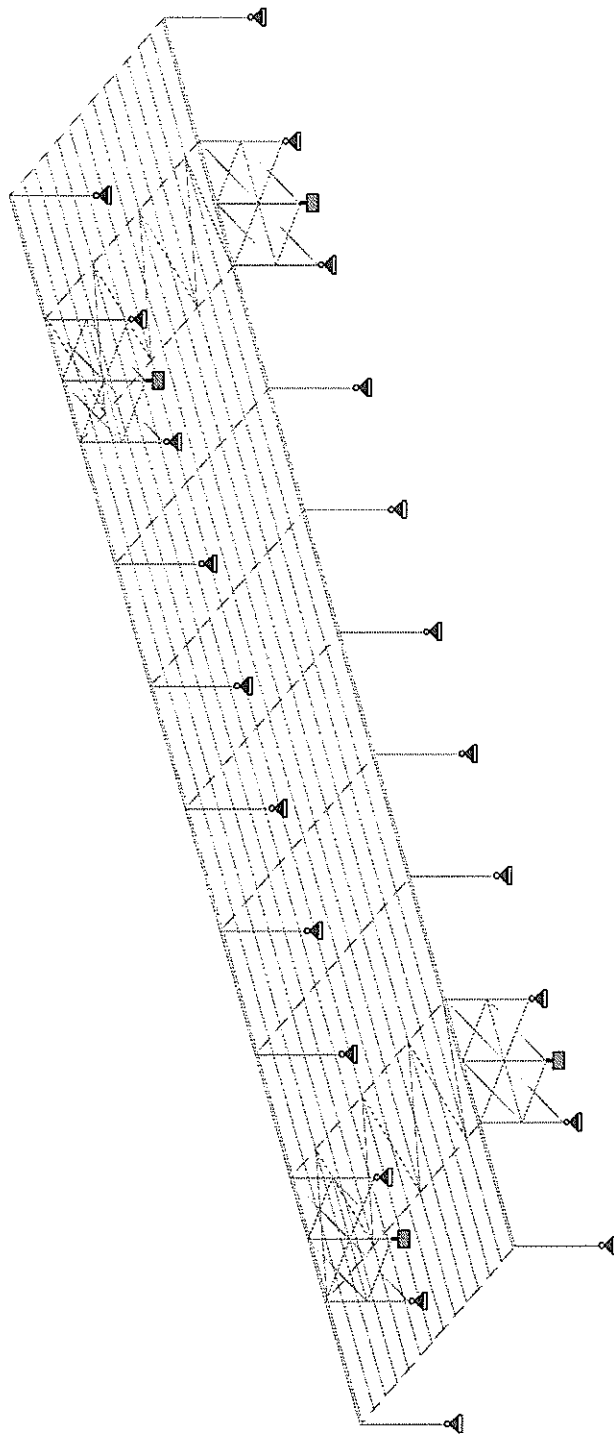


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Client - Government of Dubai _ Roads & Transport Authority

Job No	Sheet No	Rev
	1	
Part		
Ref		
By	Date	Chd
	27-Jun-06	
File	RA_BLD 2_FR2.std	Date/Time
		27-Jun-2006 20:43



Load 1

Site Dubai Metro Project - Rashidiya Bldg 2

Date

Job No. 0680/0.01

Designed by OEH

Sheet No.

Loadings :

1) Loadings on purlins

DL: s/w
sheeting + insulation $0.4 \text{ kPa} \times 1.5 = 0.6 \text{ kN/m}$

LL: $0.6 \text{ kPa} \times 1.5 = 0.9 \text{ kN/m}$

2) Loadings on main beams :

DL: s/w
M&E finishes $0.3 \text{ kPa} \times 8 = 2.4 \text{ kN/m}$

Maintenance platform, $P_3, DL = 8 \text{ kN}$

LL: Stringer system loading, $P_2 = 10 \text{ kN}$

Maintenance platform, $P_3, LL = 24 \text{ kN}$

A/C equipment $P_4 = 5 \text{ kN}$

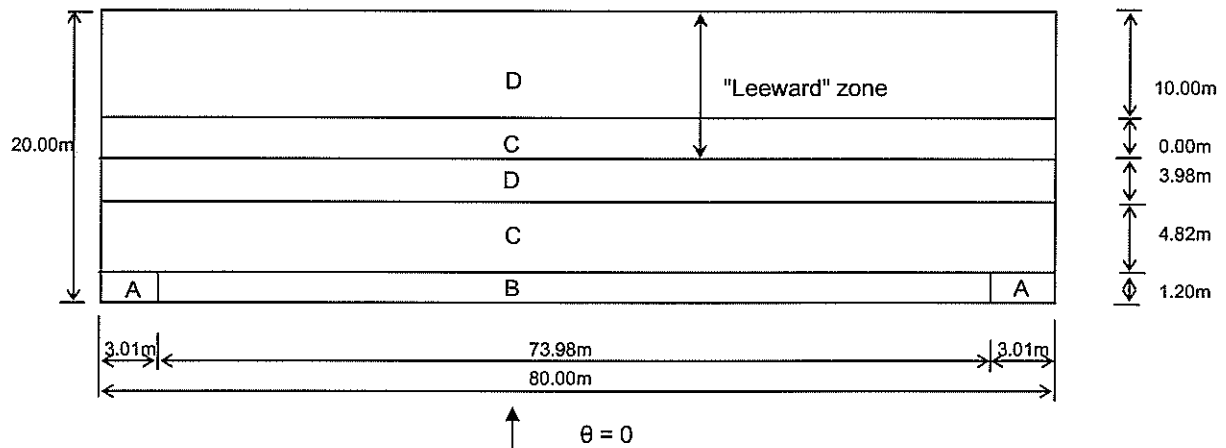
3) Loadings on columns :

DL: "Side cladding + insulation
+ side c-channel + overhang roof"
s/w $P_1, DL = 28.5 \text{ kN}$

LL: "overhang roof" $P_1, LL = 3.4 \text{ kN}$

Frame 2

Site:	Dubai Metro Project - Rashidiya Depot	Date:	27-Jun-06	Job No.:	68010.01
	Building 2 Frame 2				
	Wind Load Computation	Designed by:	OEH	Sheet No.:	

Average Cpe for roof (Wind direction normal to eave)
PLAN


Design as flat roof (with sharp eaves)

$$\begin{aligned}
 B &= 80 \text{ m} \\
 H &= 6.024 \text{ m} \\
 b &= \text{smaller of } (B, 2H) \\
 &= 2 \times 6.024 \\
 &= 12.048 \text{ m}
 \end{aligned}$$

Zone	Windward		Leeward		
	Area (m ²)	Cpe	Area	Cpe1	Cpe2
A	7.3	-2			
B	89.1	-1.4			
C	385.5	-0.7	0.0	-0.7	-0.7
D	318.1	-0.2	800.0	-0.2	0.2
Total	800.0		800.0		
Average		-0.59		-0.20	0.20